

AKADEMIKA

FOL-B-P

**FIBER OPTIC COMMUNICATION
TRAINER**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. FIBEROPTIC COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook, as they provide you with important points on safety during the installation, use and maintenance. Keep this handbook at hand for any further help.

Set all accessories in order on unpacking, and check integrity with respect to its checklist. Check the Kit for any visible damage.

Before connecting the power supply to the equipment, be sure patch cords are connected correctly, jumpers are correct as per experimental position.

In **FOL-B-P** signals are to be monitored with an oscilloscope, the scope input channel should be AC coupled, unless otherwise indicated. Ensure that X10 oscilloscope probes are used throughout.

These kits must be employed only for the use for which it has been conceived, i.e. as educational kits, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of fault or malfunctioning of the trainers, turn off the power supply and do not tamper the kits. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting patch chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on AKADEMIKA Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit /system has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on AKADEMIKA Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, AKADEMIKA bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. AKADEMIKA Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. AKADEMIKA Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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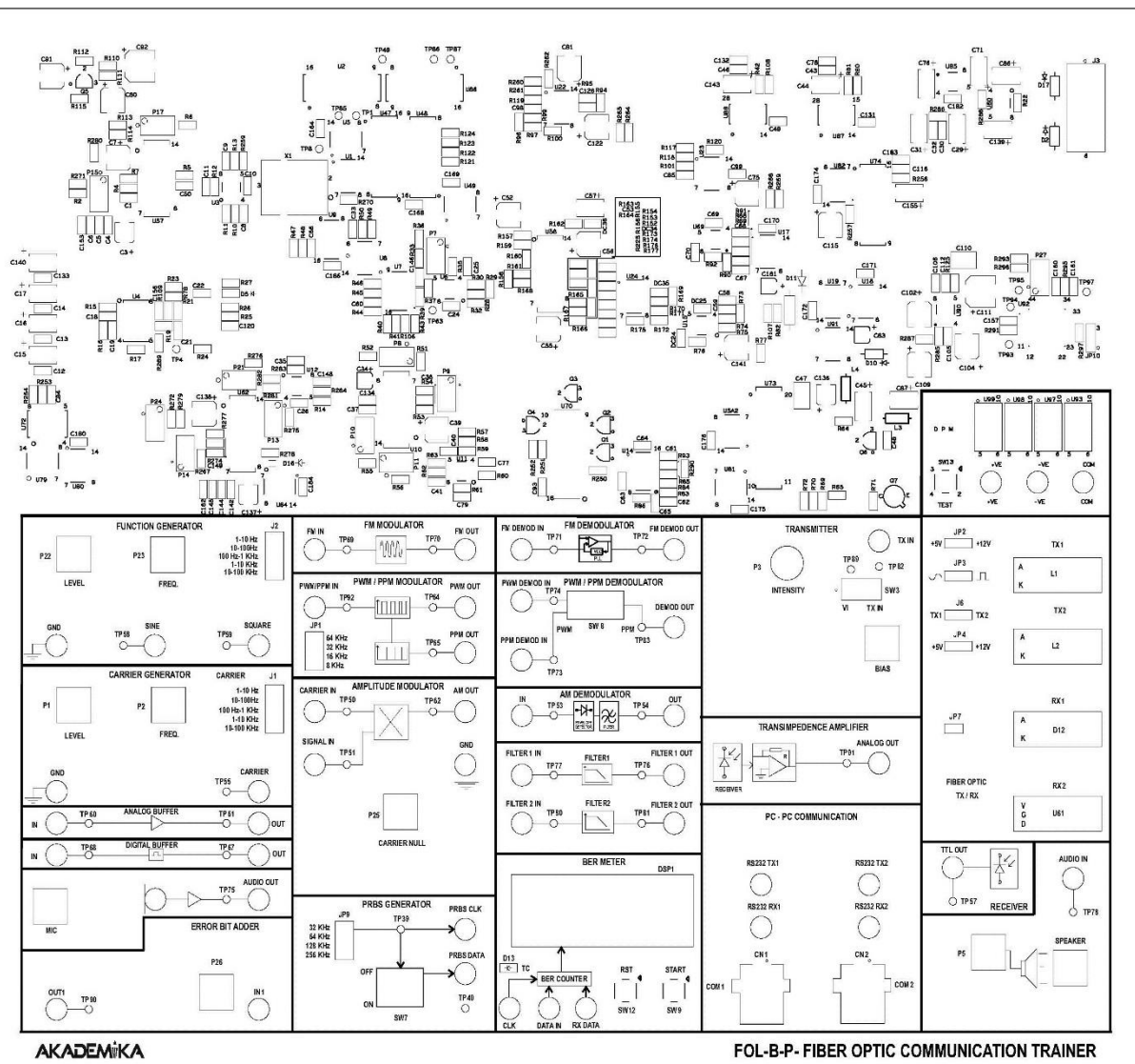
TECHNICAL SPECIFICATION

Transmitter	: 2 Nos. Siemens Fiber optics LED (i) Peak wavelength of emission 950 nm Infrared (SFH450V). (ii) Peak wavelength of emission 660 nm Red Visible (SFH756V)
Receiver	: 2 Nos. Siemens Fiber Optic Photo Detector (i) Pin Diode with responsivity of 0.3 micro Amp/micro Watt (SFH 250V). (ii) Photo Detector with TTL Logic Output (SFH551V).
Modulation Techniques	: (i) Direct Intensity Modulation (ii) Frequency Modulation (iii) Amplitude Modulation (iv) Pulse Width Modulation (PWM). (v) Pulse Position Modulation (PPM).
Analog Bandwidth	: 2 MHz.
Digital Bandwidth	: 5 MHz.
Driver Circuit	: Analog and Digital Configuration for 950 nm and 660 nm LED
Filter Circuit	: 4th Order Butter-worth Filter with 3.4 KHz Cut-off.
On-Board Function Generator	: (i) 1Hz 100 KHz, variable Amplitude Sine Wave. (ii) 1Hz -100 KHz TTL-Square waves
Voice Communication	: Fiber Optic Voice Link using dynamic MIC and Speaker
Fiber Optic Cable Type	: Plastic Optical Cable, Step index, Multimode.
Core Refractive index-n1	: 1.492.
Clad Refractive index-n2	: 1.406.
Numerical Aperture	: 0.5.
Acceptance Angle	: 60 degrees.
Fiber Diameter	: 1000 microns.
Outer Diameter	: 2.2 mm.
Fiber Lengths	: 1 and 3 Meter.
Power Supply	: $\pm 5V$, $\pm 12V$.

Accessories

FOL-B-P-II			Quantity
1	Mini Link Patch Cord	Male to Male	8
2	Short Link Patch Cord	Male to Male	8
3	FOL-B-P EXPT Manual		1
4	Power Supply		1
5	Power cord		1
6	Jumper to Crocodile 30 cm with double wire		3
7	Jumper Caps Small (5mm)		5
8	Headphone		1
9	Plastic Fiber Step Index multimode_1000um_1meter		1
10	Plastic Fiber Step Index multimode_1000um_3meter		1
11	Numerical Aperture jig ACR- 14/0		1
12	Steel Rule (15 cm /6")		1
13	Fiber polish paper 8 x 5 .5cm (0.3um (alum oxide)		1
14	USB Type A To Type-B Cable		2

FUNCTIONAL BLOCK



AKADEMIKA

FOL-B-P-FIBER OPTIC COMMUNICATION TRAINER

INTRODUCTION

Fiber Optics technology is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T; sprints have saturated the airways & print media with advertisements heralding this bright & new technology. Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

And their enthusiasm is valid. Today medical fiber optic systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologists ever come close to the fantastic growth rate that market analysts love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will pervade our homes, workplaces & recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronics newspapers become feasible. Two-way fiber optic will allow you to “attend college” in the comfort of your home.

Anyone who takes an interest in fiber optics & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few “experts” exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons exciting with the prospect of making new discoveries finding new applications for the technology.

This kit is an introduction to fiber optics communication technology for instructors, students & hobbyists. Welcome to the fascinating & expanding world of fiber optics. We hope that you find the kit challenging, stimulating & - yes – even fun.

CIRCUIT DESCRIPTION

1.1 Function Generator

The integrated circuit U62 (IC ICL8038) generates sine wave and square waveforms at their respective posts. The frequency is variable ranging from 1Hz to 100KHz. The frequency of sine wave & square wave is controlled by pot P23 and capacitors C137, C142, C144, and C145 & C167. The frequency range could be selected with the help of Jumper J2. The presets P13 and P23 adjust the symmetry of the sine signal. The amplitude of sine wave is controlled by pot P22.

1.2 Buffer

U8 (IC 74HC04) is used as TTL Buffer. IC's U4 (IC LF356) & U13 (IC LF356) are collectively used as ANALOG Buffer.

1.3 FM Modulation

The Carrier frequency is varied in accordance with the amplitude of input signal. FM Modulation is generated by using IC U10 (IC ICL8038). The frequency of the waveform generator is a direct function of the DC voltage at pin no. 8 of IC U10 (IC ICL8038), (measured from V+). By altering this voltage, frequency modulation is performed. For small deviations (e.g. $\pm 10\%$) the modulating signal can be applied directly to pin no.8 of U10, merely providing DC decoupling with a capacitor as shown in sheet 7 of FOL-B-P. The FM output obtained from U10 is fed to U11 (IC CA3140) which acts as a buffer.

1.4 Pulse Width Modulation

Pulse Width Modulation this technique of modulation controls the variation of duty cycle of the square wave (with some fundamental frequency) according to the input modulating signal. Here the amplitude variation of the modulating signal is reflected in ON period variation of square wave. Hence, it is also called as technique of V to T conversion. The circuit is divided in to two stages. First stage circuit builds around Op-Amp U6 (IC LF353). This is unity gain non-inverting amplifier designed as Level Shifter. When there is no input signal, the output of the amplifier simply follows the voltage at inverting input terminal, with inverted polarity. This is necessary because PWM chip requires only positive polarity signal. Now the input signal fed to non-inverting terminal overrides this shifted voltage. This signal is now fed to second part of the circuit consisting of U7 (IC SG 3524) & its output is fed to U8 (IC 74HC04). The IC SG 3524 is specialized for regulating Pulse Width Modulation. This primarily consists of one fixed frequency oscillator whose frequency is decided by timing register R_t and capacitor C_t . The regulation is simply given by formula

$$F = 1.18 / (R_t \times C_t)$$

Where F is in KHz, R is in $K\Omega$ and C is in μF . The four different frequencies 32 KHz, 16KHz, 10KHz & 8KHz are generated by selecting resistors R40, R41, R42, and R43 respectively. An error amplifier, comparator and flip-flop are important functioning parts internal to the IC. When no input signal is given to the error amplifier the output oscillates with the same frequency decided by the external R & C network. When input is given to the error amplifier, a linear charging of the C_t results in a linear voltage ramp, which is then fed to the comparator. This provides the linear control of the output pulse width. The amplifier output is compared with the linear charging voltage at C_t , resulting in the modulated pulse at the output of high gain comparator. Thus, we get the PWM. For more details, please refer to the IC SG 3524 data sheets.

1.5 Pulse Position Modulation

The position of the TTL pulse is changed on time scale according to the variation of input modulating signal amplitude. The circuit uses a single CMOS IC 74HC123 (U9) which is dual monostable multivibrator. The ON period of the output pulse from the monostable is

decided by the external R and C network. Here, the active low reset input pin (3) is tied to VCC. Circuit uses monostable I in the IC. Pin 2 and pin 1 are the rising and trailing edge trigger inputs of IC. Here trailing edge triggering is used. Output is taken from pin no. 13 of monostable I (Q).

Now Pulse Width Modulated signal is fed as input to this circuit. Please note that input modulating signal must be converted into Pulse Width Modulated form before applying to Pulse Position Modulator. As the signal is PWM, naturally, according to the input signal, the pulse duration is changing and this change in pulse duration causes for the delay in triggering. The input is given to trailing edge trigger input of monoshot. So finally, we get the pulses at the output, which are shifted on the time slot. This is nothing but pulse position modulation.

Thus, Pulse Positions are directly proportional to the instantaneous values of modulating signal.

1.6 Audio Pre-Amplifier

Audio Pre-Amplifier is built around Q5 (BC548) transistor in common emitter mode.

1.7 Fiber Optics Transmitter

The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. Transmitter- LED, digital, DC coupled transmitters are one of the most popular variety due to their ease of fabrication. We have used a standard TTL gate to drive a NPN transistor, which modulates the LED SFH450V source (Turns it ON and OFF).

Fiber optic transmitters are typically composed of a buffer, driver & optical source. The buffer electronics provides both an electrical connection & isolation between the transmitter & the electrical system supplying the data. The driver electronics provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally the optical source (LED) converts the electrical current to light energy with the same pattern. The LED SFH450V supplied with the FOL-B-P-II operates outside the visible light spectrum. Its optical output is centered at near infrared wavelength of 950 nm. The emission spectrum is broad, so a faint red glow can usually be seen when the LED is on in a dark room. The LED SFH450V used in the FOL-B-P-II is coupled to the transistor driver in common emitter mode. In the absence of input signal half of the supply voltage appears at the base of the transistor. This biases the transistor near midpoint within the active region for linear applications. Thus LED emits constant intensity of light at this time. When the signal is applied to the amplifier it overrides the DC level at the base of the transistor which causes the Q point of the transistor to oscillate about the midpoint. So the intensity of the LED varies about its previous constant value. This variation in the intensity has linear relation with the input electrical signal. NPN transistor (Q2) Emitter is modulated by changing Potentiometer P3 value. Optical signal is then carried over by the optical fiber. Another source used is LED SFH756V at 660nm wavelength, which is visible red light source. A standard TTL gate drives NPN transistor (Q2), which modulates the LED SFH756V source (Turns it ON and OFF).

Selection between different sources is done through jumpers provided on board.

1.8 Fiber Optics Receiver

At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter. The function of the receiver is to convert the optical energy into electrical form, which is then conditioned to reproduce the transmitted electrical signal in its original form. The detector SFH250V used in the FOL-B-P has a diode type output. The

parameters usually considered in the case of detector are its responsivity at peak wavelength & response time. SFH250V has responsivity of about $4\mu\text{A}$ per $10\mu\text{W}$ of incident optical energy at 950 nm and it has rise & fall time of $0.01\mu\text{sec}$. PIN photodiode is normally reverse biased. When optical signal falls on the diode, reverse current starts to flow, thus diode acts as closed switch and in the absence of light intensity, it acts as an open switch. Since PIN diode usually has low responsivity, a transimpedance amplifier is used to convert this reverse current into voltage formed around U12 (IC LF356). This voltage is then amplified with the help of another amplifier circuit U13 (IC LF357) & U20 (IC LF357). This voltage is the duplication of the transmitted electrical signal. There are various methods to configure detectors to extract digital data. Usually detectors are of linear nature. We have used a photo detector having TTL type output (SFH 551/V). Usually it consists of integrated photodiode, transimpedance amplifier and level shifter.

1.9 FM Demodulation

FM Demodulation is carried out using Phase Lock Loop Technique (PLL) with the help of IC U14 (IC 74HC4046). FM signal is applied to pin no. 14 & capacitor C63 & preset P16 decides the Locking frequency range of the PLL.

1.10 Pulse Width Demodulation

The PWM signal is applied to IC U15 (IC LF353) which acts as an integrator from which the signal emerges whose amplitude at any time is proportional to the pulse width at that time. This Integrated output is then filtered to get original signal.

1.11 Pulse Position Demodulation

The pulse position modulated signal is ORed with pulse generated by the rising edge of pulse width modulated signal using U18 (IC 74LS32). The o/p of the OR gate is fed to CLK I/p of flip-flop U19 (IC 74HC74). Thus flip-flop acts as a bistable multivibrator giving out high o/p for the duration between rising edge of PWM signal & PPM signal. Since PPM corresponds to the end of PWM pulse o/p of flip-flop is exactly same as that of PWM signal. This signal is then demodulated using the same technique of PWM demodulation as described in previous expt.

1.12 Audio Amplifier

Audio signal is fed to the input Pin 3 of U23 (IC LM386), through Capacitor and potentiometer P5 which controls the Volume. The gain is controlled by connecting a resistor or capacitor between Pin No. 1 and 8. The output is obtained from Pin No. 5 Which is fed to the Speaker.

1.13 Filter

Filter is formed by using R-C network and U22 (IC LM324) which forms a Low -Pass Butterworth Filter. The LOW - PASS FILTER block strongly attenuates the high-frequency ripple component at the detector's output, and also blocks the d.c. offset voltage. Consequently, the signal at the output of the LOW - PASS FILTER block should very closely resemble the original audio modulating signal.

Note that the demodulated signal has null continuous component. Vary the amplitude and frequency of the FM signal and check that the amplitude of the detected signal varies, too.

1.14 Amplitude Modulation

The integrated circuit LM1496 (U58) consists of a differential amplifier, driven by a further differential stage. The output signal consists of a constant, which multiplies the two input signals CARRIER IN and SIGNAL IN. The pot CARRIER NULL enables the balancing or unbalancing of the circuit: in the first case (balance state) the output is exactly the product of the input signals, and the circuit operates as balanced modulator; in the second case (unbalance state) the output contains a fixed component of the signal CARRIER and the circuit operates as amplitude modulator.

1.15 Amplitude Demodulation

This section consists of diode envelope detector. Positive envelope from the input signal is separated out and it is further filtered out using low pass filter circuit built around U15 (IC LF353), to recover the actual analog information signal transmitted.

1.16 Bit Error Rate Meter

Transmitted data and received data bits are passed through buffer U81(74HCT04) to D-flip-flop U5A2A (74HCT74). Output of the D-flip-flop is EX-ORed. The error pulse generated through PAL and is given to U70 (74C926), which is a 4-digit counter with multiplexed 7 segment output drivers. The error measured is displayed on the 4 digit 7 segment display with a decimal value. The counter can be reset through RST switch followed by start switch.

1.17 Error Bit Adder

Refer to the circuit diagram as shown later in this FOL-B-P manual. 555 timer is configured as astable multivibrator. An astable multivibrator, often called a *free-running* multivibrator, is a rectangular-wave-generating circuit. Unlike the monostable multivibrator, this circuit does not require an external trigger to change the state of the output, hence the name *free-running*. However, the time during which the output is either high or low is determined by the two resistors and a capacitor, which is externally connected to the 555 timer IC U29 adds these error pulses with data

1.18 Adder

Refer to the circuit diagram as shown later in this FOL-B-P manual. Adder circuit is built around U19 (IC CA3140). It adds the noise and the data and gives a noisy data at the output of the adder, whenever the noise level is high.

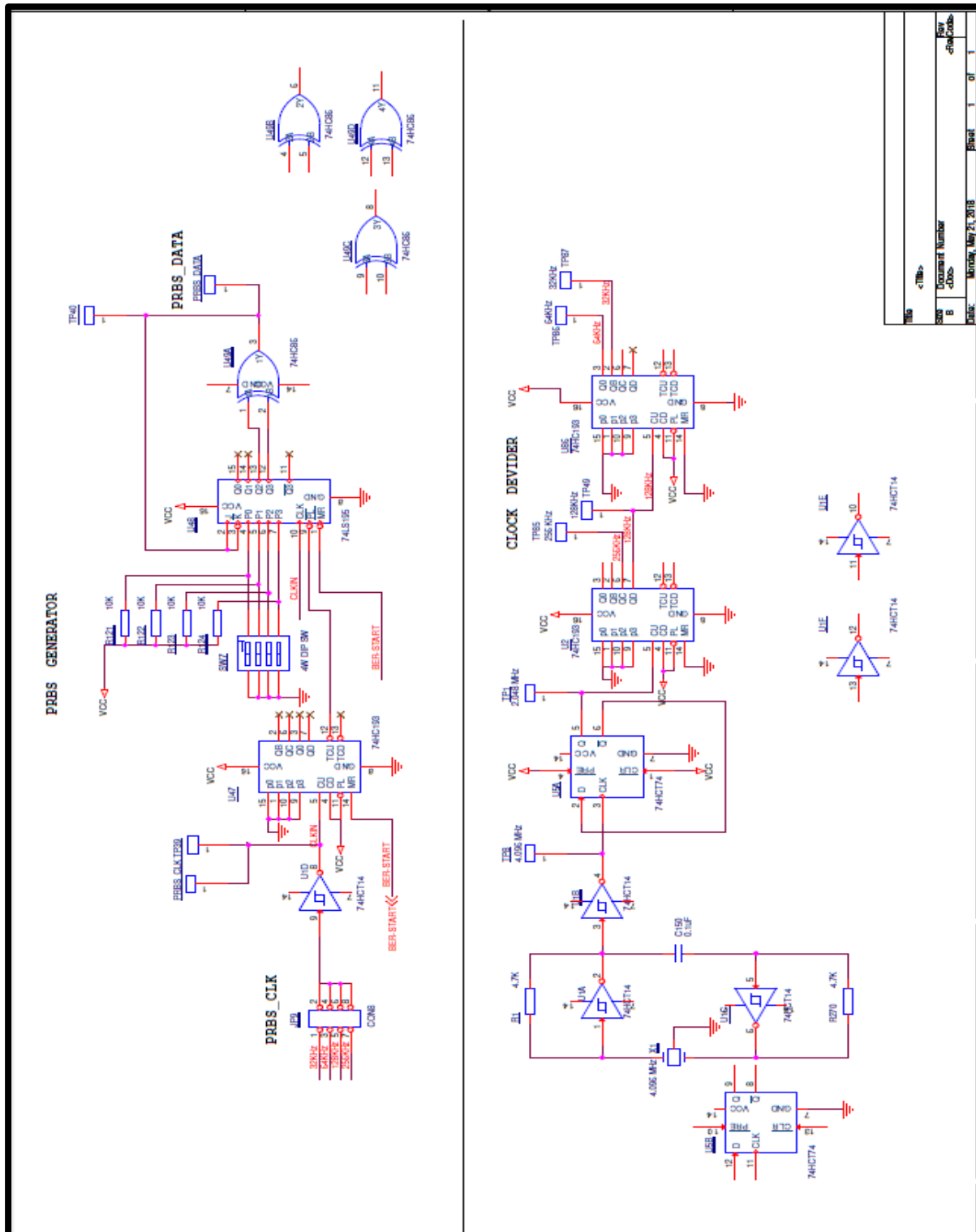
1.19 PRBS Generator

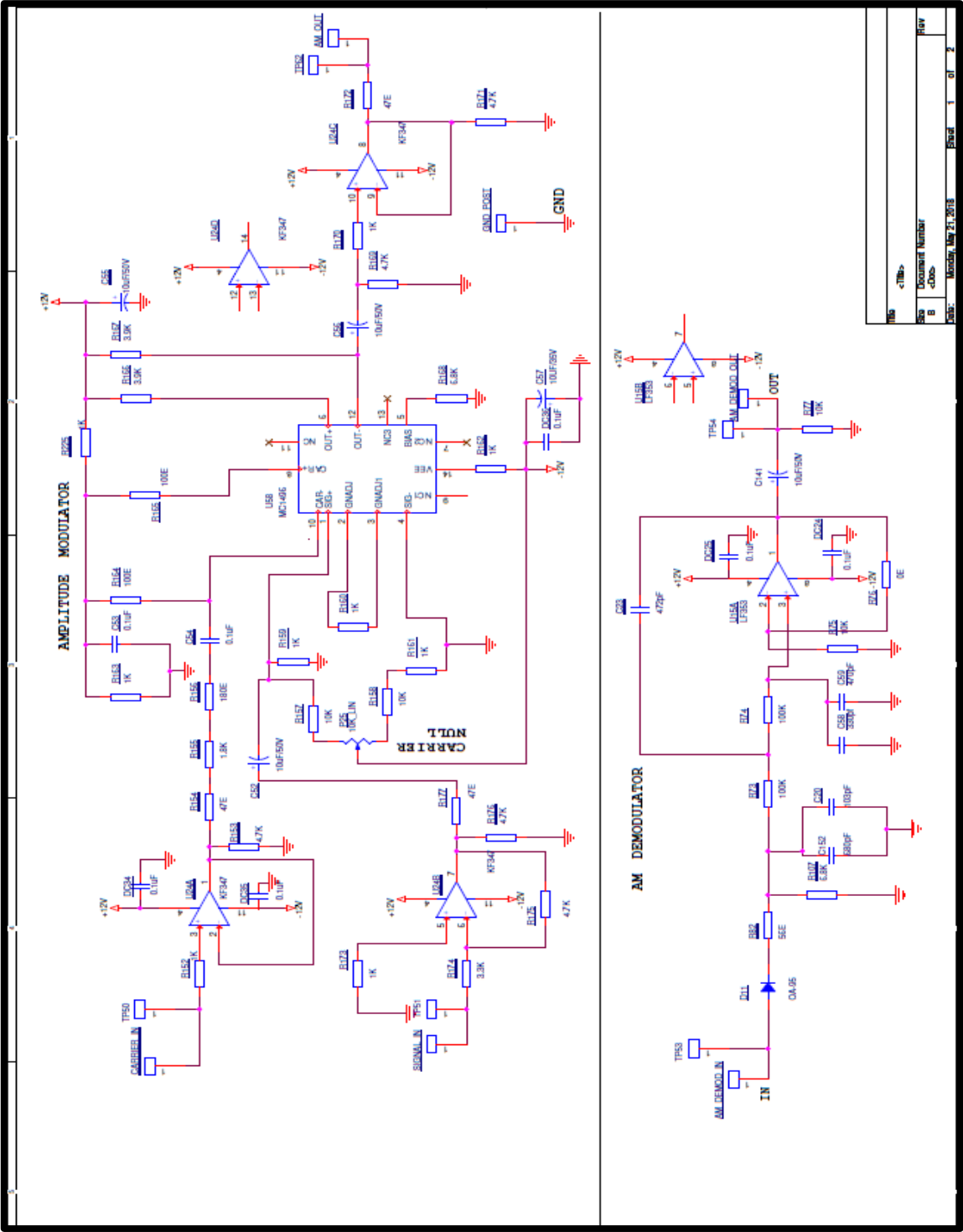
It is a 14-bit Pseudo Random sequence used as a data for BIT ERROR RATE. A 4-bit DIP switch SW7 is used to change the pattern of PRBS sequence. A 4-bit shift register U48 (74LS195) generates the pseudo Random sequence, which is a unique pattern that has minimum correlation with the data. Different clock frequencies starting from minimum 32KHz to 256KHz is fed to the PRBS generator to generate the PRBS pattern of different frequencies. The PRBS generator is a 4-bit shift register with an EX-OR U49 (IC74HC86) gate at the feedback.

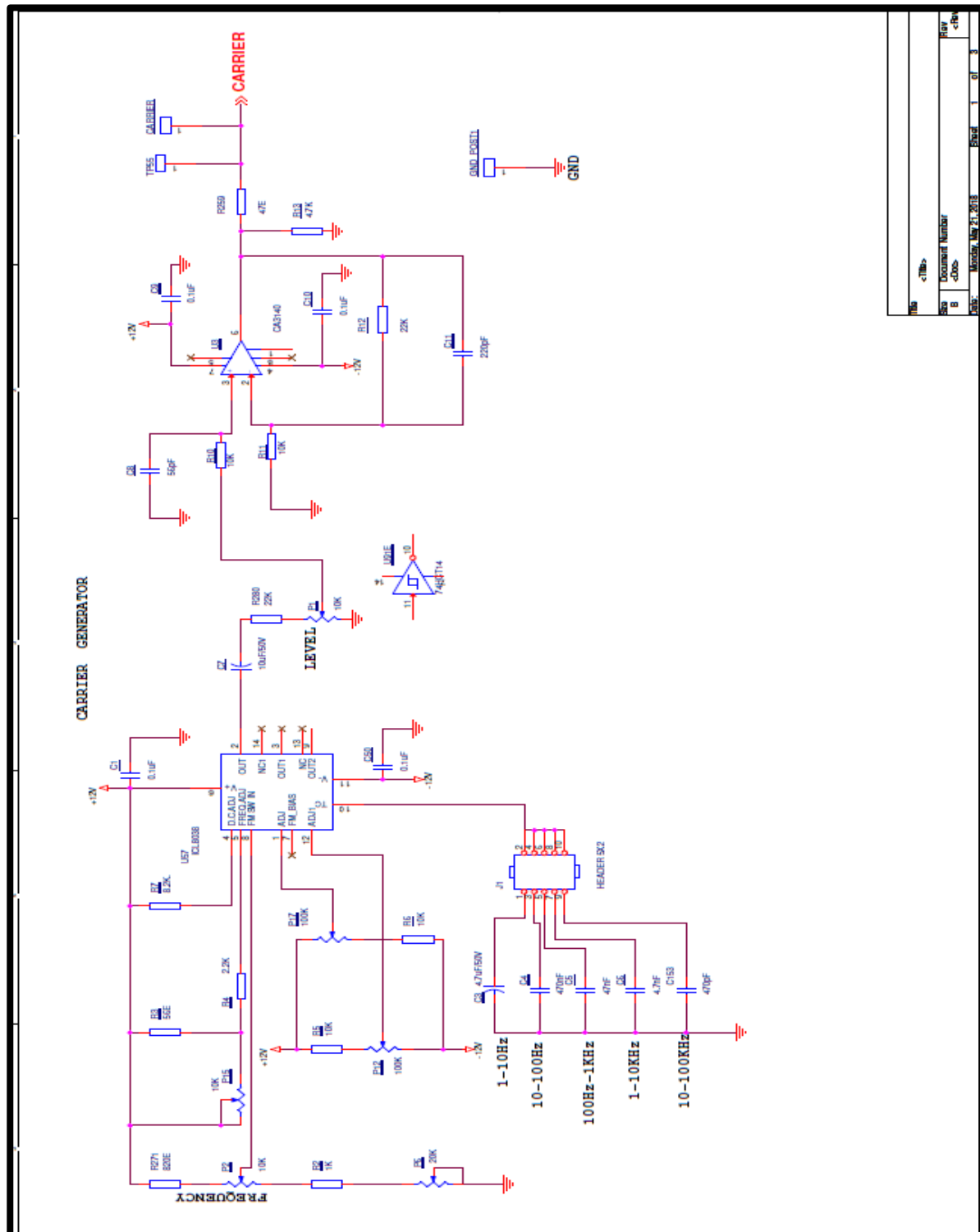
1.20 PC- PC Communication

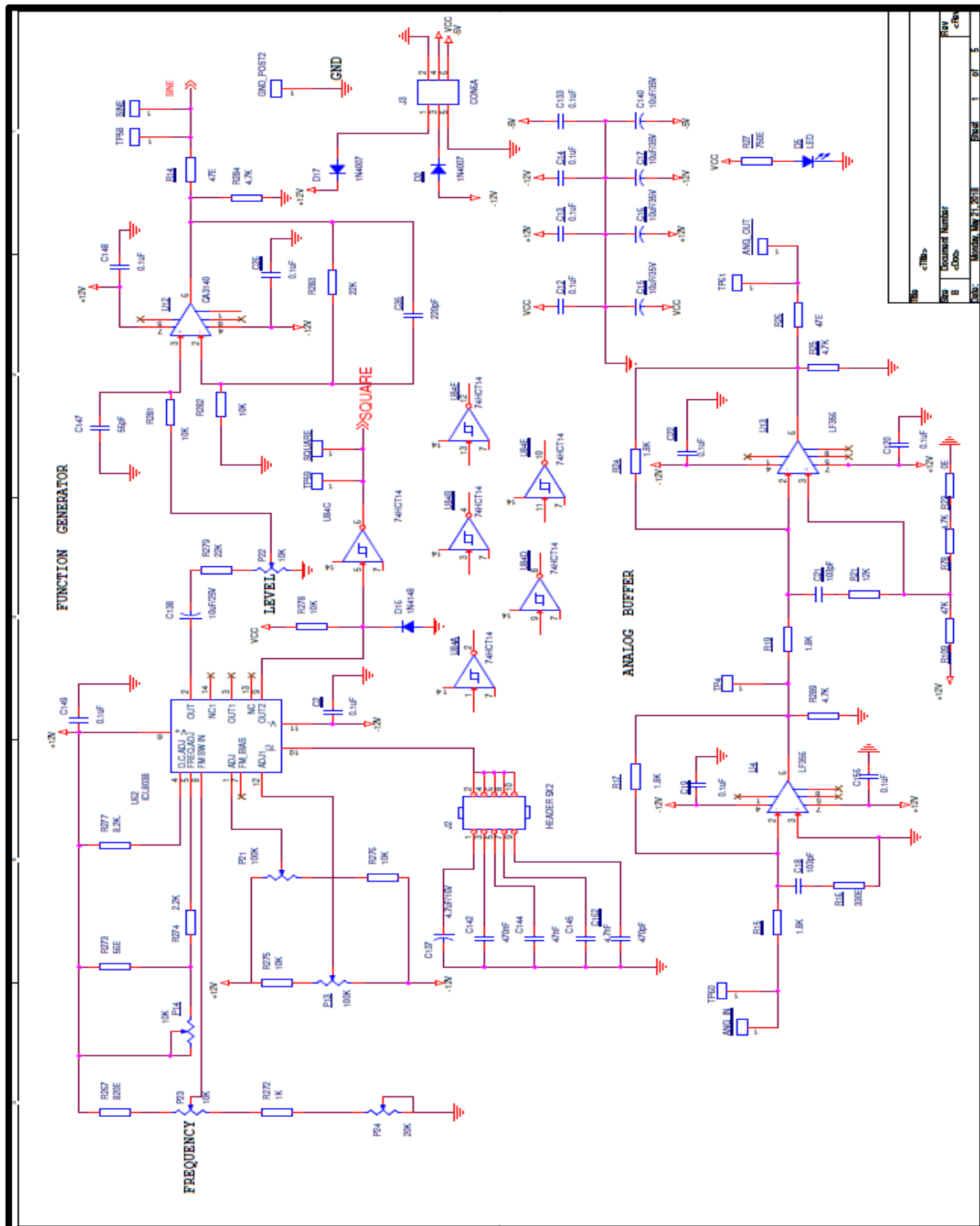
U87 & U88 (IC FT232RL) is standard IC, is used for PC-PC communication. The two USB Type-B connectors are used for Interfacing PC's. Here data Transmitted/Received by the PCs is fed to CN 1 and CN 2 connectors. RS232_TX1,RS-232_RX1,RS232_TX2 and RS232_RX2 posts are used to connect the electrical signal to the Fiber Optic Transmitter and Receiver Circuit.

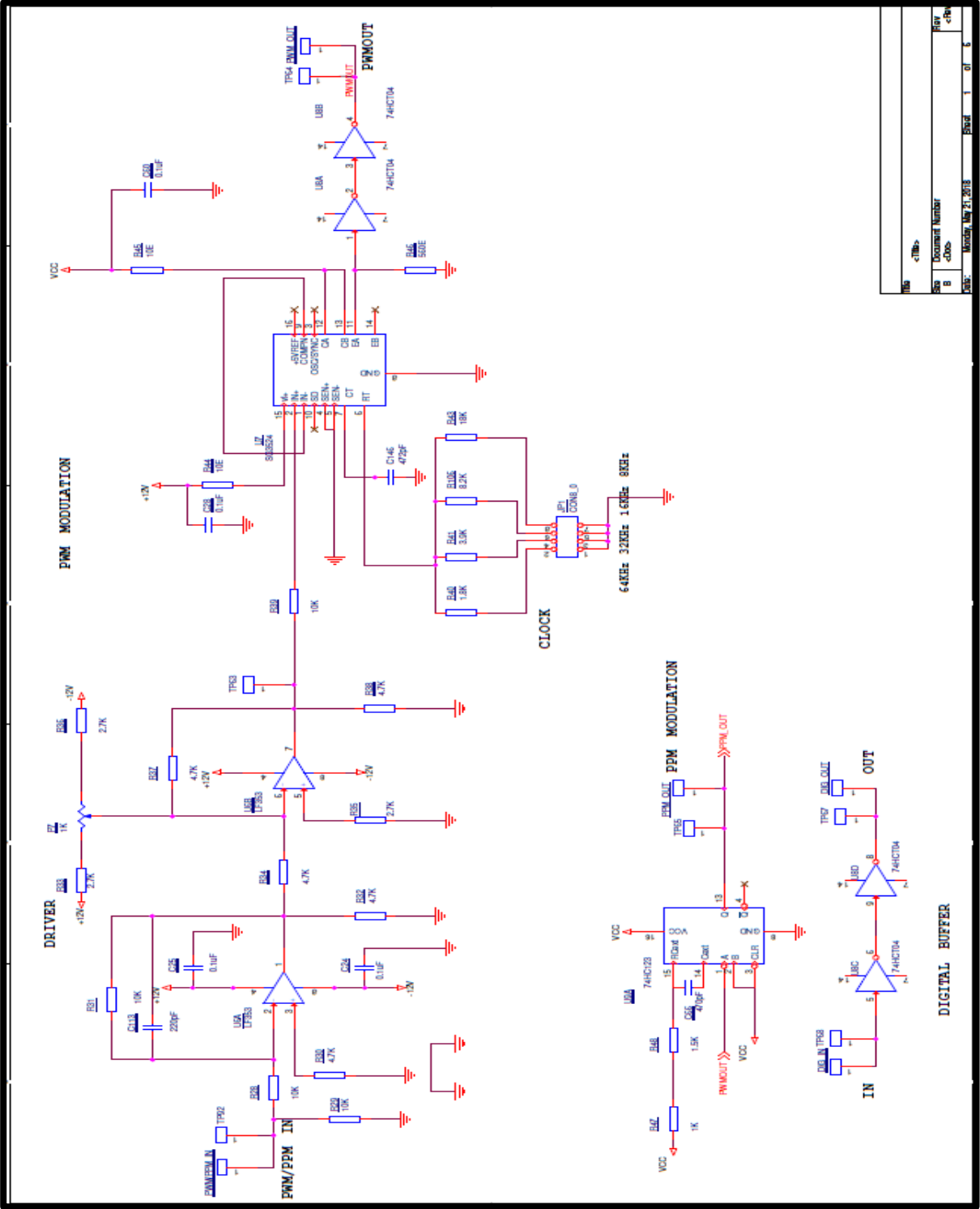
CIRCUIT DIAGRAM

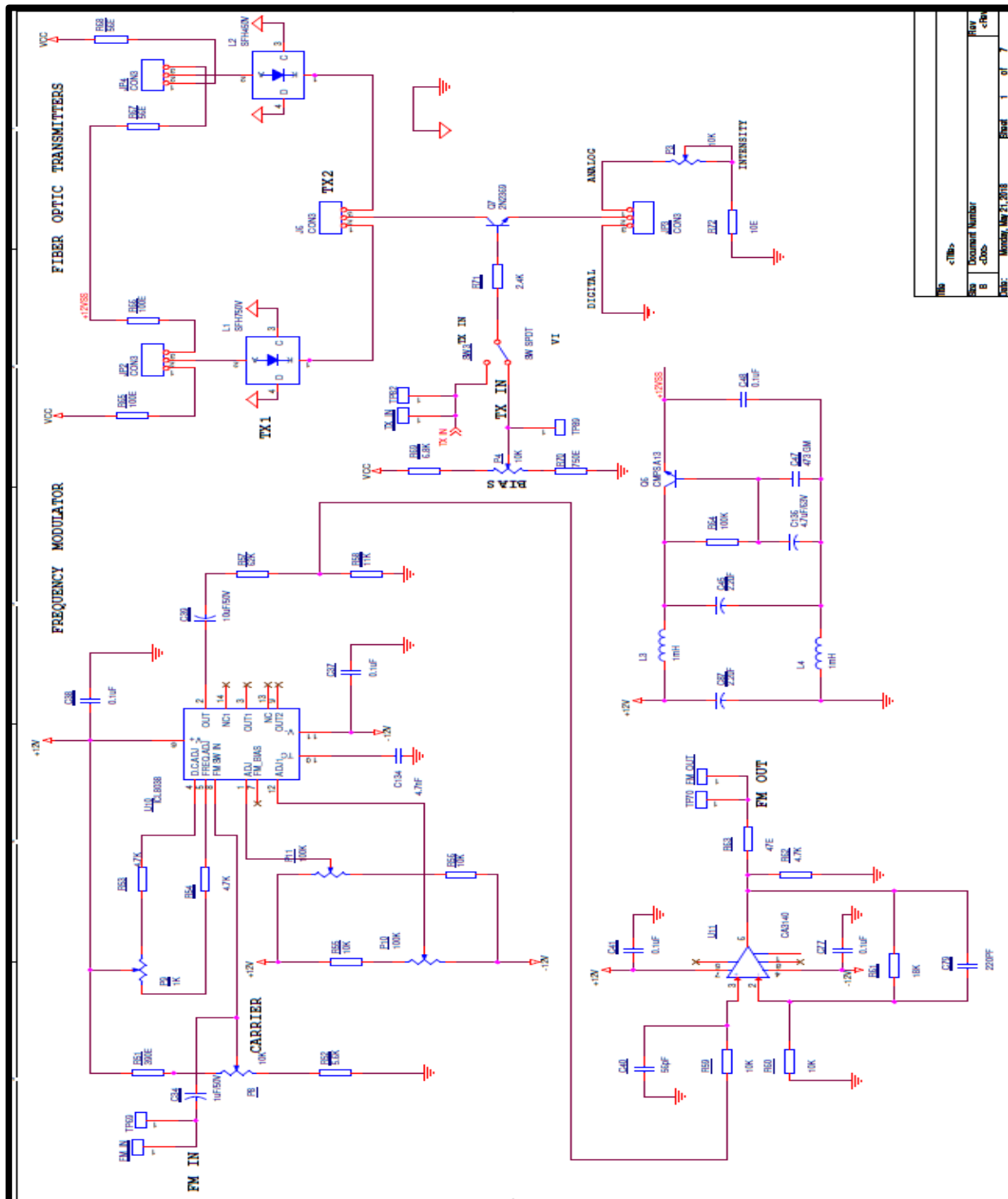


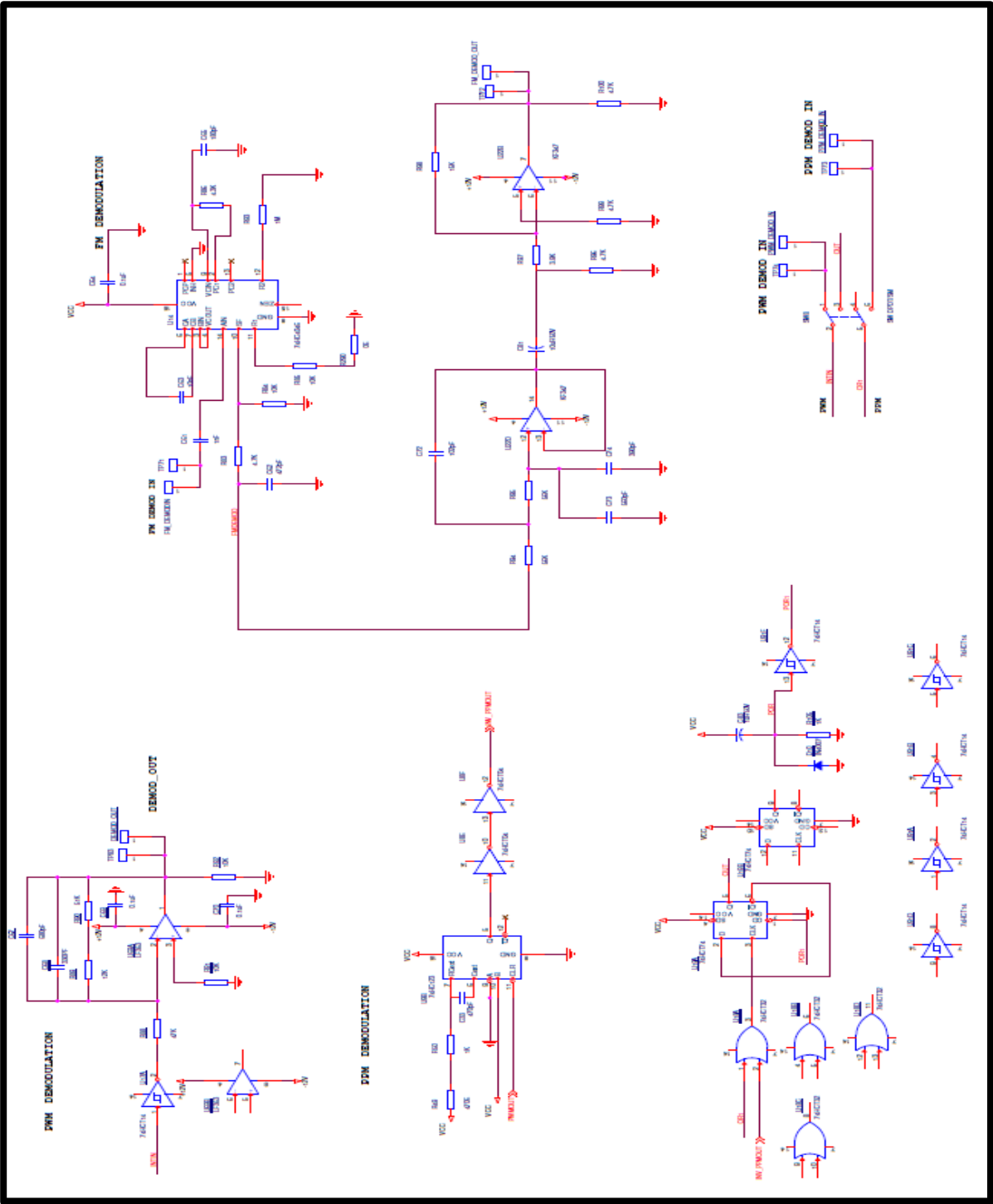


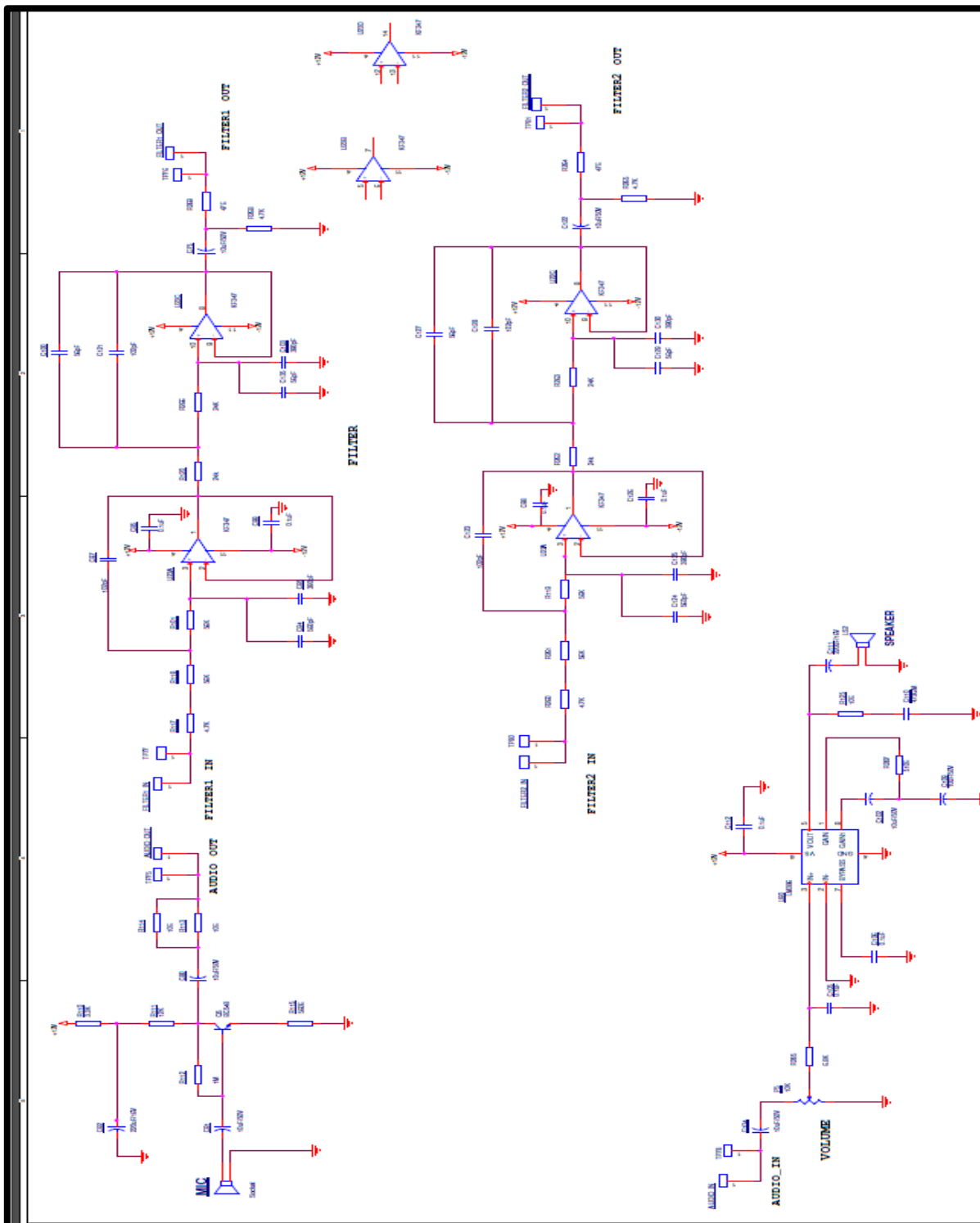


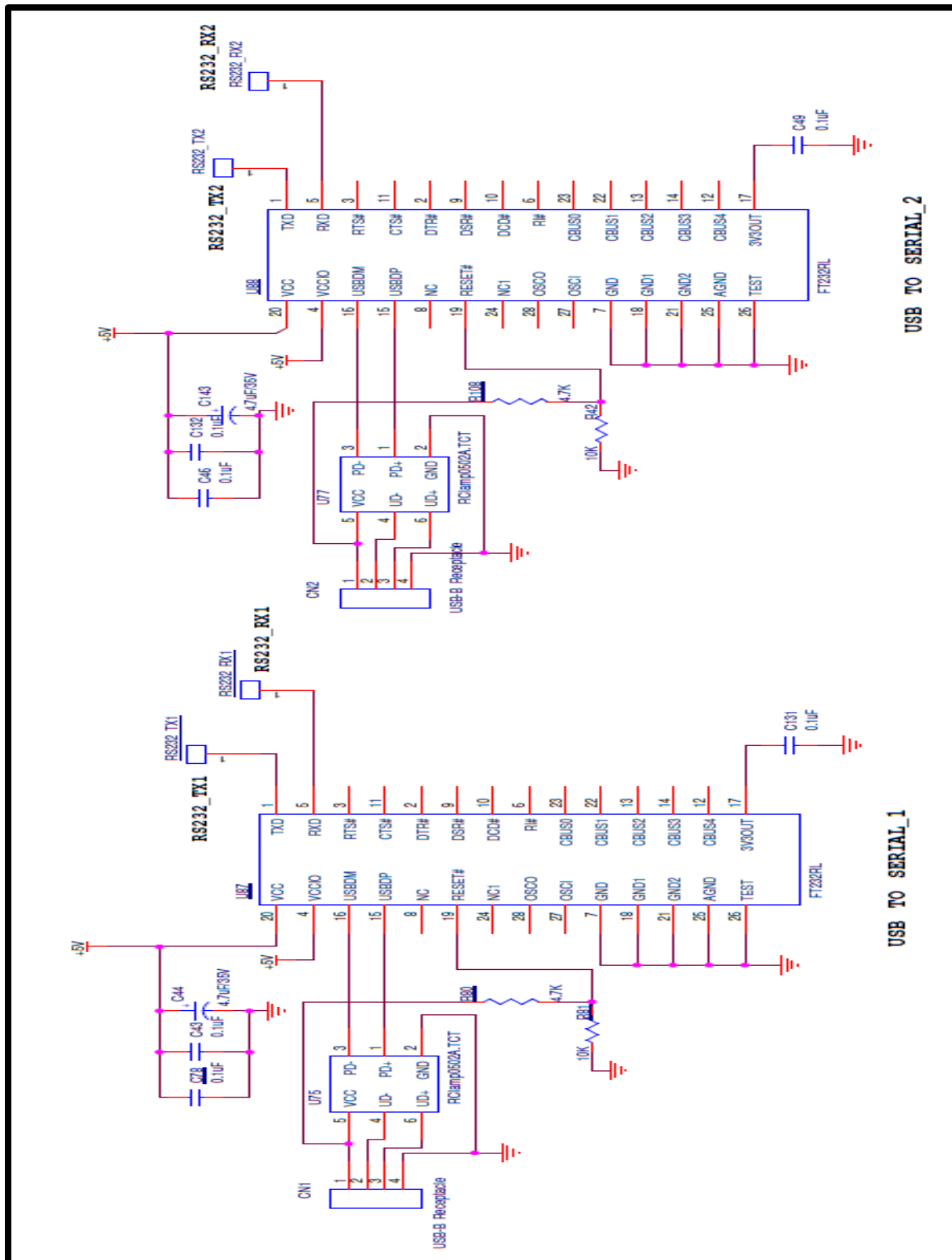


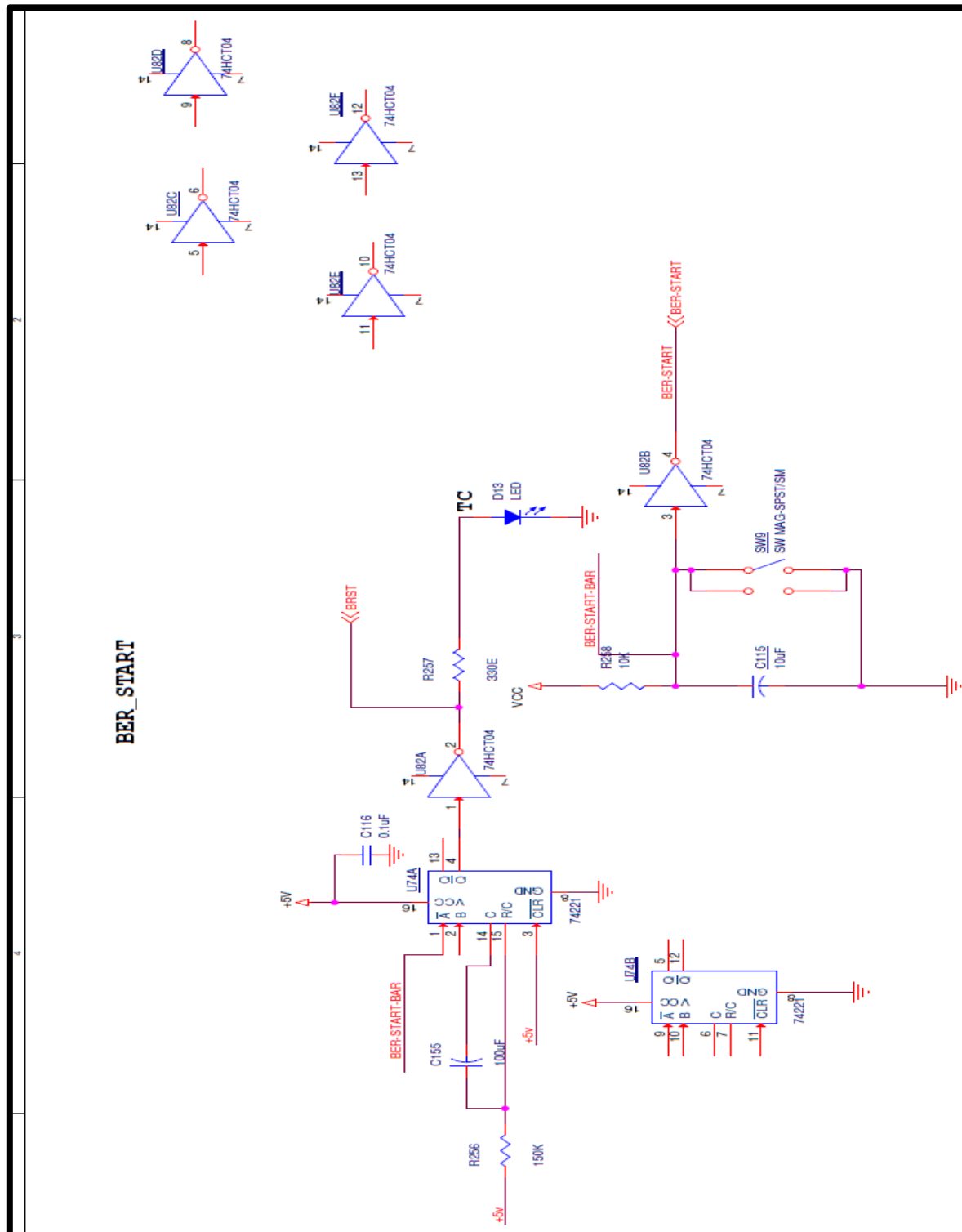




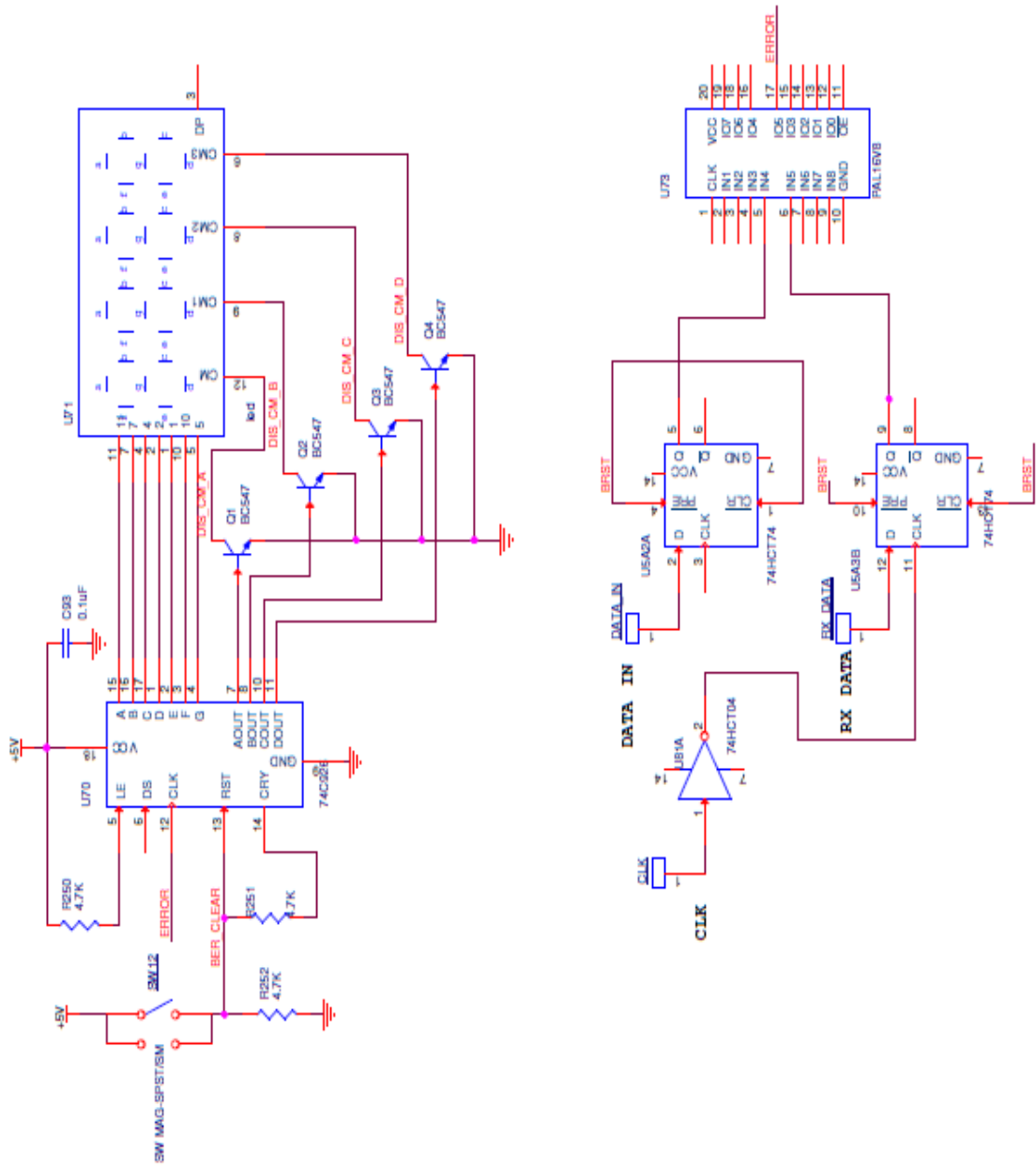


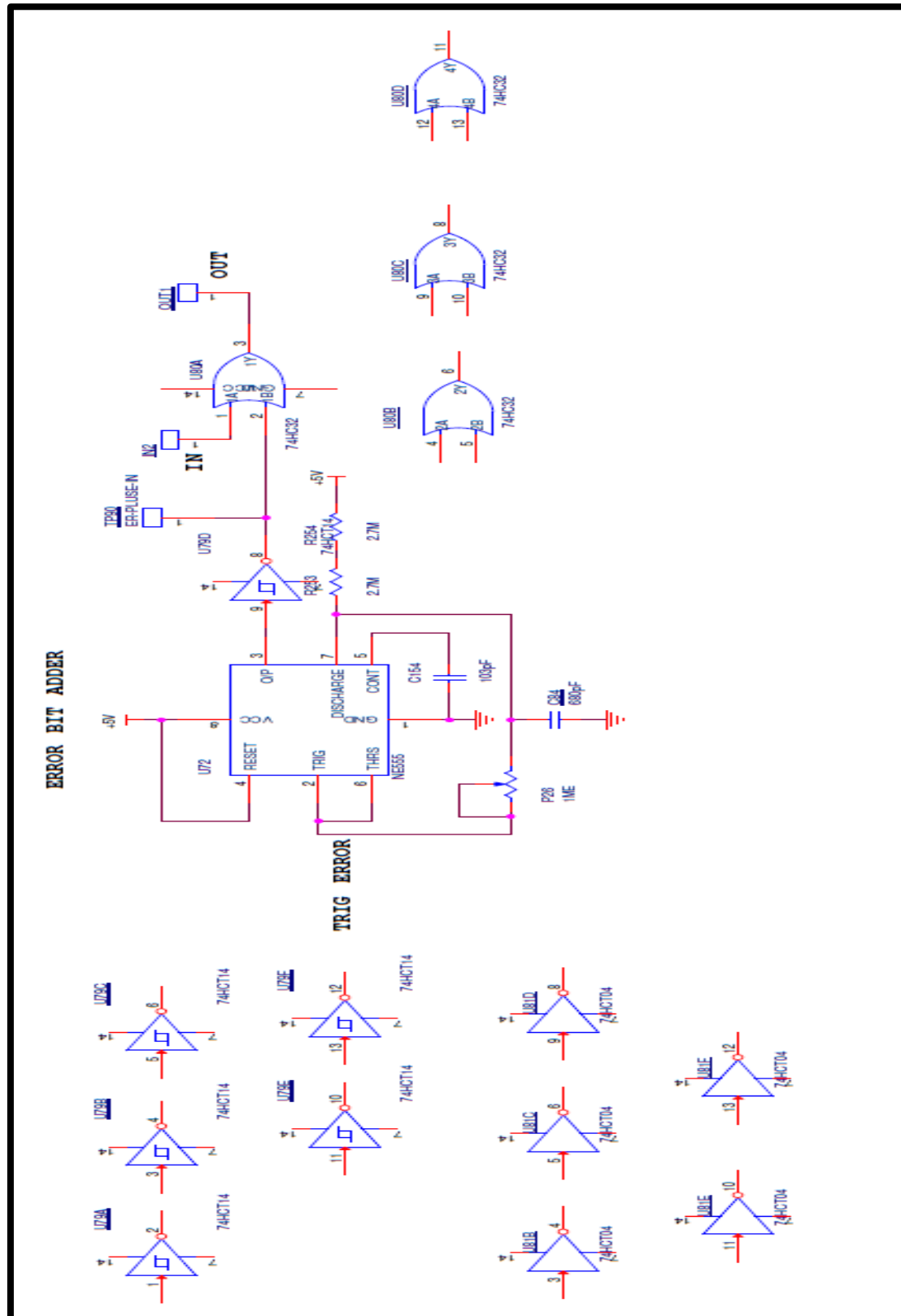


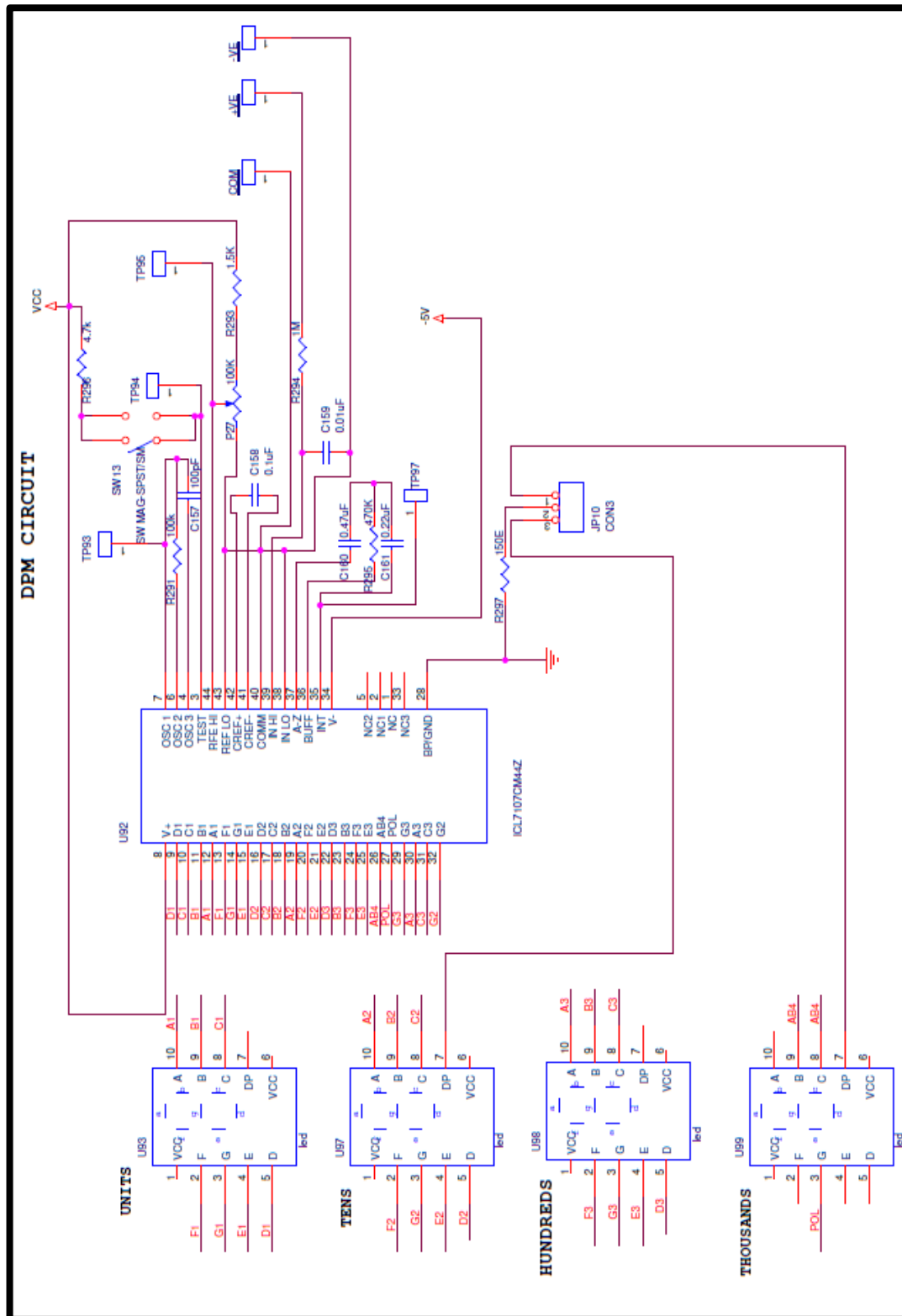




BIT ERROR RATE METER







ABOUT FIBER OPTICS

Before fiber optics came along, the primary means of real time data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic (radio) waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances – using light energy. Although initially a very controversial technology, today fiber optics has been shown to be very reliable & cost effective.

Light as utilized for communication has major advantages because it can be manipulated (modulated) at significant higher frequencies than electrical signals can. For example, a fiber optic cable can carry up to 100 million times more information than a telephone line! The fiber optic cable has lower energy loss & wider bandwidth capabilities than copper wire. As you will learn, fiber optic communication is a quite simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today. Fiber optics became reality when several technologies came together at once. It was not an immediate process, nor was it easy, but it was most impressive when it occurred. An example of one critical product, which emerged from that technological merger, was the semiconductor LED that we used in our kit.

How Optical Fiber Works?

The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Air & water are a case in point as shown in FIG. A. The light ray traveling through air actually is bent as it enters the water. The amount of bending depends on the refractive indices of the two materials involved & also on the angle of the incoming (incident) ray of light as it strikes the interface. The angle of the incident ray is measured from a line drawn perpendicular to the surface. The same is true for the angle of the outgoing (transmitted) ray of light after it has been bent.

Snell's Law explains the mathematical relationship between the incident ray & the refracted ray

$$n_1 \bullet \sin \theta_1 = n_2 \bullet \sin \theta_2$$

In which n_1 & n_2 are the refractive indices of the initial and secondary materials respectively, and θ_1 and θ_2 are the incident and transmitted angles.

Snell's Law says that refraction (bending) of light cannot take place when the angle of incidence grows too large (as when light travels from a material with a high refractive index to one with a low refractive index). If the angle of incidence exceeds a certain critical value (in which the product of n & the sine of the angle equals or exceeds one.) light cannot exit.

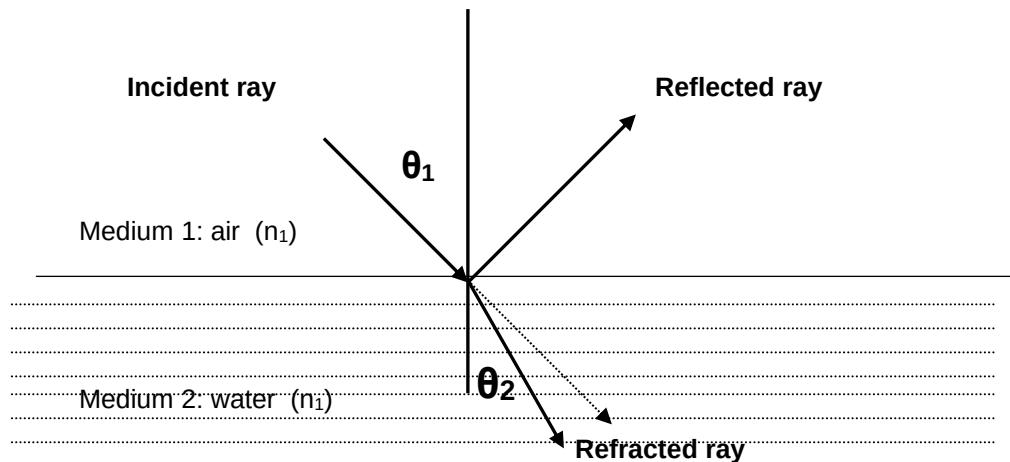


Fig.A: The Different Portions of A Light Ray At A Material Interface

If light cannot exit the material, it is reflected. The angle that is reflected is equal to the angle of incidence. The phenomenon just described is called Total Internal Reflection and it is what keeps light inside an optical fiber.

Types of Optical Fiber

The simplest fiber optic cable consists of two concentric layers of transparent materials. The inner portion (the core) transports the light. The outer covering (the cladding) must have a lower refractive index than the core, so the two are made of different materials. The cable used in this kit also has a jacket to protect the optical properties of the core & cladding.

Optical fiber is generally made from either plastic or glass. This kit uses plastic fiber that is very easy to terminate & does not require special tools. Plastic fiber is generally limited to uses involving distance of less than 100 meters because of high attenuation (higher loss). Glass fiber on other hand has very low attenuation (low loss) & hard to cut, requires special end connections & is more expensive.

The core of fiber is typically made of silica doped with impurities, which increases the refractive index relative to pure silica. The cladding, which surrounds the fiber core & creates an optical interface, is typically made from pure silica. The outer buffer coating is a plastic cover.

The first type of fiber optic cable put to use was called Step Index. In this design the cladding has a different index of refraction from the core. The light bounces off the sides & is reflected back into the fiber core. The problem with this design is that the reflected light must travel a slightly longer distances than that which travels down the center of the fiber, thus limiting the maximum transmission rate. This design was improved with the use of Graded Index Fiber. In this design, the index of refraction decreases in proportion to the distance away from the center of the fiber core. The light moves more quickly in the outer portion, thus compensating for the additional distance. The change in index also has the effect of 'bending'. The light reflects back towards the core. This change increases the transmission capacity. In the newest Single Mode design, the diameter of the fiber core is made so small that all the light travels in a straight line. Even the latest fiber optic facility in use today uses less than 5% of the maximum theoretical capacity of a Single mode Fiber.

Single Mode V/S Multimode

There are essentially two different types of fiber optic transmission schemes in use. Multimode fiber optic systems are typically used in short haul applications such as LANS & FDDI. Recent growth in multimode installations has been driven by LAN/FDDI applications. The term Multimode means that the diameter of the fiber optic core is large enough to propagate more than one mode (electromagnetic wave). Because of the multiple modes the pulse that is transmitted down the fiber tends to become stretched over distance. This is referred to as Modal Dispersion & has the effect of reducing available bandwidth.



Fig. B: Modal Dispersion Of Light

Typical wavelengths used in multimode applications are 850 nm & 1300 nm. There are other wavelengths used (e.g. 660 nm for plastic fiber, 820/870 nm) in specialized applications such as the military. The vast majority of multimode applications are 850 or 1300 nm wavelengths. Most multimode cables in use today are graded index in type with a cladding diameter of 125 microns & fiber core diameter of 62.5 or 50 microns (Typically referred to as 62.5/125 or 50/125 fiber)

Single Mode Systems are generally used in long haul applications (10's or 100's of kilometers) & local loop applications in both the carrier & CATV markets. Some large utilities such as electrical power companies also utilize single mode fiber optic backbones. As the name implies single mode fiber is designed to propagate only one mode of light. It is therefore not affected by modal dispersion & has higher bandwidth capacity. It is considered to be higher quality transmission system. Wavelengths used in single mode applications are 1300 & 1550 nm. The majority of today's installed systems uses 1300 nm wavelength technology & is designed for higher speed applications. These systems are more sensitive to back reflections from connectors & sharp cable bends. They often require additional tests, such as Optical Return Loss, to be run. Most single mode cable in used today are step index in type with cladding diameter of 125 micron & fiber core diameter of 9 micron (Typically referred to as 9/125 fiber).

Advantages of Fiber Optics over Conventional Copper Cables

- Much greater bandwidth (Several orders of magnitude are theoretically possible).
- Immunity to electrical disturbances, ground loops, cross talks etc. in addition, no EMI radiation is generated.
- Much lighter. A 3-inch bundle of 900 twisted copper pairs can carry about 21,000 channels of traffic & weighs about 25,000 pounds/mile. A 1/2 inch fiber link with 12 fiber strands can carry about 3,00,000 channels & weighs about 200 pounds/mile. In addition less duct space is required to make the cable within building.
- Better in hostile environment, not affected as much by temperature, water etc.
- Lower transmission loss. Typical loss on a fiber link is 0.2 dB/Km. On a copper based facility, one can usually expect at least 5 db/km. For longer links, fewer repeaters are required.

- Better security as it is not possible to simply bridge onto the facility & monitor the traffic.

With all these properties of optical fiber, it is becoming very important in sensing field also. Most of the properties of optical fiber are useful for sensors. Further optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

ABOUT FIBER OPTIC TRAINER KIT FOL-B-P

The purpose of FIBER OPTIC TRAINER KIT is to provide your hands on experience on the various Fiber Optic & Digital Communication Techniques. In preparing this manual, we assumed you have a basic grasp of digital & transistor circuits. The experiments that can be performed with this Kit have been designed in a ladder fashion.

You have to start from the first step & then go for the next. Experiments start with basic principles of fiber optics & then go on to more sophisticated techniques. Every advanced experiment uses concept, procedures, and results of the previous experiment. So you are advised not to skip in between experiments. In this manual before starting the procedure of every experiment, objective of the experiment & basic theory related to that is given. You are advised to read it before starting the experiment because it will help you to understand the experiment very well. Also wherever necessary refer the datasheets of various components provided on the backside of the manual.

Before starting the experiments check following things in the kit.

- Trainer Kit FOL-B-P
- Plastic fibers of 1 & 3 Meter length.
- NA JIG
- Steel Rule
- Speaker and Microphone
- 1 No. Power Supply & Power Chord
- 8 Nos. Shorting Links
- 8 Nos. Mini Links
- 3 Nos Jumper to Crocodile
- Experimental Manual

Optical Fiber Preparation Instructions

The fibers provided with this Lab are multimode, step index, 1000-micron plastic fiber & are factory prepared for use. They are not require to prepared again even after several operations, if handled carefully, In case the fiber is broken or damaged you have to prepare each end of the optical fiber cable very carefully so it transmits light effectively.

- Cut off the ends of the cable with a single-edge razor or sharp knife. Try to obtain a precise 90-degree angle.
- Wet the polishing paper with water or light oil & place it on a flat, firm surface. Hold the optical fiber upright, at right angle to the paper & polish the fiber tip with a gentle "figure 8 " motion. You may get the best result by supporting the upright fiber against some flat object such as a portion of the printed circuit board.
- Using an 18-gauge wire stripper, remove 3 mm of the jacket from the end of the fiber. Be careful not to nick the buffer in the process or you will be losing light later.

EXPERIMENT NO. 1

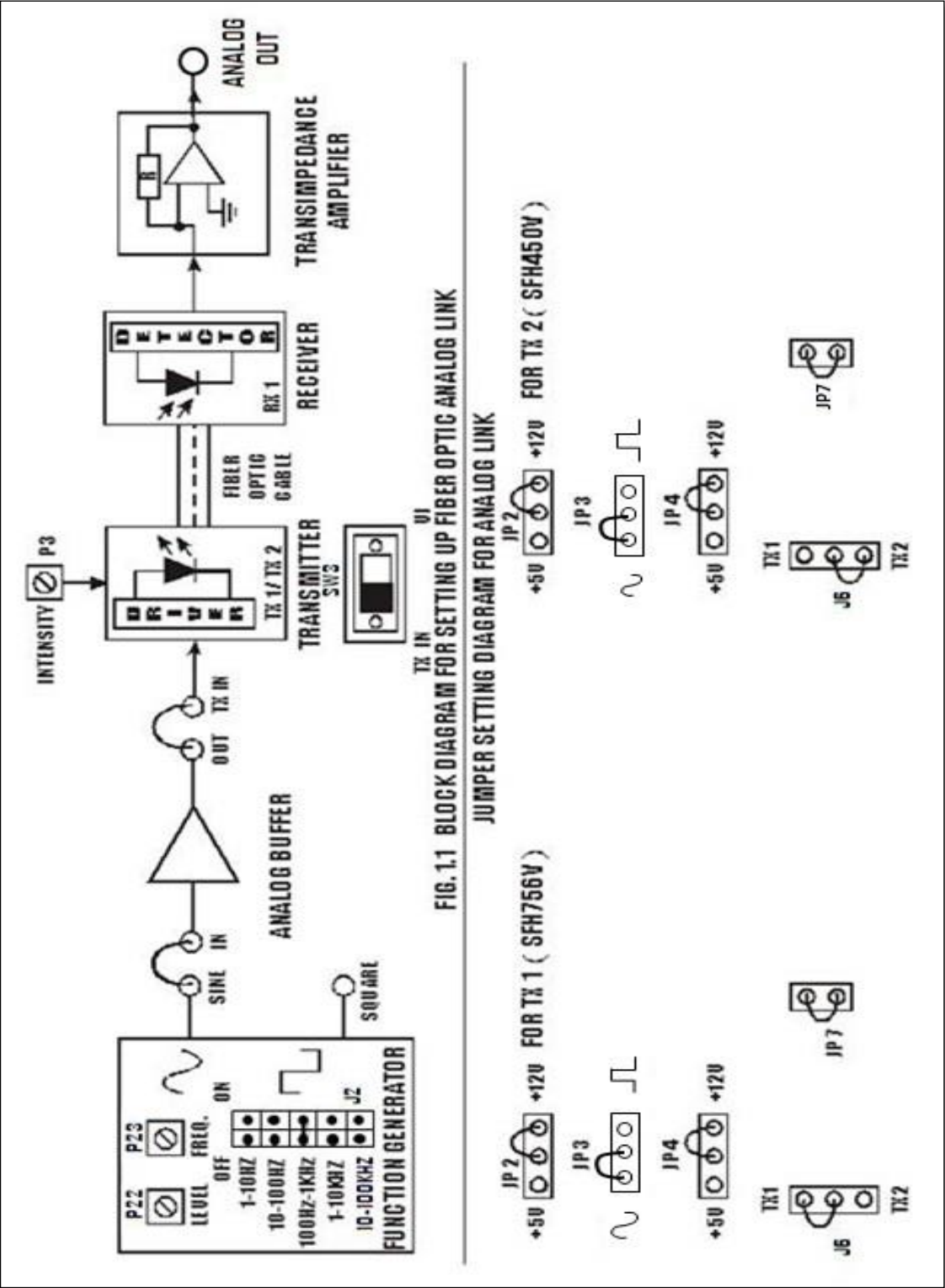


FIG. 1.1 BLOCK DIAGRAM FOR SETTING UP FIBER OPTIC ANALOG LINK

JUMPER SETTING DIAGRAM FOR ANALOG LINK

EXPERIMENT NO.1

NAME:

Setting Up a Fiber Optic Analog Link

OBJECTIVE:

The objective of this experiment is to study a 660 nm/ 950 nm Fiber Optic Analog Link. In this experiment you will study the relationship between the input signal & received signal.

THEORY:

Fiber Optic Links can be used for transmission of digital as well as analog signals. Basically, a fiber optic link contains three main elements, a transmitter, an optical fiber & a receiver. The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter.

Transmitter

Fiber optic transmitters are typically composed of a buffer, driver & optical source. The buffer electronics provides both an electrical connection & isolation between the transmitter & the electrical system supplying the data. The driver electronics provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally, the optical source (LED) converts the electrical current to light energy with the same pattern. The LED SFH756V supplied with the kit operates inside the visible light spectrum. Its optical output is centered at near visible wavelength of 660 nm. The emission spectrum is broad, so a dark red glow can usually be seen when the LED is on. The LED SFH450V supplied with the kit operates outside the visible light spectrum. Its optical output is centered at near infrared wavelength of 950 nm.

Receiver

The function of the receiver is to convert the optical energy into electrical form, which is then conditioned to reproduce the transmitted electrical signal in its original form. The detector SFH250V used in the kit has a diode type output. The parameters usually considered in the case of detector are its responsivity at peak wavelength & response time. SFH250V has responsivity of about 4 μA per 10 μW of incident optical energy at 950 nm and it has rise & fall time of 0.01 μsec .

PIN photodiode is normally reverse biased. When optical signal falls on the diode, reverse current starts to flow, thus diode acts as closed switch and in the absence of light intensity, it acts as an open switch. Since PIN diode usually has low responsivity, a trans impedance amplifier is used to convert this reverse current into voltage. This voltage is then amplified with the help of another amplifier circuit. This voltage is the duplication of the transmitted electrical signal.

EQUIPMENTS:

- FOL-B-P kit
- Dual Channel Oscilloscope
- 1 Meter Fiber cable
- Power Supply

PROCEDURE:

- Slightly unscrew the cap of LED SFH 756V TX1 (660 nm) from kit. Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap and assure that the fiber is properly fixed. Now tight the cap by screwing it back. Keep **INTENSITY** pot P3 at minimum position i.e. fully anticlockwise.
- Make the connections and Jumper settings as shown in **FIG. 1.1**. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
- Switch on the power supply.
- Select the frequency range of Function Generator with the help of Range Selection Jumper **J2**, frequency can be varied with Pot **FREQ**. Adjust the voltage level of the Sine Wave with Pot **LEVEL** as per following setting FREQUENCY: 1KHz, LEVEL: 1Vp-p.
- Connect **SINE** post of the Function Generator section to **IN** post of Analog Buffer Section.
- Connect **OUT** post of the Analog Buffer Section to **TX IN** post of TRANSMITTER.
- Connect the other end of the fiber to detector SFH 250V (RX 1) in kit very carefully as per the instructions in step 1.
- Check the output signal of the Analog Buffer at its OUT post in Kit. It should be same as that of the applied input signal.
- Observe the output signal from the detector at **ANALOG OUT** post on DSO by adjusting **INTENSITY** (Optical Power Control) Pot **P3** in kit and you should get the reproduction of the original transmitted signal.
- To measure the analog bandwidth of the link, connect the external Signal Generator with 1Vp-p sine wave to **IN** post of Analog Buffer Section and vary the frequency of the input signal from 100Hz onwards. Measure the amplitude of the received signal for each frequency reading.
- Plot a graph of gain / Frequency. Measure the frequency range for which the response is flat.
- Repeat the procedure 1 to 11 for IR LED 450V (950 nm) TX2.

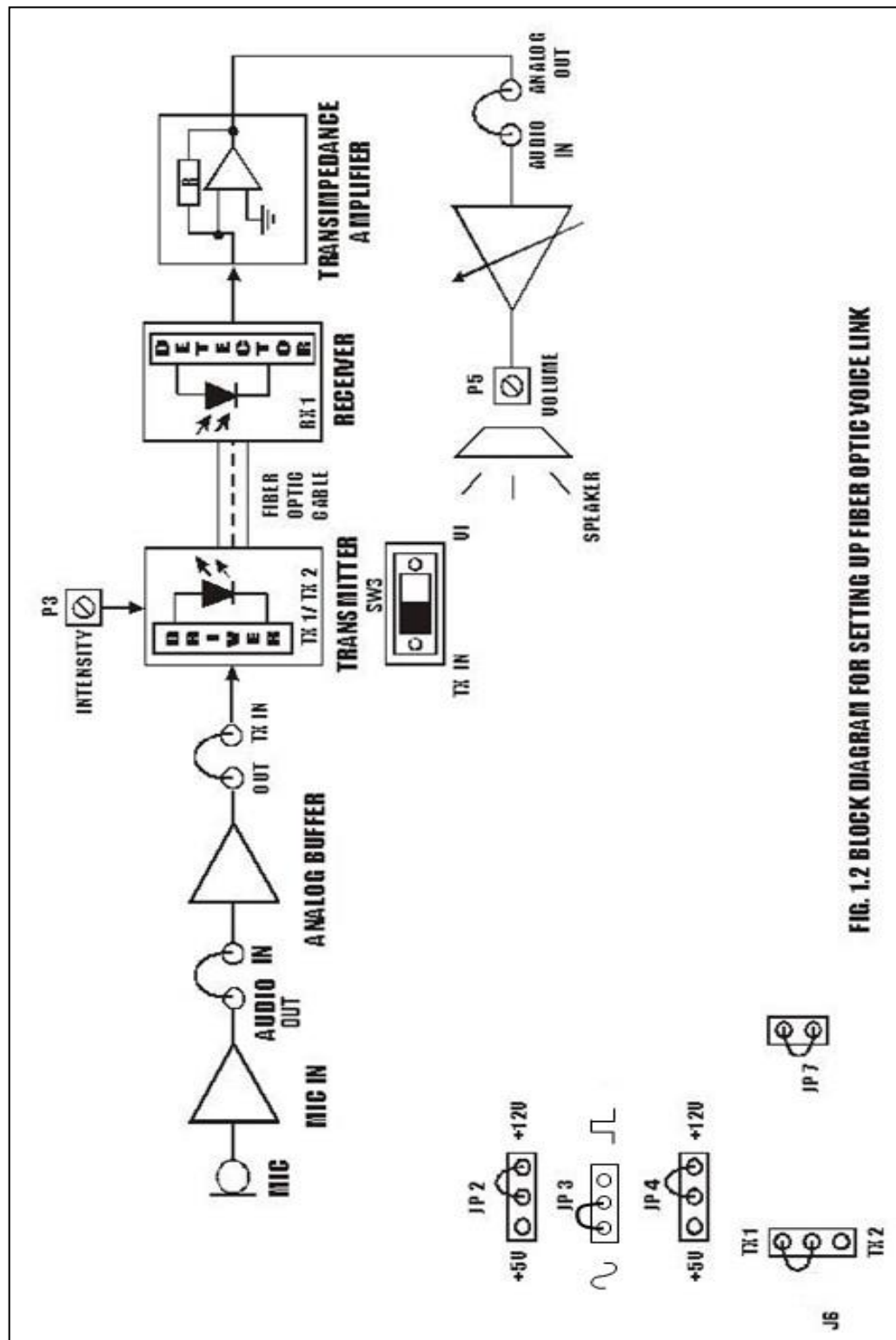


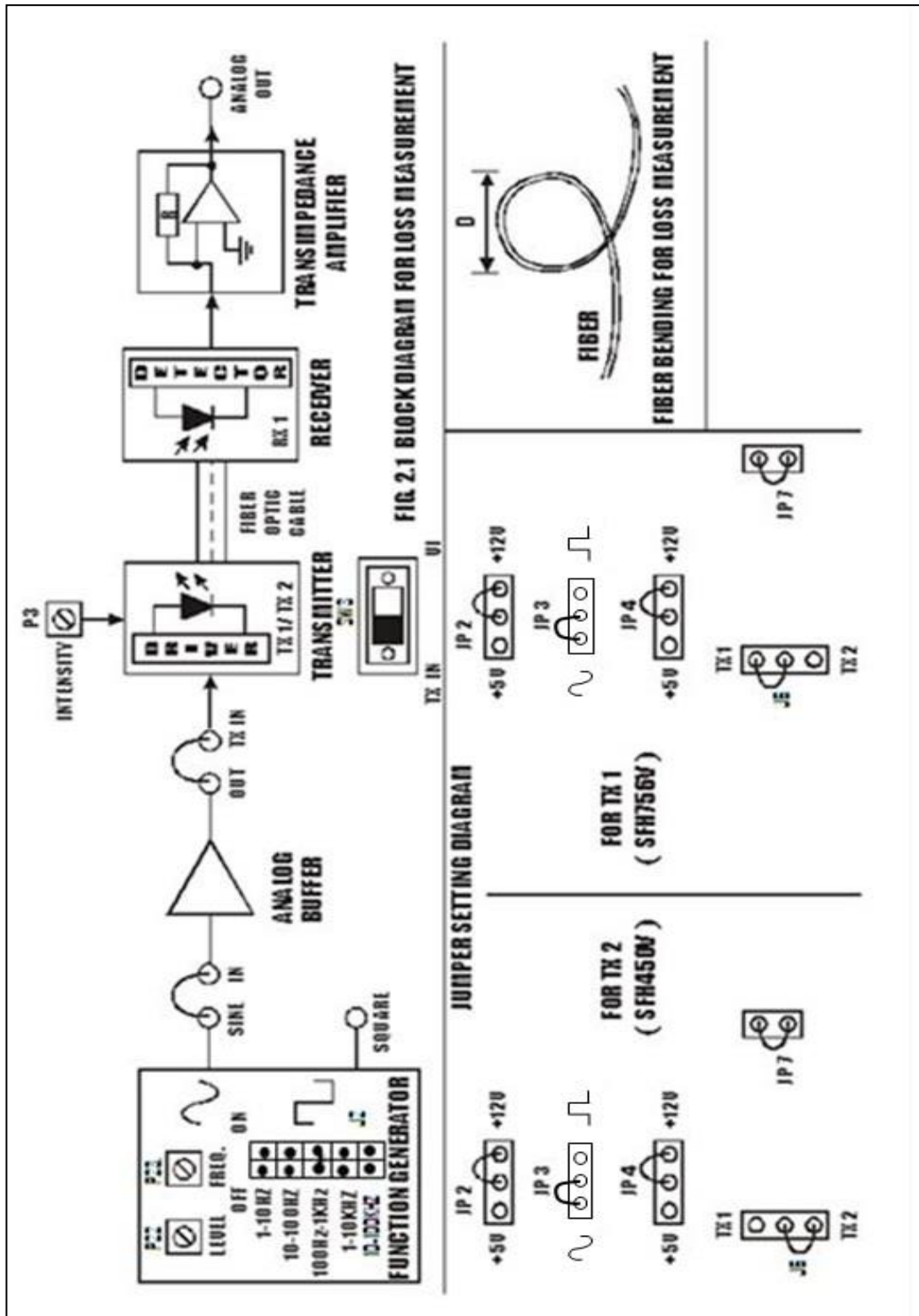
FIG. 1.2 BLOCK DIAGRAM FOR SETTING UP FIBER OPTIC VOICE LINK

Setting Up Simple Fiber Optic Voice Link

PROCEDURE

- Connect the microphone provided with the kit to the socket marked **MIC** on the kit.
- Make the connection on the Kit as per the Connection Diagram **FIG. 1.2**. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
- Connect **AUDIO OUT** post **IN** post of Analog Buffer Section.
- Connect **OUT** post of the Analog Buffer Section to **TX IN** post of TRANSMITTER.
- Connect **ANALOG OUT** in Receiver Trans-impedance Amplifier Section post to **AUDIO IN** post.
- Connect speakers provided with the kit to the socket marked **SPEAKER** on the kit.
- Switch on the power supply.
- Speak from the Microphone and you could hear your sound from the speaker. Adjust **INTENSITY** (Optical Power Control) Pot **P3** and **VOLUME** Pot to setup fiber optic audio link.

EXPERIMENT NO. 2



EXPERIMENT NO.2

NAME:

Study of Losses In Optical Fiber

OBJECTIVE:

The objective of this experiment is to measure propagation loss in plastic fiber provided with the Lab for three different wavelengths of radiation as 950 nm, 660 nm and also to measure the bending loss.

THEORY:

Optical fibers are available in different variety of materials. These materials are usually selected by taking into account their absorption characteristics for different wavelengths of light. In case of optical fiber, since the signal is transmitted in the form of light, which is completely different in nature as that of electrons, one has to consider the interaction of matter with the radiation to study the losses in fiber. Losses are introduced in fiber due to various reasons. As light propagates from one end of fiber to another end, part of it is absorbed in the material exhibiting absorption loss. Also, part of the light is reflected back or in some other directions from the impurity particles present in the material contributing to the loss of the signal at the other end of the fiber. In general terms it is known as propagation loss. Plastic fibers have higher loss of the order of 180 dB/Km. Whenever the condition for angle of incidence of the incident light is violated the losses are introduced due to refraction of light. This occurs when fiber is subjected to bending. Lower the radius of curvature more is the loss. Another losses are due to the coupling of fiber at LED & photo detector ends.

EQUIPMENTS:

- FOL-B-P kit
- Dual Trace Oscilloscope
- 1 & 3 Meter Fiber Cable
- Power Supply

PROCEDURE:

- Slightly unscrew the cap of IR LED SFH 450V (950 nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap and assure that the fiber is properly fixed. Now tight the cap by screwing it back. Keep pot **P3** at minimum position (fully anticlockwise).
- Make connections and Jumper settings as shown in Connection block diag. **FIG. 2.1**. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
- Set the sine wave with 1KHz, 1Vp-p amplitude and connect it to **IN** Post of Analog Buffer.
- Connect the other end of the fiber to detector SFH 250V RX1 (Analog Detector) very carefully as per the instructions in step 1.
- Switch on the power supply.
- Observe the output signal from the detector at **ANALOG OUT** post on CRO by adjusting **INTENSITY** (Optical Power Control) Pot **P3** in kit and you should get the reproduction of the original transmitted signal. Mark this amplitude level as V_1 .
- Now replace 1-meter fiber by 3-meter fiber without disturbing any of the previous settings. Measure the amplitude level at the receiver side again. You will notice that it is less than the previous one. Mark this as V_2 .

If α is the attenuation of the fiber then we have,

$$\alpha_{dB} = (10/L_1 - L_2) \log_{10} (V_2/V_1)$$

where

α = dB / Km,

L_1 = fiber length for V_1

L_2 = fiber length for V_2

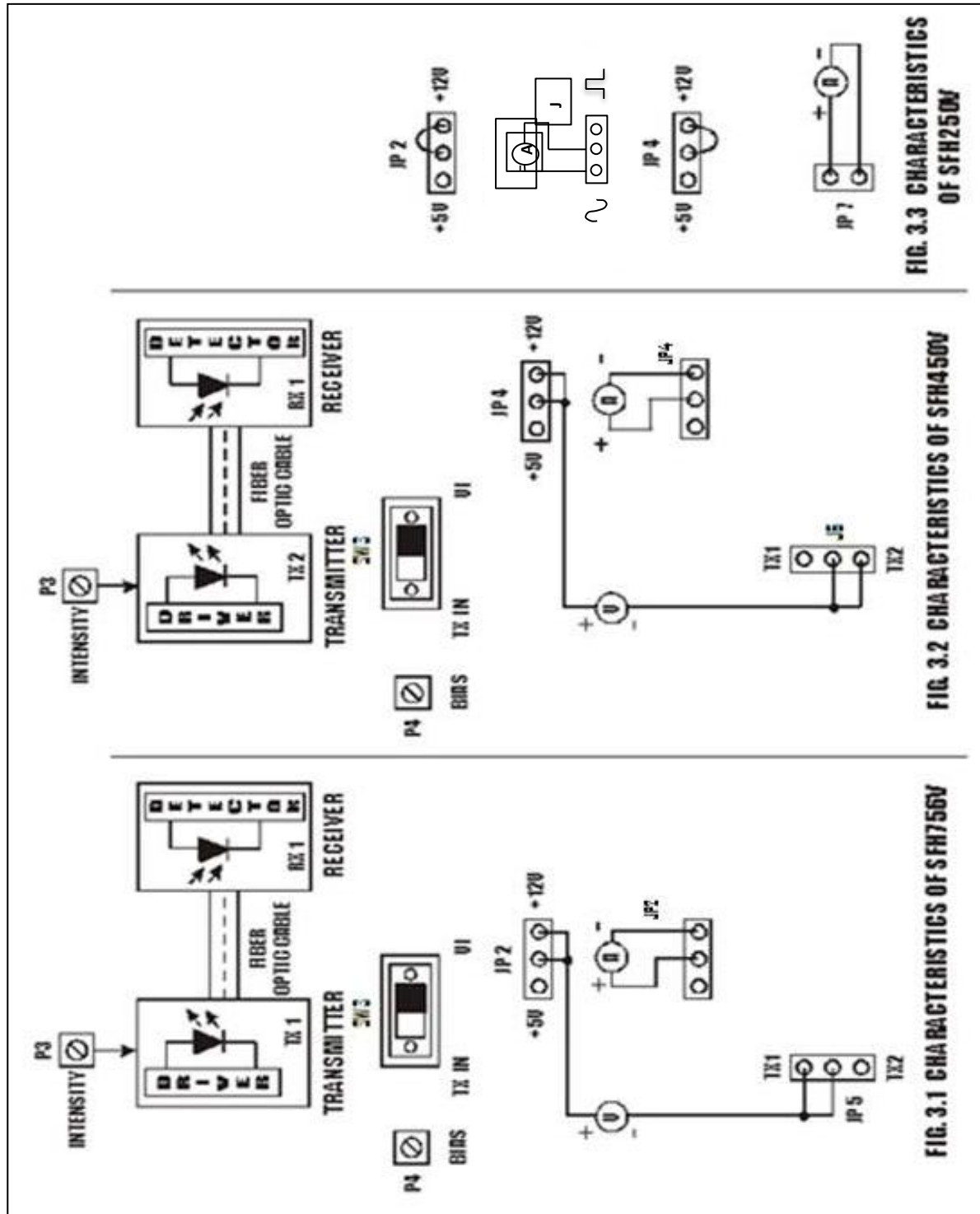
This α is for the wavelength of 950 nm. To get the α for 660 nm wavelength proceed as follows.

- Repeat steps 1 to 12 replacing SFH 450V (950 nm) by SFH756V (660 nm).
- Make the jumper settings as shown in the block diagram **FIG. 2.1**.
- Compare the values of α and find out the wavelength, which has less attenuation in the fiber.

Measurement Of Bending Losses

- Repeat all the Steps from 1 to 10 as above.
- Bend the fiber in a loop as shown in **FIG. 2.1**. Measure the amplitude of the received signal.
- Keep reducing the diameter to about 2 cm & take corresponding output voltage readings. (Do not reduce loop diameter less than 2 cm.)
- Plot a graph of the received signal amplitude versus the loop diameter.

EXPERIMENT NO. 3



EXPERIMENT NO.3

NAME:

Study of Characteristics Of Fiber Optic LED And Detector

OBJECTIVE:

To study the VI characteristics of fiber optic LEDs

THEORY:

In optical fiber communication system, electrical signal is first converted into optical signal with the help of E/O conversion device as LED. After this optical signal is transmitted through optical fiber, it is retrieved in its original electrical form with the help O/E conversion device as photo detector.

Different technologies employed in chip fabrication lead to significant variation in parameters for the various emitter diodes. All the emitters distinguish themselves in offering high output power coupled into the plastic fiber. Data sheets for LEDs usually specify electrical and optical characteristics, out of which the important ones are the peak wavelength of emission, conversion efficiency (usually specified in terms of power launched in optical fiber for specified forward current), optical rise and fall times which put the limitation on operating frequency, maximum forward current through LED and typical forward voltage across LED.

Photo detectors usually come in variety of forms like photoconductive, photovoltaic, transistor type output and diode type output. Here also characteristics to be taken into account are response time of the detector, which puts the limitation on the operating frequency, wavelength sensitivity and responsivity.

EQUIPMENTS:

- FOL-B-P kit
- Dual Channel Oscilloscope
- 1-Meter Fiber Cable
- Power Supply
- Ammeter-1
- Voltmeter-1
- Jumper Connecting Wires-3

PROCEDURE:

A. Characteristics Of Fiber Optic LED

- Make the jumper and switch settings as shown in the jumper diagram **FIG. 3.1**. Keep pot **P4** in fully clockwise position.
- Connect the ammeter with the jumper connecting wires (provided along with the kit) in jumpers **JP2** as shown in the diagram **FIG. 3.1**.
- Connect the voltmeter with the jumper wires to **J6** and **JP2** at positions as shown in the diagram.
- Switch on the power supply. Keep the potentiometer **P3** in its minimum position (fully anticlockwise position), **P4** is used to control biasing voltage of the LED. To get the **VI** characteristics of LED, rotate **P3** slowly and measure forward current and corresponding forward voltage of the LED, Take number of such readings for forward voltage, forward current & optical power and note down in tables as shown in **FIG. 3.4** (for SFH756V) **FIG. 3.5** (for SFH450V). **FIG. 3.6** & **FIG. 3.7** shows the table with readings taken for VI characteristics & optical power of SFH 756V LED & SFH 450. Graph for VI characteristics of LED SFH 756V & SFH 450 is shown in **FIG. 3.7** & **FIG. 3.9**.
- For each reading taken above, find out the power, which is product of V and I. This is the electrical power supplied to the LED. Data sheets for the LED specify optical power coupled into plastic fiber when forward current was 10mA as 200 μ W. This means that the electrical power at 10mA current is converted into 200 μ W of optical energy. Hence the efficiency of the LED comes out to be approx. 1.15%.
- With this efficiency assumed, find out optical power coupled into plastic optical fiber for each of the reading in step 4. Plot the graph of forward current v/s output optical power of the LED. **FIG. 3.8** shows graph of forward current v/s output optical power of the LED SFH 756V.
- Now refer the diagram as shown in **FIG. 3.2** for SFH 450V (950 nm) LED. Repeat the above procedure by using SFH 450V (950 nm) LED. Graph for VI characteristics of SFH 450V is shown in **FIG. 3.9**. **FIG. 3.10** shows graph of forward current v/s output optical power of the LED SFH 450V.

B. Characteristics Of Detector

- Make the jumper and switch settings as shown in the jumper diagram **FIG. 3.3**. Keep pot **BIAS** in fully clockwise position.
- Connect the ammeter with the jumper connecting wires (provided along with the kit) in jumpers **JP7** as shown in the diagram **FIG. 3.3**.
- Connect 1 Meter fiber optic cable between LED (TX1) SFH756V and detector (RX1) SFH250V.
- Switch on the power supply and measure corresponding forward current of LED (TX1) as per table shown in **FIG. 3.4**. Measure the current flowing through Detector (RX1) SFH250V at corresponding optical power output (Normally in μ A).
- We can observe that as incident optical power on detector increases, current flowing through the detector increases.

Fig. 3.4: Table for VI Characteristics of Fiber Optic Led SFH 756v & Detector

SR. NO.	V _f Forward Voltage of LED SFH756V (V)	I _f Forward Current of LED SFH756V (mA)	Electrical Power $P_i = V * I$ (mW)	Optical Power of LED 756V $P_0 = P_i * 1.15\%$ (μ W)	O/P Current of SFH250V (μ A)
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					

Fig. 3.5: Table for VI Characteristics of Fiber Optic Led SFH 450v & Detector

SR. NO.	V _f Forward Voltage of LED SFH450V (V)	I _f Forward Current of LED SFH450V (mA)	Electrical Power $P_i = V * I$ (mW)	Optical Power of LED 756V $P_0 = P_i * 1.15\%$ (μW)	O/P Current of SFH250V (μA)
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					

Fig. 3.6 : Table for VI Characteristics of Fiber Optic Led SFH 756v & Detector With Readings

SR. NO.	Vf Forward Voltage of LED SFH756V (V)	If Forward Current of LED SFH756V (mA)	Electrical Power $P_i = V * I$ (mW)	Optical Power of LED 756V $P_0 = P_i * 1.15\%$ (μ W)	O/P Current of SFH250V (μ A)
1	1.25	0	0	0	0
2	1.29	0.001	0.00129	0.0148	0
3	1.54	0.1	0.154	1.771	0
4	1.57	0.2	0.314	3.611	1
5	1.59	0.5	0.795	9.1425	2
6	1.62	1.0	1.62	18.63	7
7	1.64	1.5	2.46	28.29	12
8	1.65	2.04	3.366	38.709	15
9	1.67	3.04	5.0768	58.3832	26
10	1.69	5.1	8.619	99.1185	44
11	1.70	6.0	10.2	117.3	50
12	1.72	8.2	14.104	162.196	71
13	1.73	10.0	17.3	198.95	80
14	1.76	15.0	26.4	303.6	119
15	1.78	18.0	32.04	368.46	146
16	1.79	20.0	35.8	411.7	168
17	1.80	22.0	39.6	455.4	182
18	1.81	25.0	45.25	520.375	196
19	1.82	27.0	49.14	565.11	212
20	1.84	31.0	57.04	655.96	246

(Table given for reference only actual values may differ from above)

FIG. 3.7: Table for VI Characteristics of Fiber Optic Led SFH 450v & Detector with Readings

SR. NO.	V _f Forward Voltage of LED SFH450V (V)	I _f Forward Current of LED SFH450V (mA)	Electrical Power $P_i = V * I$ (mW)	Optical Power of LED 756V $P_0 = P_i * 1.15\%$ (μ W)	O/P Current of SFH250V (μ A)
1	0.68	0	0	0	0
2	0.69	0.001	0.00069	0.007935	0
3	0.86	0.02	0.0172	0.1978	0
4	0.92	0.05	0.046	0.529	0
5	0.94	0.1	0.094	1.081	0
6	0.98	0.2	0.196	2.254	0
7	0.99	0.3	0.297	3.4155	0
8	1.00	0.4	0.4	4.6	0
9	1.02	0.5	0.51	5.865	0
10	1.05	1.0	1.05	12.075	1
11	1.06	1.5	1.59	18.285	2
12	1.07	2.0	2.14	24.61	3
13	1.09	3.0	3.27	37.605	5
14	1.11	5.0	5.55	63.825	8
15	1.12	8.1	9.072	104.328	12
16	1.14	12.0	13.68	157.32	20
17	1.15	15.0	17.25	198.375	24
18	1.17	20.0	23.4	269.1	38
19	1.18	25.0	29.5	339.25	44
20	1.19	33.0	39.27	451.605	45

(Table given for reference only actual values may differ from above)

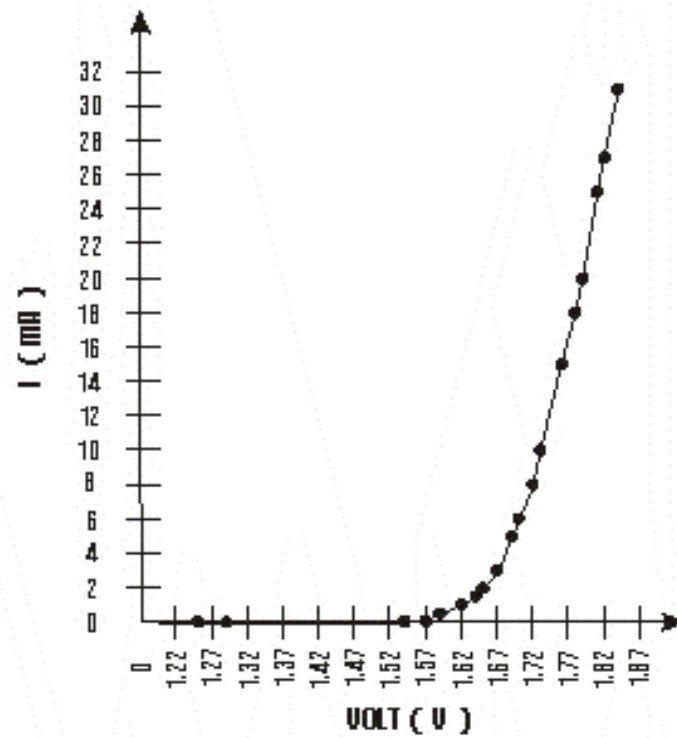


FIG. 3.7 VI CHARACTERISTICS GRAPH OF LED SFH756V

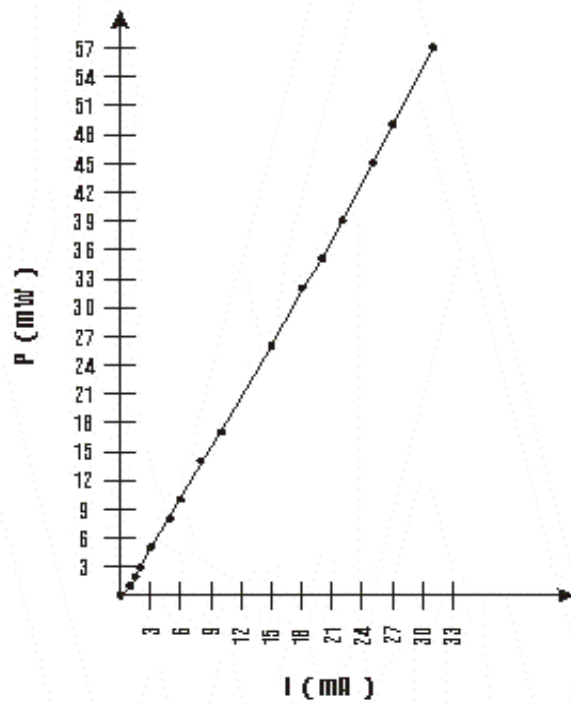


FIG. 3.8 OPTICAL POWER VERSUS CURRENT GRAPH OF LED SFH756U

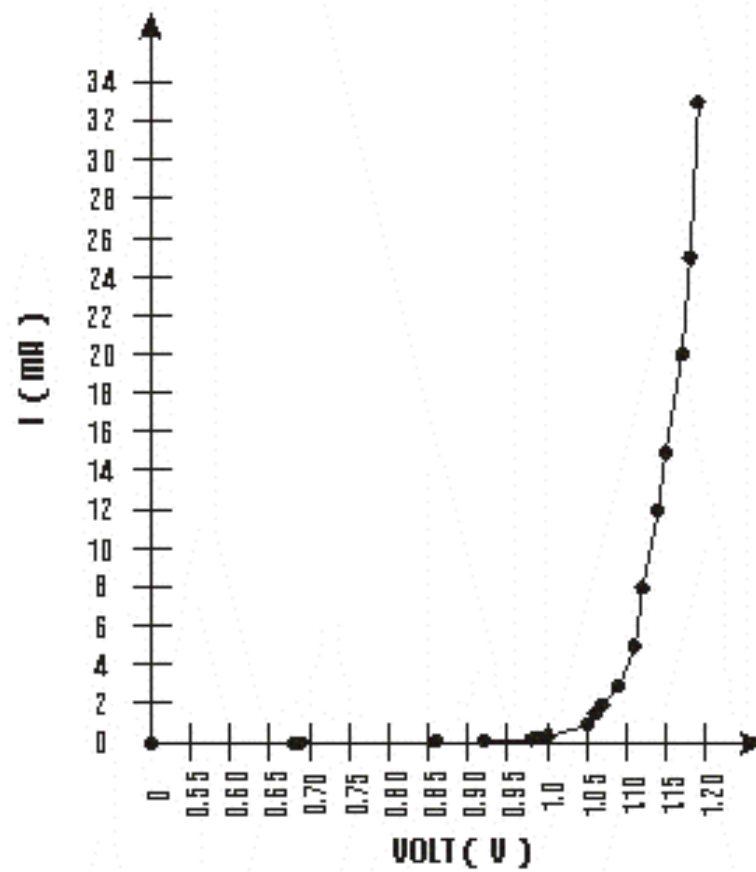


FIG. 3.9 VI CHARACTERISTICS GRAPH OF LED SFH450V

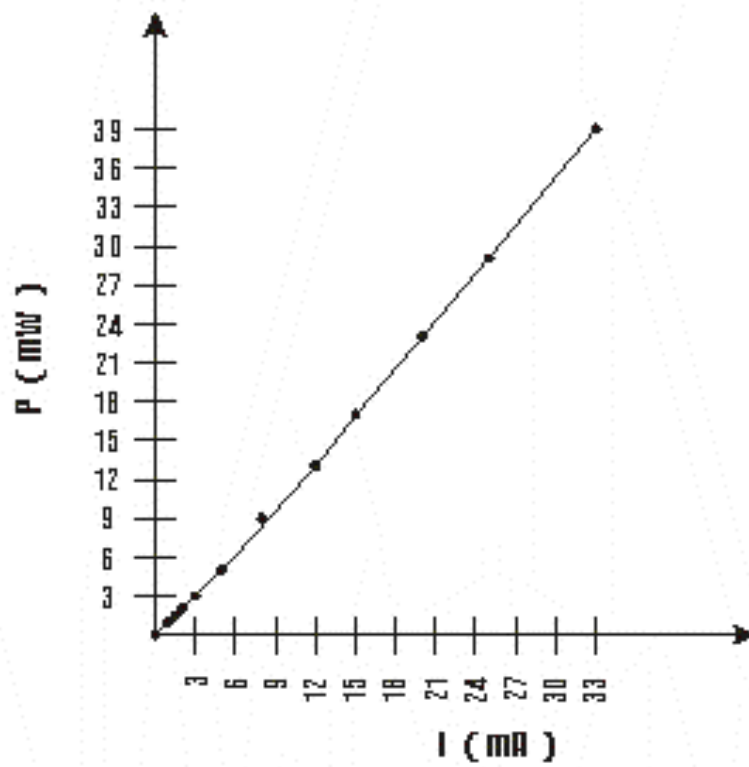


FIG. 3.10 OPTICAL POWER VERSUS CURRENT GRAPH OF LED SFH450V

**EXPERIMENT
NO. 4**

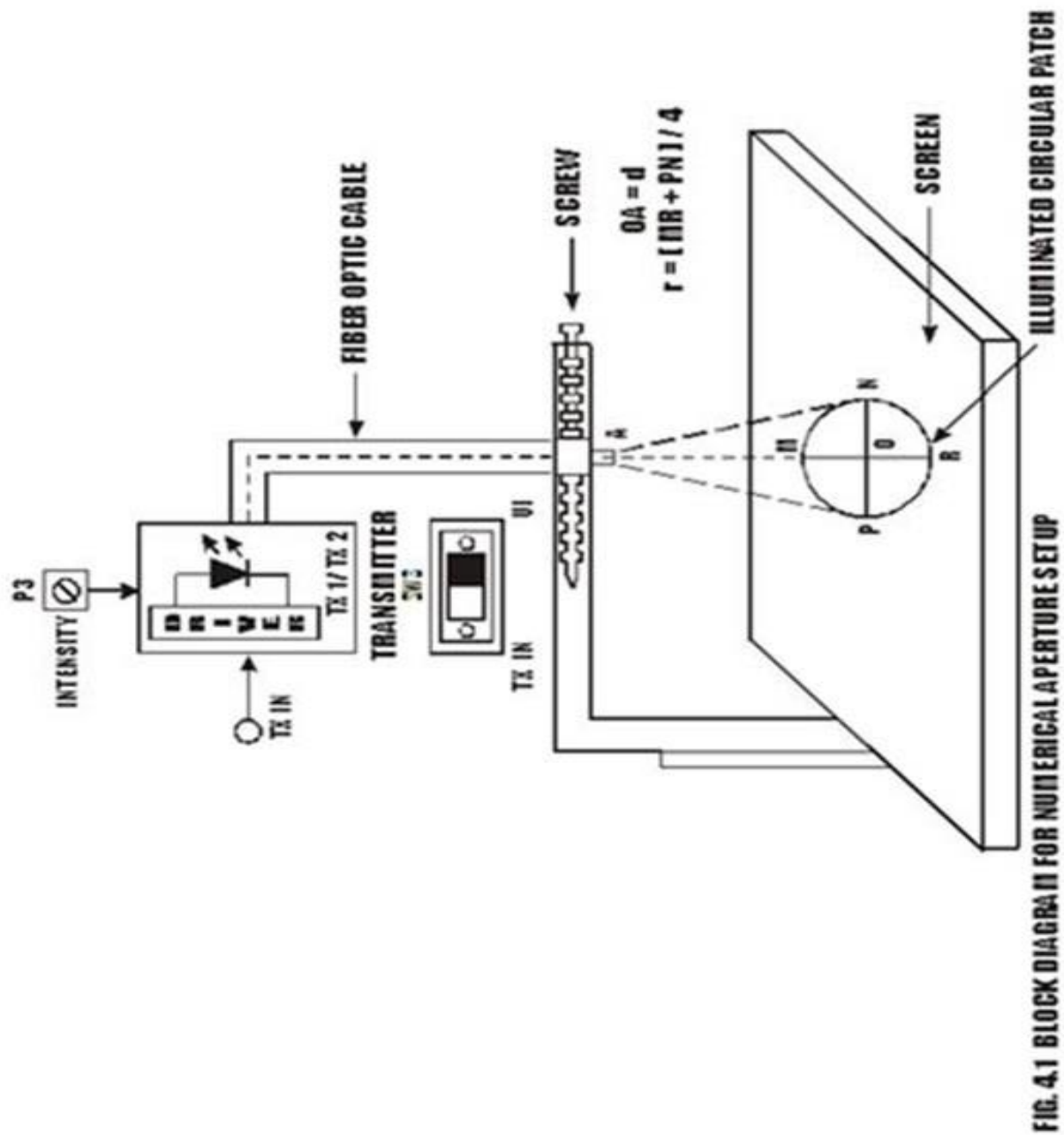
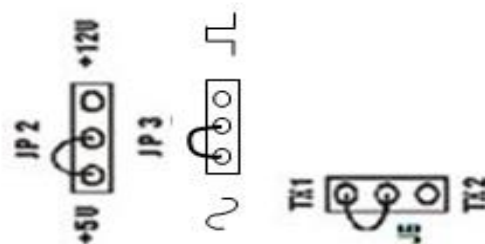


FIG. 4.1 BLOCK DIAGRAM FOR NUMERICAL APERTURE SETUP



EXPERIMENT NO. 4

NAME:

Study of Numerical Aperture of optical Fiber

OBJECTIVE:

The objective of this experiment is to measure the numerical aperture of the plastic fiber provided with the kit using 660 nm wavelength LED.

THEORY:

Numerical aperture refers to the maximum angle at which the light incident on the fiber end is totally internally reflected and is transmitted properly along the fiber. The cone formed by the rotation of this angle along the axis of the fiber is the cone acceptance of the fiber. The light ray should strike the fiber end within its cone of acceptance, else it is refracted out of the fiber core.

Considerations in a NA Measurement

- It is very important that the optical source should be properly aligned with the cable & distance from the launched point & the cable is properly selected to ensure that the maximum amount of optical power is transferred to the cable.
- This experiment is best performed in a less illuminated room.

EQUIPMENTS:

- FOL-B-P-II kit
- 1 meter fiber cable
- NA JIG
- Steel Ruler
- Power Supply

PROCEDURE:

- Slightly unscrew the cap of LED SFH756V (660 nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
- Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF. Do not apply any TTL signal from Function Generator. Make the connections as shown in **FIG. 4.1**.
- Keep Pot P3 fully Clockwise Position and P4 fully anticlockwise position.
- Switch on the power supply.
- Insert the other end of the fiber into the numerical aperture measurement jig. Hold the white sheet facing the fiber. Adjust the fiber such that its cut face is perpendicular to the axis of the fiber.
- Keep the distance of about 10 mm between the fiber tip and the screen. Gently tighten the screw and thus fix the fiber in the place.
- Now adjust Pot P4 fully Clockwise Position and observe the illuminated circular patch of light on the screen.
- Measure exactly the distance **d** and also the vertical and horizontal diameters **MR** and **PN** indicated in the **FIG. 4.1**.
- Mean radius is calculated using the following formula

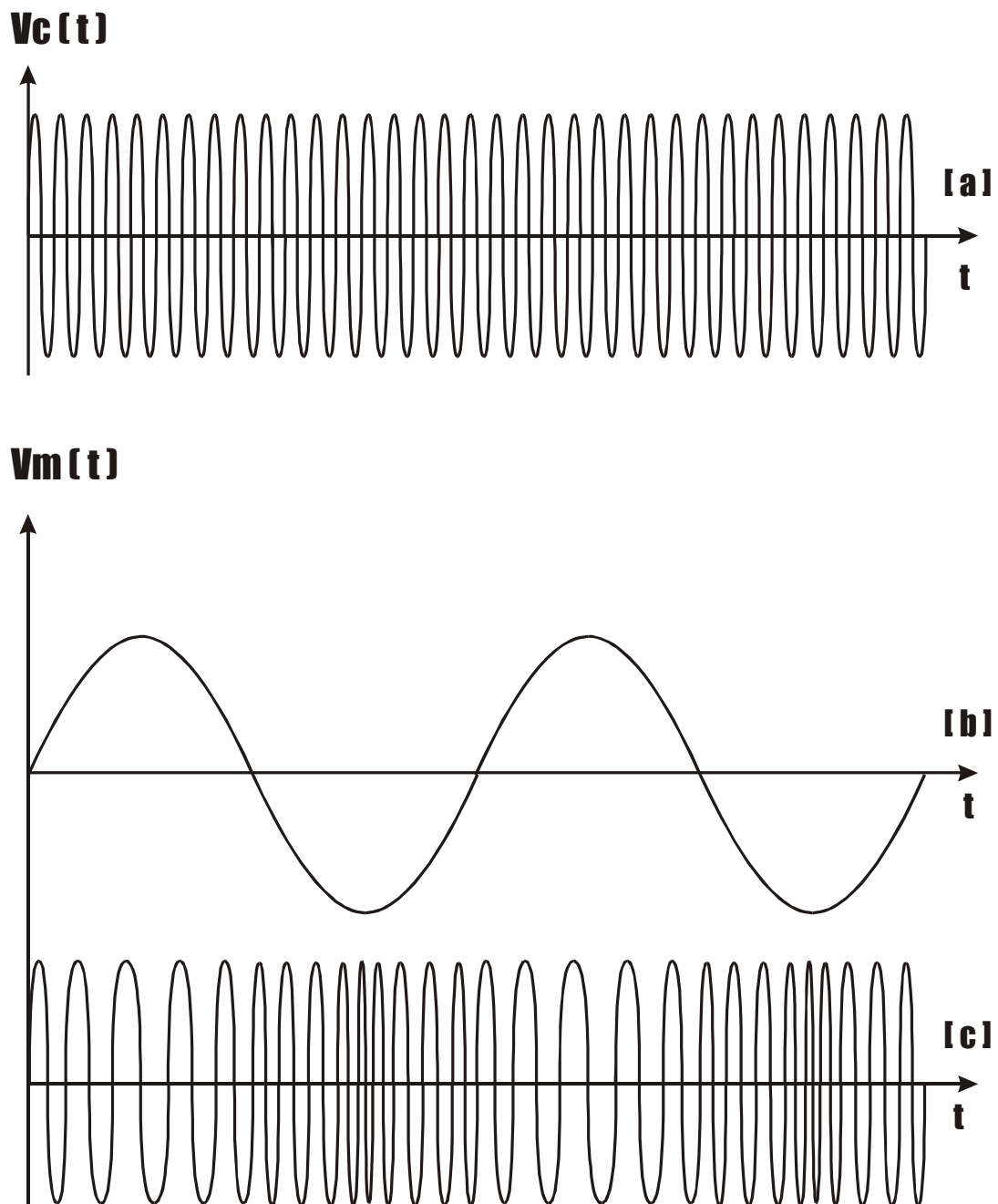
$$r = (MR + PN) / 4$$

- Find the numerical aperture of the fiber using the formula

$$NA = \sin\theta_{\max} = r / \sqrt{d^2 + r^2}$$

Where $\theta_{\max}$ is the maximum angle at which the light incident is properly transmitted through the fiber.

EXPERIMENT NO. 5



**FIG. 5.1 WAVE FORMS FOR A] CARRIER SIGNAL, B] MODULATING SIGNAL,
C] MODULATED SIGNAL**

EXPERIMENT NO. 5

NAME:

Study of Frequency Modulation & Demodulation Using Fiber Optic Link.

OBJECTIVE:

The objective of this experiment is to study the circuit action of Frequency Modulation & Demodulation over Fiber Optic Link using 660 nm and 950 nm LED.

THEORY:

Frequency Modulation

It is a type of modulation in which the frequency of the high frequency (Carrier) is varied in accordance with the instantaneous value of the modulating signal.

Main Aspects

Consider a sine wave signal $v_m(t)$ with pulse w . Refer **FIG. 5.1**.

$$v_m(t) = B \cdot \sin(w \cdot t)$$

and another sine wave $v_c(t)$ with upper Ω pulse:

$$v_c(t) = A \cdot \sin(\Omega \cdot t)$$

The signal $v_m(t)$ is called modulating signal, the signal $v_c(t)$ is called carrier signal.

Vary the frequency of the carrier $v_c(t)$ in a way proportional to the amplitude of the modulating signal $v_m(t)$. You obtain a $v_m(t)$ frequency modulated diagonal, which can be expressed by the relation:

$$v_m(t) = A \cdot \sin [\theta(t)]$$

with $\theta(t)$ instantaneous angle function of $v_m(t)$.

Mathematical Expression Of The Frequency Modulated Signal

The instantaneous pulse $\Omega(t)$ of the FM signal by definition:

$$\Omega(t) = \Omega + K \cdot v_m(t)$$

with $h\Omega$ = carrier pulse

K = modulation sensitivity

The instantaneous angle $\Omega(t)$ to be used as subject of the sine to obtain the mathematical operation of the FM signal, is detected by integrating $\Omega(t)$:

$$\theta(t) = \int \Omega(t) dt$$

In the case of modulating sine wave signal [$v_m(t) = B \cdot \sin(w \cdot t)$], $\theta(t)$ it results:

$$\theta(t) = \Omega \cdot (t) - (K \cdot B / w) \cdot \cos(w \cdot t)$$

The expression of the frequency modulated signal $v_m(t)$ becomes:

$$v_m(t) = A \cdot \sin[\Omega \cdot (t) - (K \cdot B / w) \cdot \cos(w \cdot t)]$$

Frequency Deviation ΔF And Modulation Index M_f

The instantaneous frequency $F(t)$ of the carrier modulated by a sine wave, results:

$$F(t) = \frac{\Omega(t)}{2\pi} = \frac{\Omega}{2\pi} + K \cdot B \cdot \sin(w \cdot t)$$

and oscillates between a minimum F_{min} and a maximum value F_{max} :

$$F_{min} = \frac{\Omega}{2\pi} - \frac{K \cdot B}{2\pi} \qquad F_{max} = \frac{\Omega}{2\pi} + \frac{K \cdot B}{2\pi}$$

The frequency deviation ΔF represents the maximum shift between the modulated signal frequency, over and under the frequency of the carrier:

$$\Delta F = \frac{F_{max} - F_{min}}{2}$$

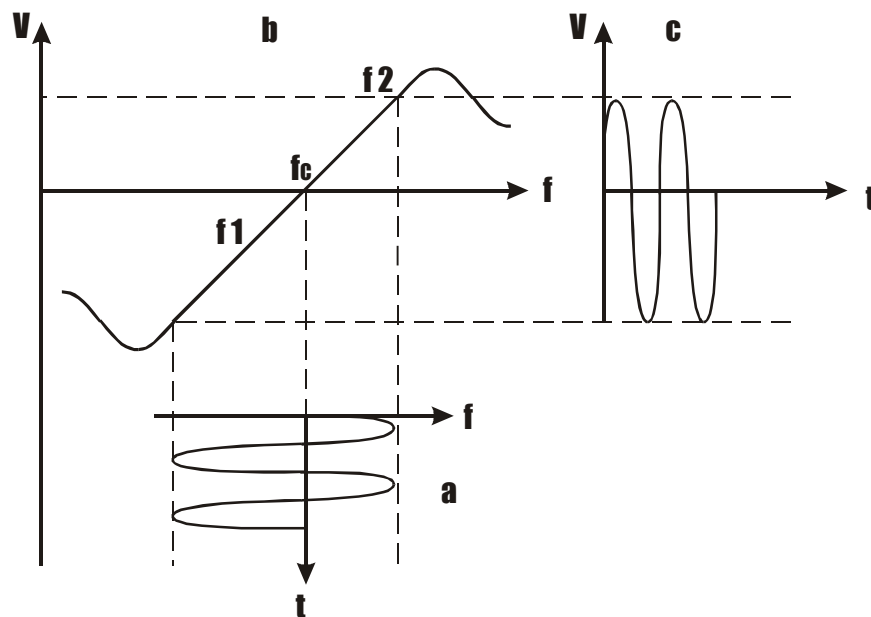


FIG. 5.2

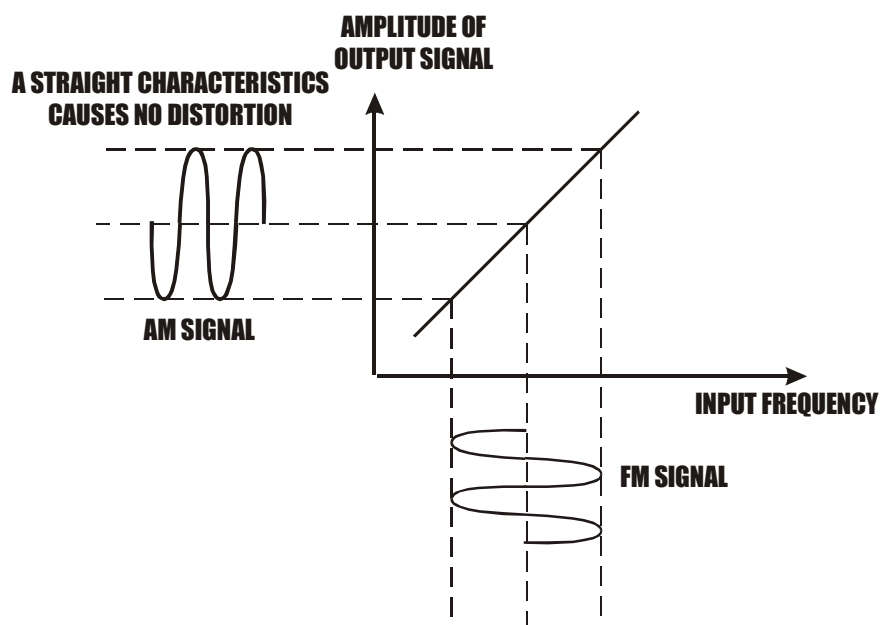


FIG. 5.3 AN IDEAL LINEAR VOLTAGE/ FREQUENCY CHARACTERISTICS

We define as modulation index m_f the ratio between ΔF and the modulating frequency f :

$$m_f = \frac{\Delta F}{f}$$

Frequency Modulation Generation

The circuits used to generate a frequency modulation must vary the frequency of a high frequency signal (carrier) as function of the amplitude of a low frequency signal (modulating signal). In practice there are two main methods used to generate the FM:

Direct Method

An oscillator is used in which the reactance of one of the elements of the resonant circuit depends on the modulating voltage. The most common device with variable reactance is the Varactor or Varicap, which is a particular diode which capacitance varies as function of the reverse bias voltage (the Varicap is described in the next chapter). The frequency of the carrier is established with AFC circuits (Automatic Frequency Control) or PLL (Phase Locked Loop). In this KIT FM is generated by using FM generation IC. This generated FM is then sent via Fiber Optic and is received by optical detector. The output of the detector is then fed to the FM demodulation section.

The frequency of the carrier is established with AFC circuits (Automatic Frequency Control) or PLL (Phase Locked Loop).

Indirect Method

The FM is obtained in this case by a Phase Modulation, after the modulating signal has been integrated. In the phase modulator, the carrier can be generated by a quartz oscillator and so its frequency stabilization is easier.

In the circuit used for the exercise, the frequency modulation is generated by a Hartley oscillator, whose frequency is determined by a fixed inductance and by a capacity (variable) supplied by Varicap diodes.

Frequency Demodulation

To demodulate a frequency modulated signal a circuit is necessary which supplies the output with proportional voltage to the frequency deviation of the input modulated signal.

The ideal characteristic of the demodulator is a straight line, also if it is actually sufficient to obtain characteristics as the one of **FIG. 5.2**, which presents a linear behavior only for a certain frequency range (demodulator usage range). The figure reports:

- a) The instantaneous frequency f of the modulated signal, oscillating between F_1 and F_2 (F_c frequency of the carrier).
- b) The voltage/frequency characteristic curve of the demodulator.
- c) The detected signal.

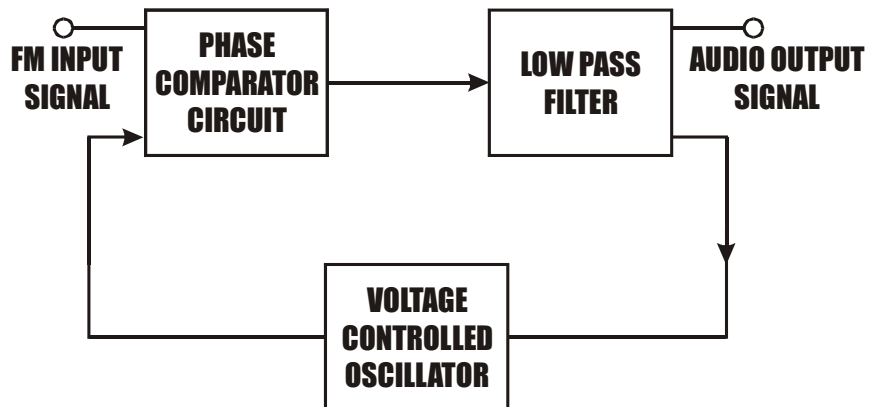


FIG. 5.4 BLOCK DIAGRAM FOR PHASE LOCK LOOP DETECTOR

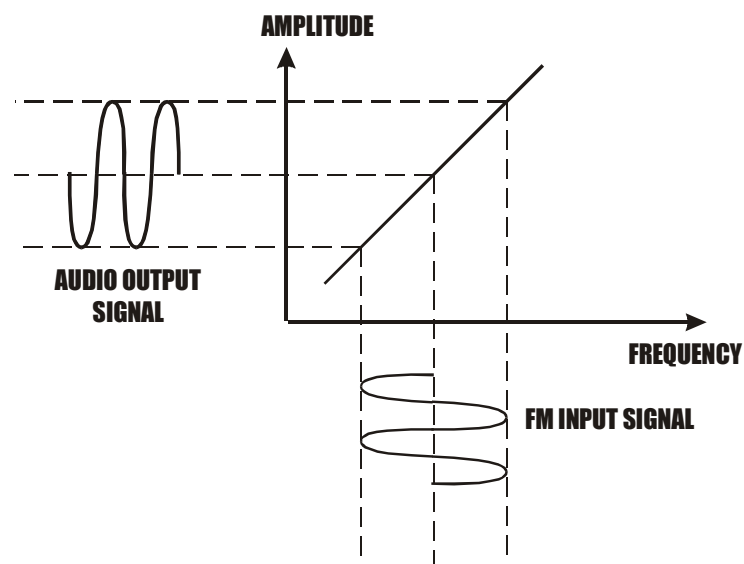


FIG. 5.5 CHARACTERISTICS OF PHASE LOCK LOOP DETECTOR

Frequency Demodulation

To demodulate a frequency modulated signal a circuit is necessary which supplies the output with proportional voltage to the frequency deviation of the input modulated signal.

The ideal characteristic of the demodulator is a straight line, also if it is actually sufficient to obtain characteristics as the one of **FIG. 5.2**, which presents a linear behavior only for a certain frequency range (demodulator usage range). The figure reports:

- d) The instantaneous frequency f of the modulated signal, oscillating between F_1 and F_2 (F_c frequency of the carrier).
- e) The voltage/frequency characteristic curve of the demodulator.
- f) The detected signal.

Demodulation of FM Signals

An FM receiver is very similar to an AM receiver. The most significant change is that the demodulator must now extract the information signal from a frequency, rather than an amplitude modulated wave.

The basic requirement of any FM demodulator is therefore to convert frequency changes into changes in voltage, with the minimum amount of distortion.

To achieve this, it should ideally have linear voltage/frequency characteristics, similar to that shown in **FIG. 5.3**.

A 'demodulator' can also be called a 'discriminator' or a 'detector'.

Any design of circuit that has a linear voltage/frequency characteristic would be acceptable and we are going to consider the five most popular types.

In each case the main points to look for are:

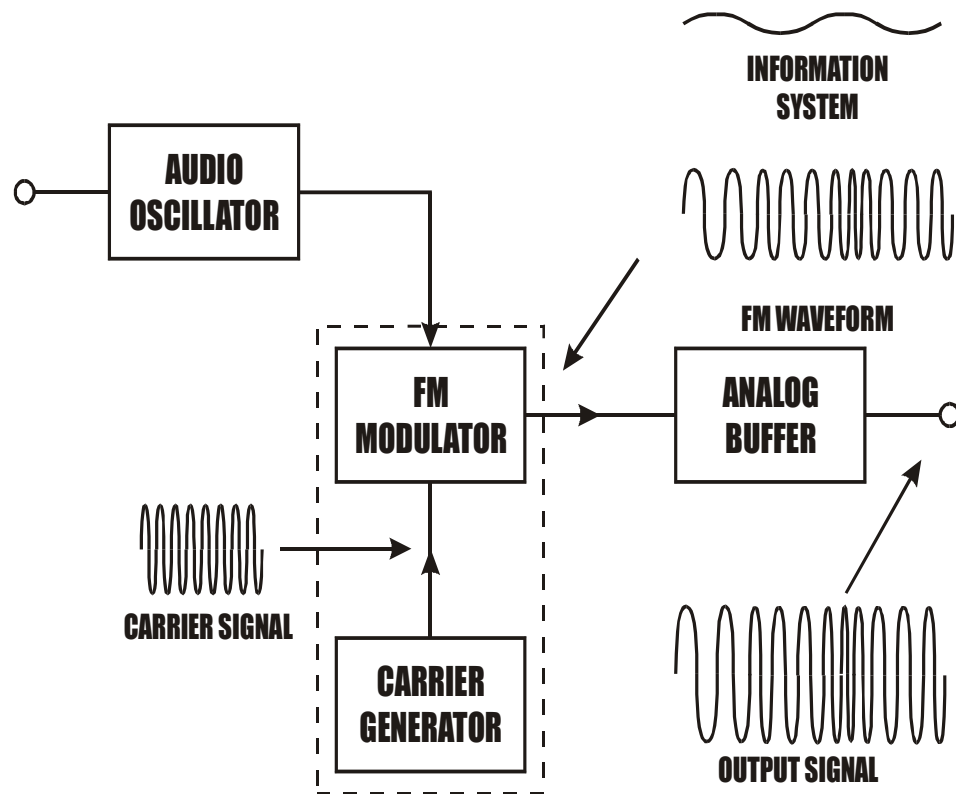
- How do they convert FM signals into AM signals?
- How linear is their response - this determines the amount of distortion in the final output?
- How good are they at rejecting noise signals?

Frequency Demodulator Circuits

For the detection of the frequency-modulated signals different circuit solutions have been used, some are out of use and others are used at the moment. Among the first ones, we mention:

PLL Detector

It constitutes one of the applications of the Phase Locked Loop and is less sensitive to noise.

**FIG. 5.6 FM TRANSMITTER**

The Phase – Locked Loop (PLL) Detector

This is another demodulator that employs a phase comparator circuit. It is a very good demodulator and has the advantage that it is available as a self-contained integrated circuit so there is no setting up required-you plug it in and it works. For these reasons it is often used in commercial broadcast receivers. It has very low levels of distortion and is almost immune to external noise signals and provides very low levels of distortion.

The overall action of the circuit may, at first, seem rather pointless. The incoming FM signal is taken to one input of the PHASE COMPARATOR CIRCUIT, where its phase is compared with the square-wave output from the VOLTAGE- CONTROLLED OSCILLATOR (VCO). If the central frequency of the detector and the carrier frequency of the FM signal and local oscillator frequency coincide, you will get the original reproduction of the modulating signal. As we can see in **FIG. 5.4** there is a voltage-controlled oscillator (VCO). The frequency of this oscillator is controlled by the DC output voltage from the output of the low pass filter. Now, this DC voltage keeps the oscillator running at the same frequency as the original input signal and 90° out of phase.

The question often arises as to why we would want the oscillator to run at the same frequency and 90° out of phase. And if we did, then why not just add a phase shifting circuit at the input to give the 90° phase shift?

The answer can be seen by imagining what happens when the input frequency changes – as it would be with an FM signal.

If the input frequency increases and decreases, the VCO frequency is made to follow it. To do this, the input control voltage must increase and decrease. It is these changes of DC voltage level that form the demodulated signal.

The AM signal then passes through a signal buffer to prevent any loading effects from disturbing the VCO and then through an audio amplifier if necessary.

The frequency response is highly linear as shown in **FIG. 5.5**.

The FM Demodulation section consists of Phase Lock Loop (PLL) IC which demodulates the FM signal. This demodulated signal is then fed to the filter section to get the exact original signal.

FM Transmitter

As we can see from **FIG. 5.6**, the audio oscillator supplies the information signal and could, if we wish, be replaced by a microphone and AF amplifier to provide speech and music instead of the sine wave signals.

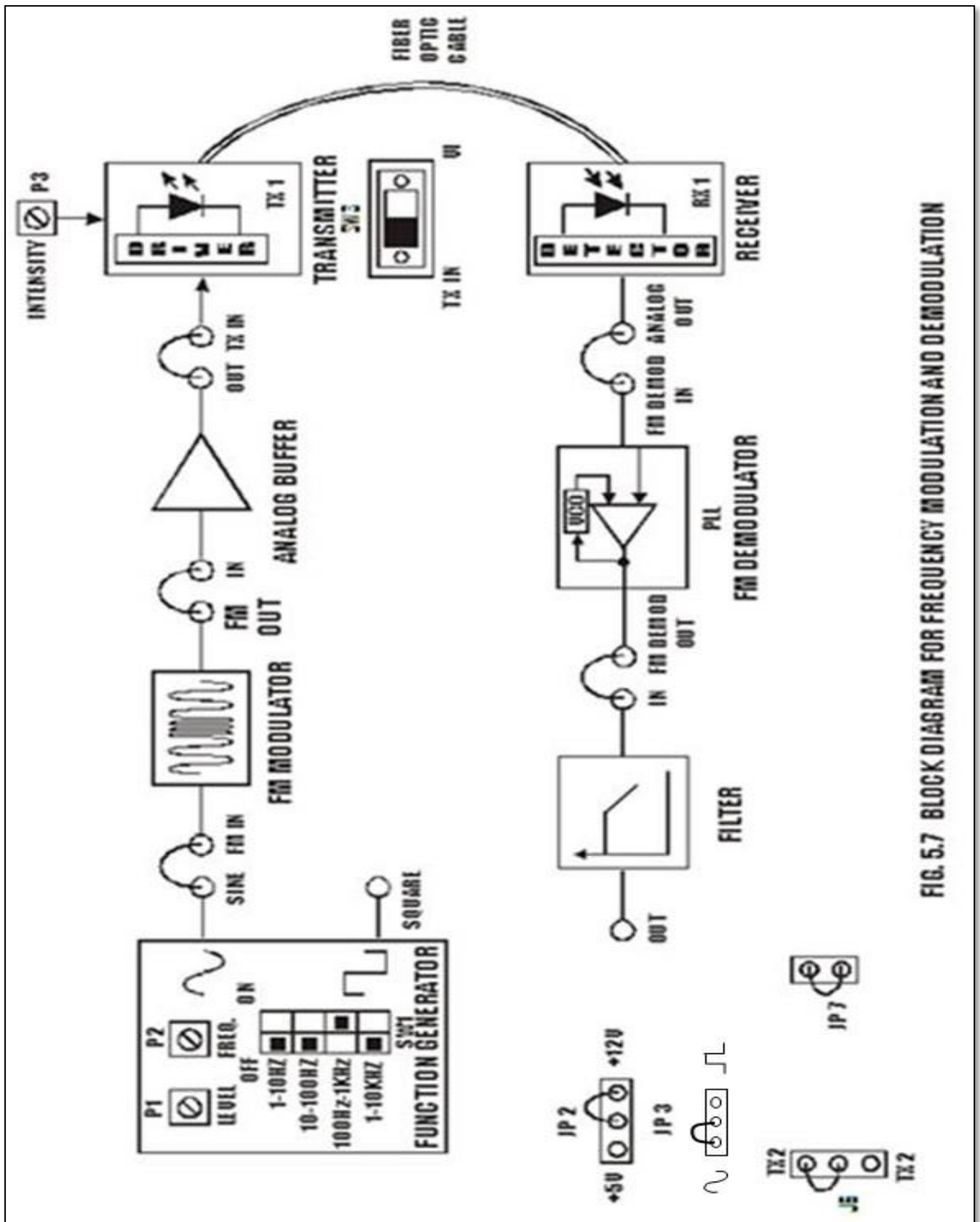


FIG. 5.7 BLOCK DIAGRAM FOR FREQUENCY MODULATION AND DEMODULATION

EQUIPMENTS:

- FOL-B-P kit
- Dual Channel Oscilloscope
- 1 Meter Fiber Cable
- Power Supply

PROCEDURE:

- Slightly unscrew the cap of LED SFH 756V TX1 (660 nm) from kit. Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap and assure that the fiber is properly fixed. Now tight the cap by screwing it back. Keep INTENSITY pot P3 at minimum position i.e. fully anti clockwise.
- Make the connections and Jumper settings as shown in **FIG. 5.7**. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
- Select the frequency range of about 1 KHz from Function Generator with the help of Range Selection Jumper **J2**; frequency can be varied with Pot **FREQ**. Adjust the voltage Level of the Sine Wave to 1Vp-p with Level Pot **LEVEL**.
- Connect **SINE** post of the Function Generator section to **FM IN** post of FM Modulator Section.
- Connect **FM OUT** post section of FM Modulator section to **IN** post of Analog Buffer Section.
- Switch on the power supply and observe the signal on the CRO at the **OUT** post of Analog Buffer, FM waveform from **FIG. 5.8** could be observed.
- Increase the time period on the oscilloscope and you could obtain the waveform as shown in the **FIG. 5.8**. to obtain this figure keep your voltage level to **0.5V** p-p.
- The frequency deviation ΔF can be calculated as follows (Refer to **FIG. 5.9**). From the oscilloscope evaluate F_M and F_m , detecting the periods of the respective sine waves.
The frequency deviation ΔF is defined as: $\Delta F = (F_M - F_m) / 2$.
- You can note that if the modulator operates in a linear zone so F_M and F_m are over and under the central frequency F of the same quantity ΔF , otherwise this does not occur.
- The value of the modulation index m_f is calculated by the relation, $M_f = \Delta F / f$, where f is the frequency of the modulating signal.
- Then observe the FM signal as shown in **FIG. 5.10**.
- Connect **OUT** post of the Analog Buffer Section to **TX IN** post of TRANSMITTER.
- Connect the other end of the fiber to detector SFH 250V (RX 1) in kit very carefully as per the instructions in step 1.
- Observe the output signal from the detector at ANALOG OUT post on CRO by adjusting **INTENSITY** pot **P3**, and you should get the reproduction of the original transmitted signal.
- Connect **ANALOG OUT** in Receiver Trans-impedance Amplifier Section to **FM DEMOD IN** post of FM Demodulator Section. Then connect **FM DEMOD OUT** post to **IN** post of Filter Section.
- Observe demodulated signal at **FM DEMOD OUT** post. and then observe output at **Filter OUT** post which is same as Input signal.

- The fact that there is still some high-frequency ripple at the output of the FM Demodulator block, we use the LOW PASS FILTER block to overcome this problem.

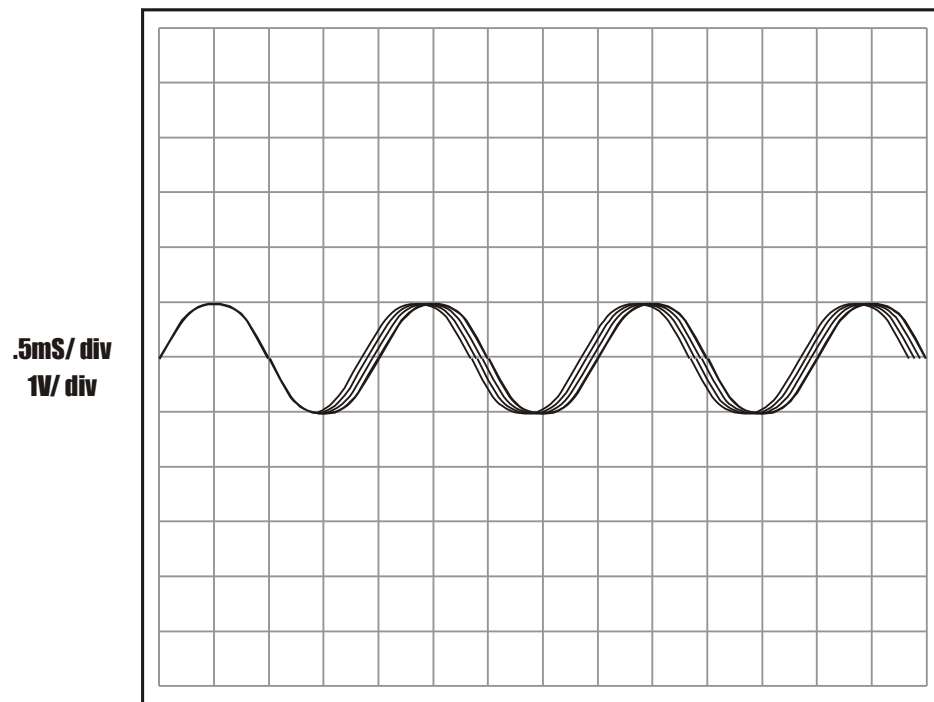


FIG. 5.8

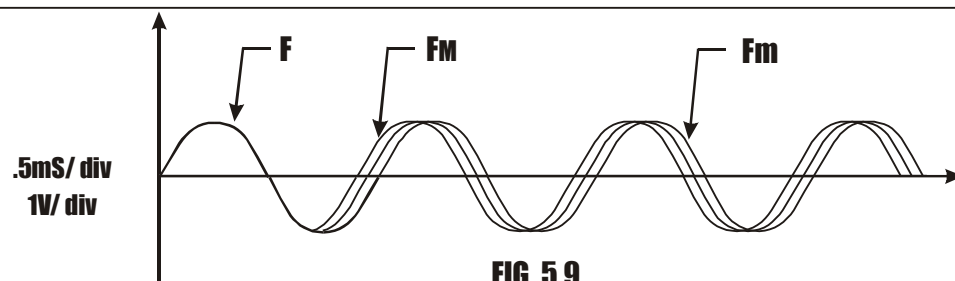


FIG. 5.9

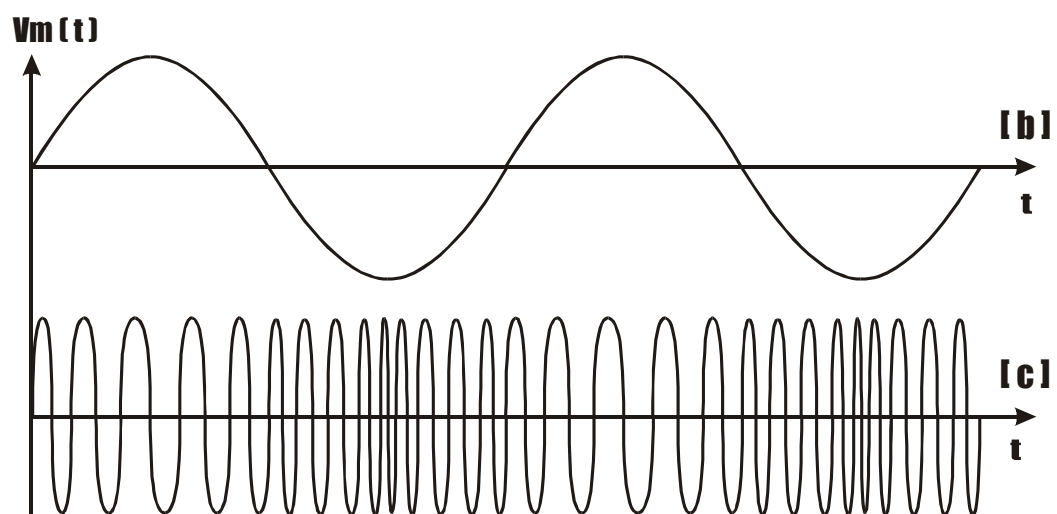


FIG. 5.10

Setting Up Fiber Optic Voice Link Using Frequency Modulation & Demodulation

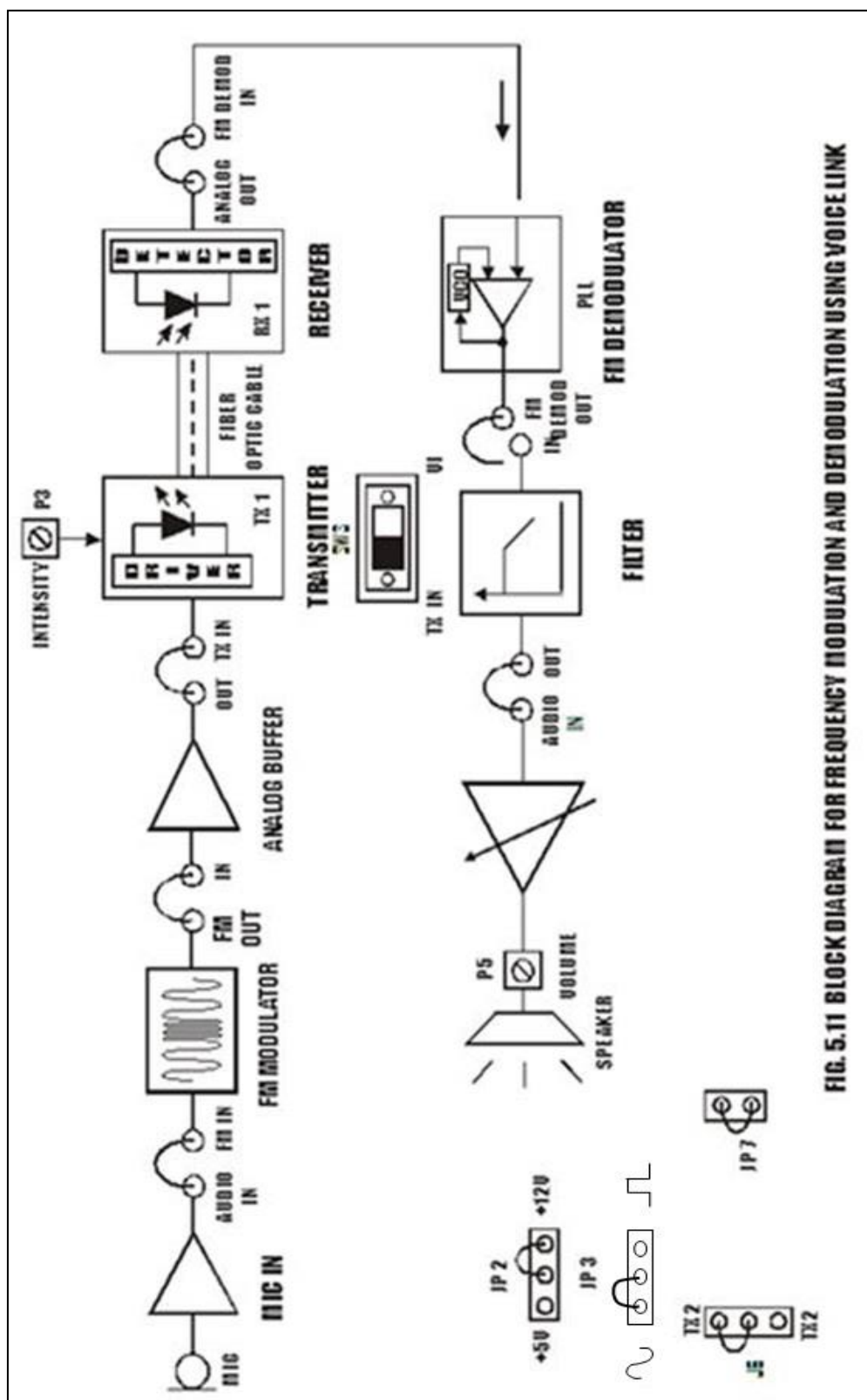
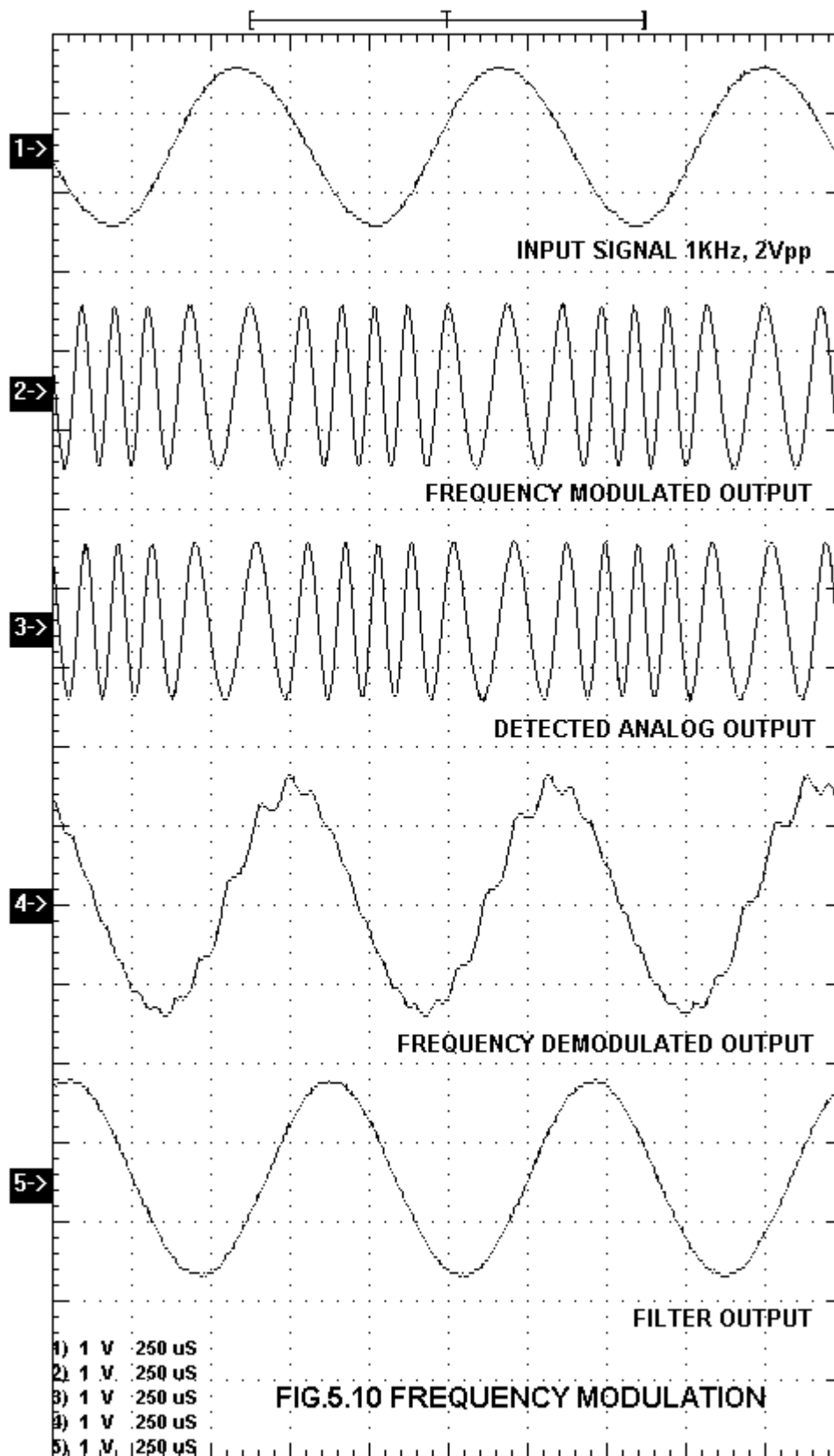


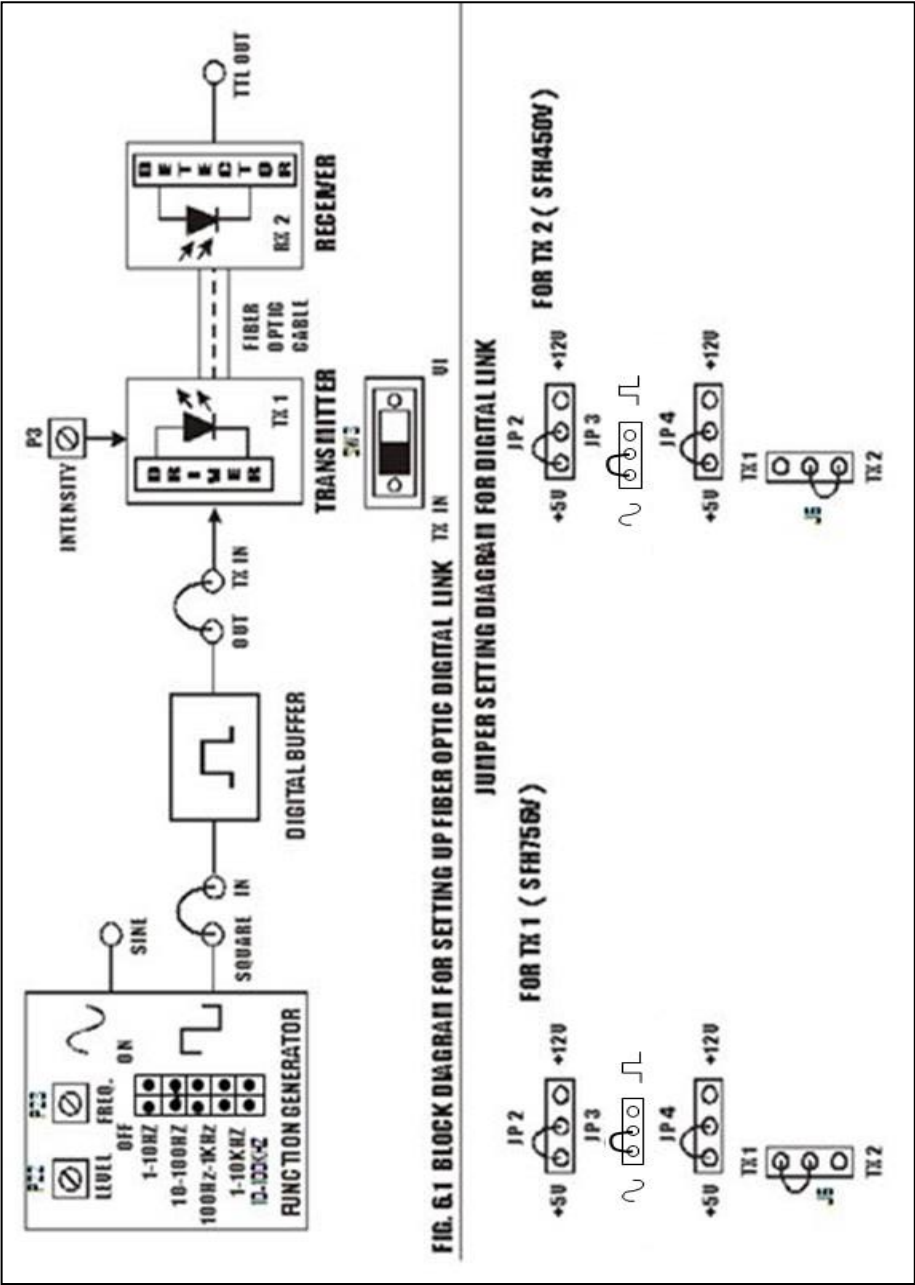
FIG. 5.11 BLOCK DIAGRAM FOR FREQUENCY MODULATION AND DEMODULATION USING VOICE LINK

PROCEDURE:

- Make the connection on the Kit as per the Connection Diagram **FIG. 5.11**. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
- Connect the microphone provided with the kit to the socket marked **MIC** on the kit.
- Connect **AUDIO OUT** post to **IN** post of FM Modulator Section.
- Connect OUT post in Filter Section to **AUDIO IN** post.
- Connect speaker provided with the kit to the socket marked **SPEAKER** on the kit.
- Switch on the power supply.
- Speak from the Microphone and you could hear your sound from the speaker. Adjust INTENSITY (Optical Power Control) Pot **P3** and Volume Pot **P5** to setup fiber optic audio link.



**EXPERIMENT
NO. 6**



EXPERIMENT NO. 6

NAME:

Setting Up A Fiber Optic Digital Link

OBJECTIVE:

The objective of this experiment is to study 950 and 660 nm fiber optic digital link. Here you will study how digital signal can be transmitted over fiber cable and reproduce at the receiver end.

THEORY:

In the experiment no. 1, we have seen how analog signal can be transmitted and received using LED, fiber and detector. The same LED, fiber and detector can be configured for the digital applications to transmit binary data over fiber. Thus basic elements of the link remain same even for digital applications.

Transmitter

LED digital, DC coupled transmitters are one of the most popular variety due to their ease of fabrication. We have used a standard TTL gate to drive NPN transistor, which modulates the LED SFH450V OR SFH756V source (Turns it ON and OFF).

Receiver

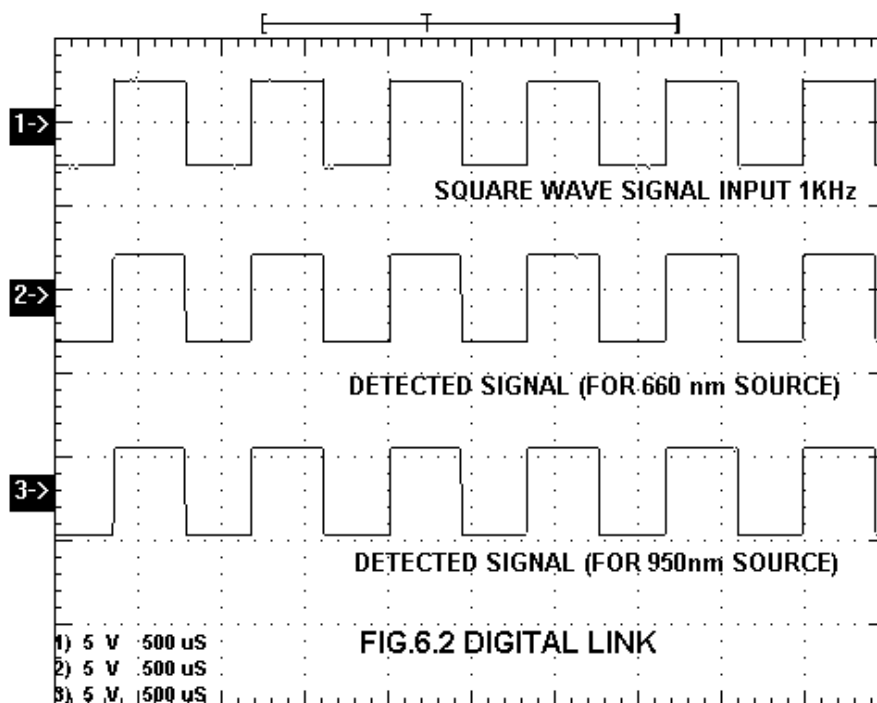
There are various methods to configure detectors to extract digital data. Usually detectors are of linear nature. We have used a photodetector SFH551V having TTL type output. Usually it consists of PIN photodiode, transimpedance amplifier and level shifter.

EQUIPMENTS:

- FOL-B-P kit
- Dual Trace Oscilloscope
- 1-Meter Fiber Cable
- Power Supply

PROCEDURE:

- Slightly unscrew the cap of LED SFH 756V (660 nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
- Make the connections and Jumper settings as shown in **FIG. 6.1**. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF. Now switch on the power supply.
- Feed the Onboard Square (TTL) signal of about 1 KHz to **IN** post of Digital Buffer Section and observe the signal at its **OUT** post. It should be same as that of the input signal.
- Connect OUT post of the Digital Buffer Section to **TX IN** post of TRANSMITTER.
- Connect the other end of the fiber to detector SFH 551V **RX2** (Digital Detector) very carefully as per the instructions in step 1.
- Observe the received signal on CRO at **TTL OUT** post. The transmitted signal and received signal are same. Vary the frequency of the input signal and observe the output response.
- To set up a digital link using SFH 450 (950 nm) change jumper connections as shown in jumper block diagram **FIG. 6.2** & repeat steps 1 to 8 using SFH 450V (950 nm) instead of SFH 756V (660 nm).



EXPERIMENT NO. 7

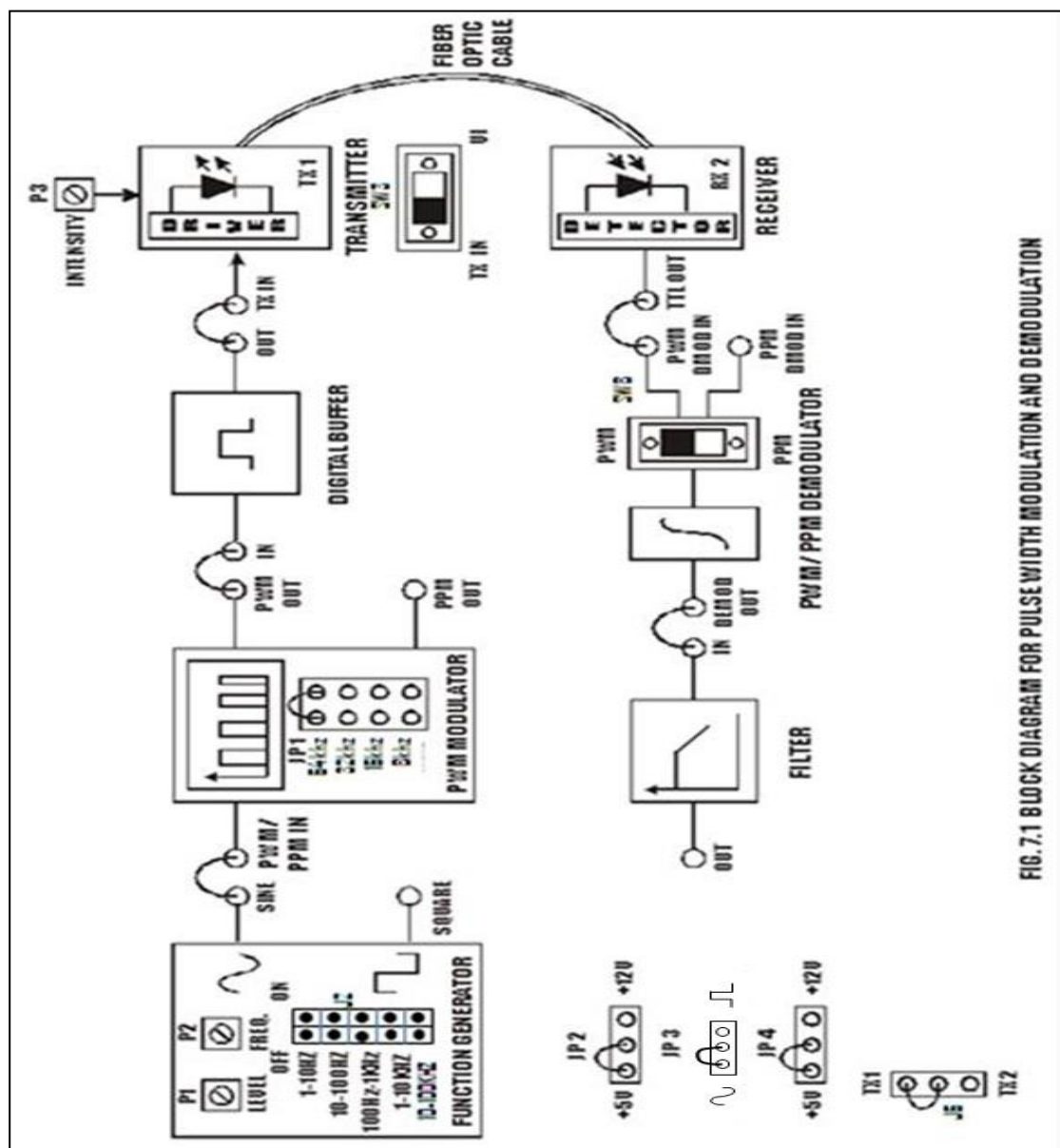


FIG. 7.1 BLOCK DIAGRAM FOR PULSE WIDTH MODULATION AND DEMODULATION

EXPERIMENT NO. 7

NAME:

Study of Pulse Width Modulation And Demodulation

OBJECTIVE:

The objective of this experiment is to study the circuit action of Pulse Width Modulation and Demodulation over Fiber Optic Digital Link.

THEORY:

Pulse Width Modulation

This technique of modulation controls the variation of duty cycle of the square wave (With some fundamental frequency) according to the input modulating signal. Here the amplitude variation of the modulating signal is reflected into ON period variation of square wave. Hence, it is also called as technique of V to T conversion.

Pulse Width Demodulation

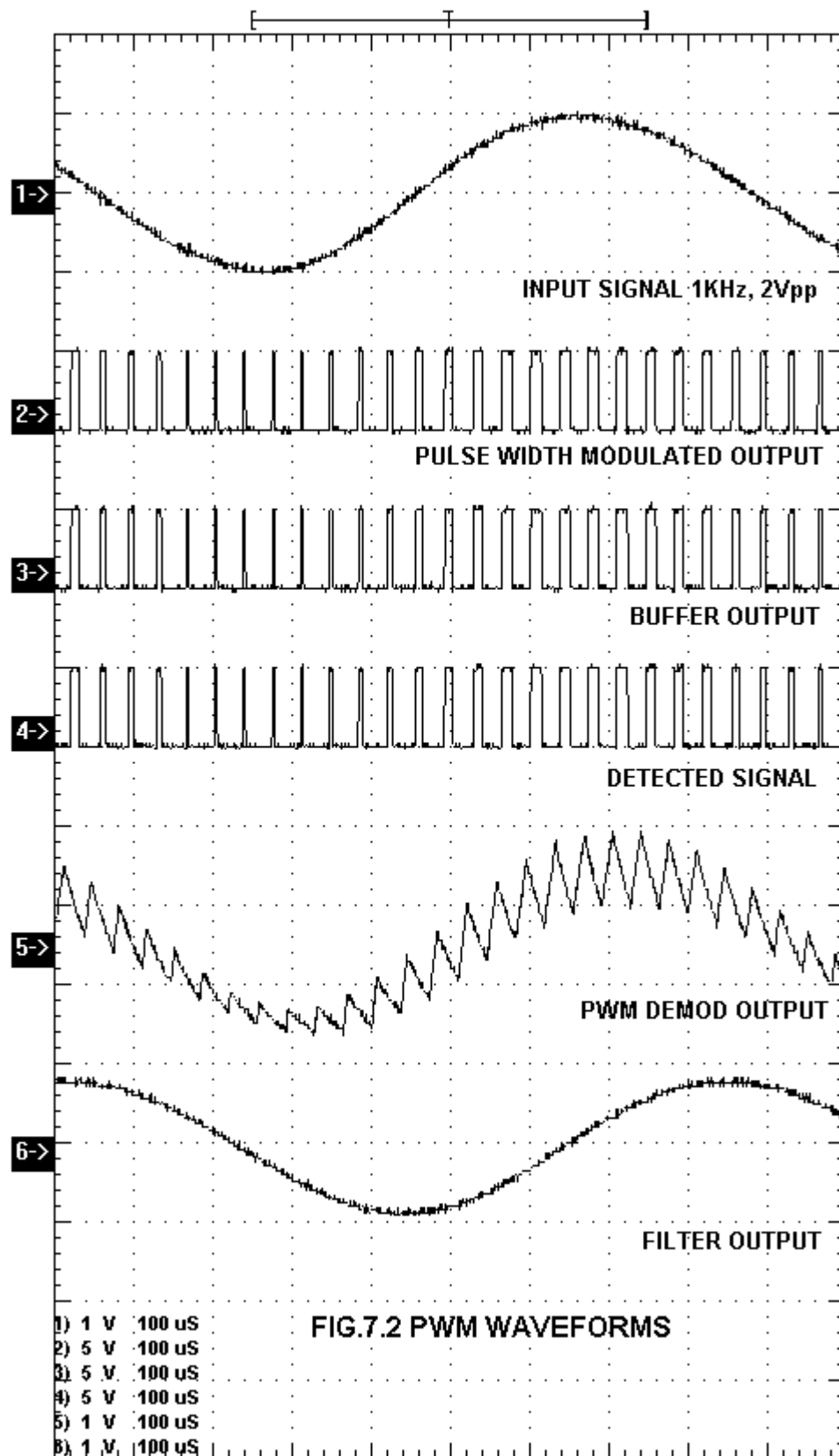
The input signal is pulse width modulated, so the ON time of the signal is changing according to the modulating signal. In this demodulation technique, the PWM signal is applied to an Integrator, whose output is then Filtered to obtain original signal.

EQUIPMENTS:

- FOL-B-P kit
- Dual Channel oscilloscope
- 1 Meter Fiber cable
- Power Supply

PROCEDURE:

- Slightly unscrew the cap of SFH756V (660 nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
- Make the connections and jumper settings as shown in **FIG. 7.1** Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF. Now switch on the power supply.
- Connect **SINE** post (**FIG. 7.2a**) of the Function Generator section to **PWM IN** post of PWM/ PPM Modulator Section.
- Select the frequency 100Hz-1kHz with amplitude of 1V p-p (Max) for proper observation of phenomena.
- Observe PWM signal at **PWM OUT** Post (**FIG. 7.2b**). Variation in width of square wave is seen clear. Then keep frequency range upto 1KHz, 1Vp-p, if the frequency is high then due to persistence of vision, only blurt band in the waveform will be observed. If the Function generator is OFF, only square wave of fundamental frequency and fixed ON time will be observed and no width variations are present.
- Connect **PWM OUT** post of PWM/PPM Modulator Section to **IN** post of Digital Buffer Section.
- Connect **OUT** post of the Digital Buffer Section to **TX IN** post of TRANSMITTER.
- Connect the other end of fiber to detector SFH551V (Digital Detector) very carefully as per the instructions in step 1.
- Observe the received signal over fiber at **TTL OUT** post it should be exactly similar to the signal available at **PWM OUT** post.
- Slide the switch SW 8 to PWM position.
- Connect this **TTL OUT** post to **PWM DEMOD IN** Post in PWM / PPM Demodulator Section.
- Vary input freq. POT **P23** and observe demodulated signal at **DEMOD OUT** post (**FIG. 7.2c**) of PWM / PPM Demodulator Section. Connect **PWM / PPM DEMODOUT** post to **IN** post of Filter Section and observe output at its **OUT** post (**FIG. 7.2d**), which is same as Input signal.
- For Different Sampling frequencies change the jumper cap of **JP1** from 8 KHz to the desired value of frequency. You can observe the PWM output clearly at lower sampling frequency demodulated **PWM OUT** is more distorted at lower sampling frequency.
- Observe the waveforms at different test points as shown in **FIG. 7.2**
- Repeat the above all procedures for SFH450V.



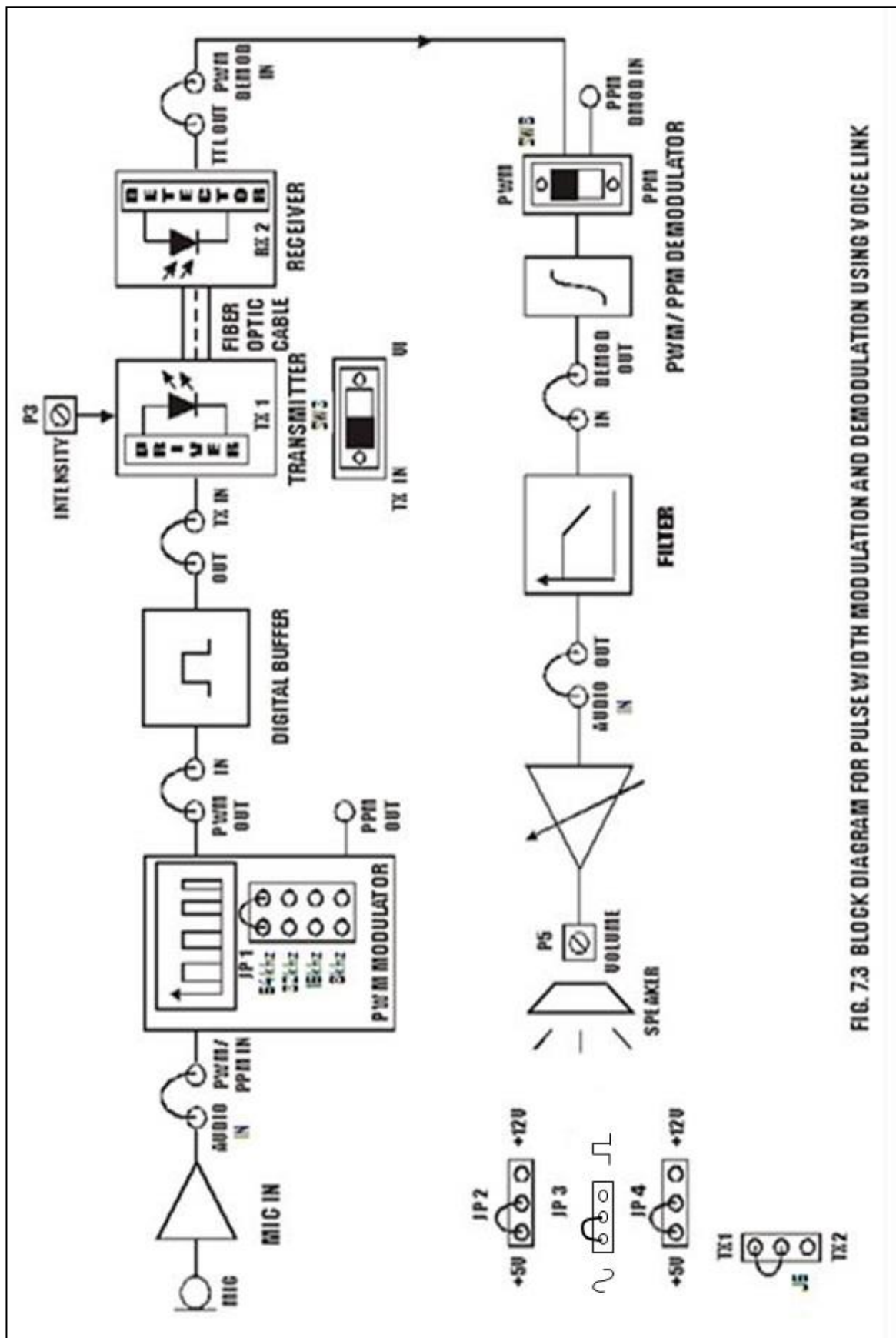


FIG. 7.3 BLOCK DIAGRAM FOR PULSE WIDTH MODULATION AND DEMODULATION USING VOICE LINK

Setting Up Fiber Optic Voice Link Using Pulse Width Modulation

PROCEDURE

- Make the connection on the Kit as per the Connection Diagram **FIG. 7.3**. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
- Connect the microphone provided with the kit to the socket marked **MIC** on the kit.
- Connect **AUDIO OUT** post to **PWM / PPM IN** post.
- Connect **OUT** post in Filter Section to **AUDIO IN** post.
- Connect speaker provided with the kit to the socket marked **SPEAKER** on the kit.
- Switch on the power supply.
- Speak from the Microphone and you could hear your sound from the speaker. Adjust Volume Pot **P5** to setup fiber optic audio link.

EXPERIMENT NO. 8

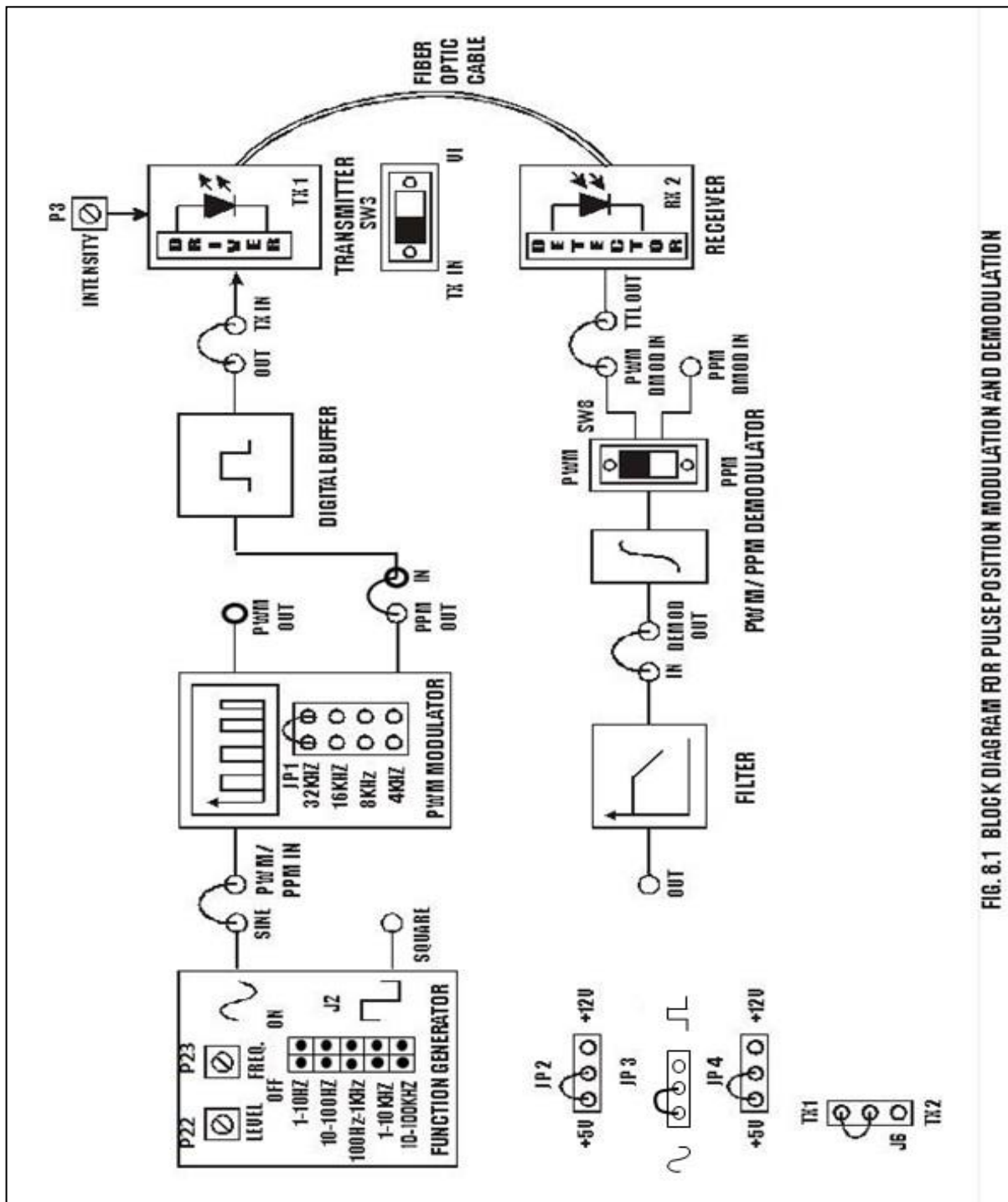


FIG. 8.1 BLOCK DIAGRAM FOR PULSE POSITION MODULATION AND DEMODULATION

EXPERIMENT NO. 8

NAME:

Study of Pulse Position Modulation & Demodulation

OBJECTIVE:

The objective of this experiment is to study the circuit action of Pulse Position Modulation & Demodulation over Fiber Optic Digital Link.

THEORY:

Modulation

The position of the TTL pulse is changed on time scale according to the variation of input modulating signal amplitude.

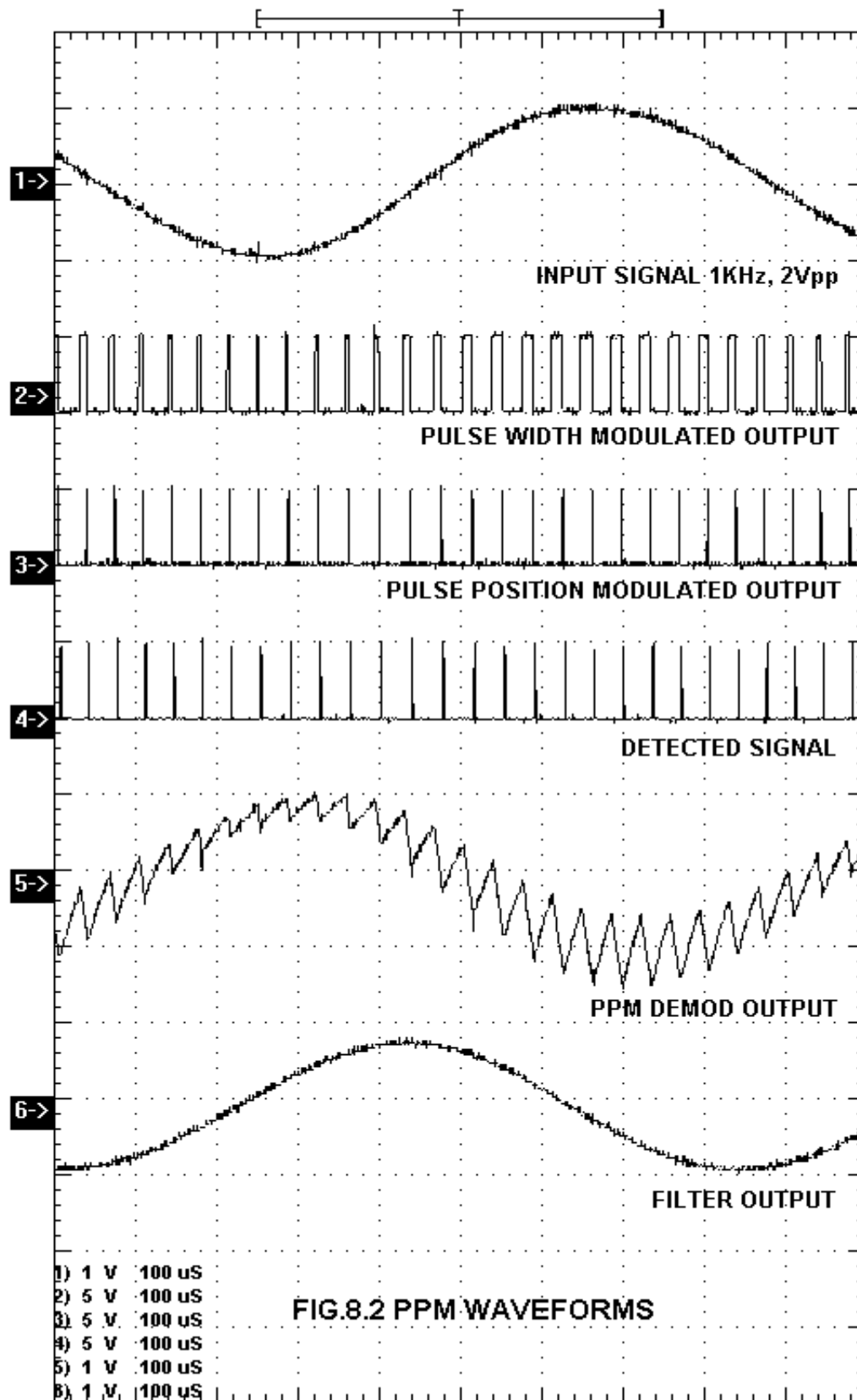
Now pulse width modulated signal is fed as input to this circuit. Please note that input modulating signal must be converted into pulse width modulated form before applying to pulse modulator. As the signal is PWM, naturally, according to the input signal, the pulse duration is changing and this change in pulse duration causes for the delay in triggering. The input is given to trailing edge trigger input of monoshot. So finally we get the pulses at the output, which are shifted on the time slot. This is nothing but pulse position modulation. The Pulse Positions are directly proportional to the instantaneous values of modulating signal.

Demodulation

The pulse position modulated signal is ORed with pulse generated by the rising edge of pulse width modulated signal. The o/p of the OR gate is fed to clk I/p of flip-flop. Thus flip-flop acts as a bistable multivibrator giving out high o/p for the duration between rising edge of PWM signal & PPM signal. Since PPM corresponds to the end of PWM pulse, o/p of flip-flop is exactly same as that of PWM signal. This signal is then demodulated using the same technique of PWM demodulation as described in previous experiment.

EQUIPMENTS:

- FOL-B-P kit
- Dual Channel Oscilloscope
- 1-Meter Fiber Cable
- Power Supply



PROCEDURE:

- Slightly unscrew the cap of SFH 756V (660 nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
- Make the connections and jumper settings as shown in **FIG. 8.1** Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
- Connect **SINE** post of the Function Generator section to **PPM IN** post of PWM/PPM Modulator Section.
- Keep the Function Generator in sine wave mode & select the frequency 1-10Hz with amplitude of 2V p-p (Max) for proper observation of phenomena.
- Connect **PPM OUT** post of PWM/PPM Modulator section to **IN** post of Digital Buffer Section.
- Switch on the power supply.
- Observe PPM signal at **PPM OUT** post by connecting IST Channel of CRO at **PPM OUT** post. Refer **FIG. 8.2c**. Variation in width of square wave is seen clear by connecting IIND Channel of CRO at **PWM OUT** post. If the frequency is high i.e. frequency is 1 KHz having Level 2Vp-p then due to persistence of vision, only blurt band in the waveform will be observed. If the Function generator is **OFF**, only square wave of fundamental frequency and fixed ON time will be observed and no width position variations are present.
- Connect **OUT** post of the Digital Buffer Section to **TX IN** post of TRANSMITTER.
- Connect the other end of the fiber to detector SFH551V (Digital Detector) very carefully as per the instructions in step 1.
- Observe the received signal over fiber at **TTL OUT** Post. It should be exactly similar to the signal available at **PPM OUT** post.
- Connect this **TTL OUT** post to **PPM DEMOD IN** Post in PWM / PPM Demodulator Section.
- Vary input freq. (not more than 3 KHz) & observe demodulated signal at **DEMOD OUT** post (**FIG. 8.2d**).
- Connect **DEMOD OUT** post to **FILTER IN** post & observe output at **FILTER OUT** post (**FIG. 8.2e**), which is same as Input signal (**FIG. 8.2a**).
- For Different Sampling frequencies change the jumper cap of **JP1** from 8 KHz to the desired value of frequency.
- Observe the waveforms at different test points as shown in **FIG. 8.2**
- Repeat the above all procedures for SFH450V.

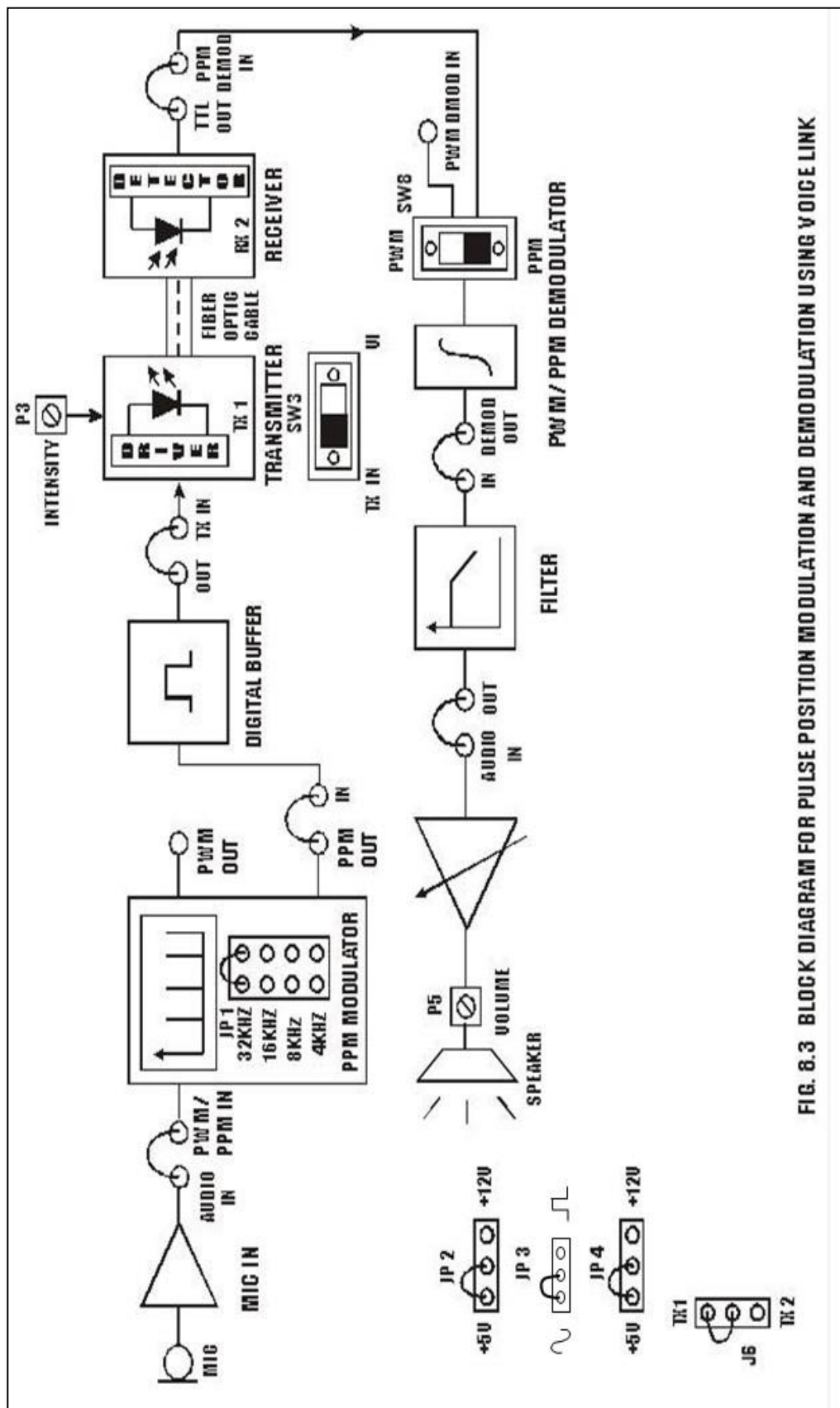


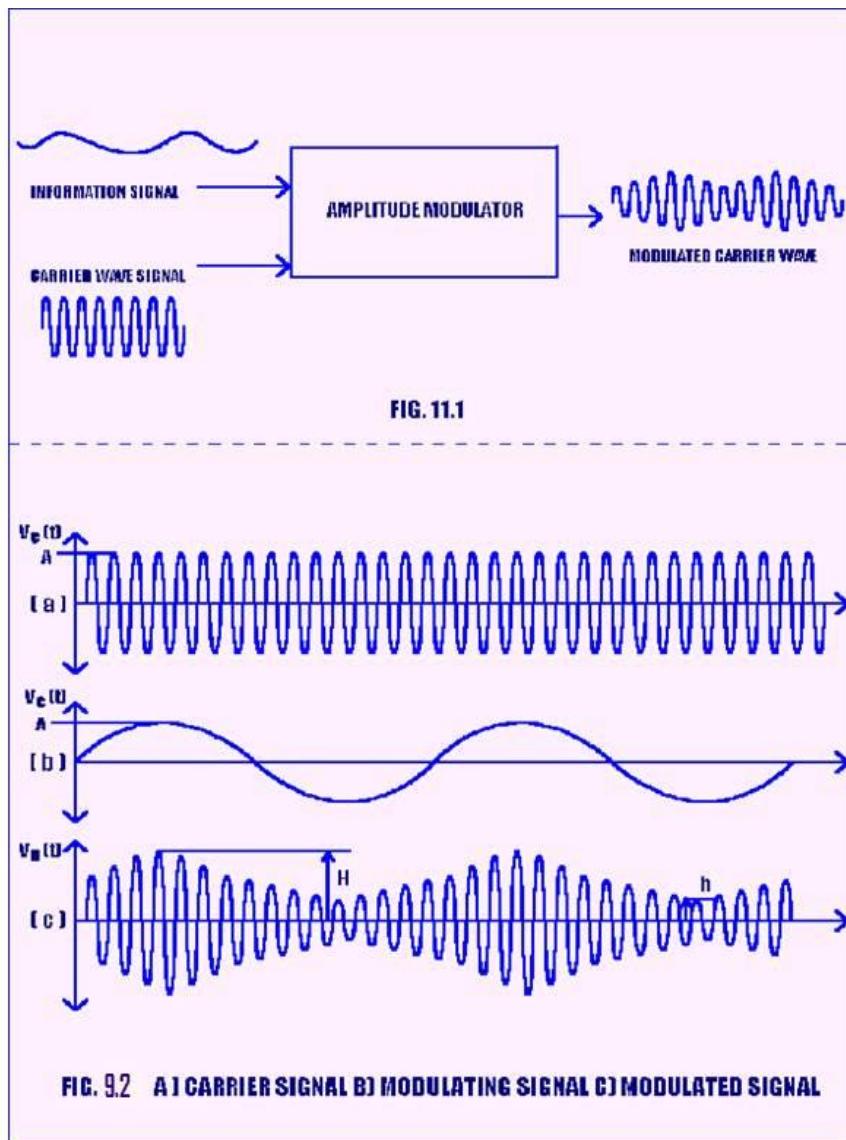
FIG. 8.3 BLOCK DIAGRAM FOR PULSE POSITION MODULATION AND DEMODULATION USING VOICE LINK

Setting Up Fiber Optic Voice Link Using Pulse Position Modulation

PROCEDURE

- Make the connection on the Kit as per the Connection Diagram **FIG. 8.3**. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
- Connect the microphone provided with the kit to the socket marked **MIC** on the kit.
- Connect **AUDIO OUT** post to **PWM / PPM IN** post.
- Connect **PPM OUT** post in Filter Section to **AUDIO IN** post.
- Connect speaker provided with the kit to the socket marked **SPEAKER** on the kit.
- Switch on the power supply.
- Speak from the Microphone and you could hear your sound from the speaker. Adjust Volume Pot **P5** to setup fiber optic audio link.

EXPERIMENT NO. 9



EXPERIMENT NO: 9

NAME:

Study of Amplitude Modulation & Demodulation using Fiber Optic Cable

OBJECTIVE:

The objective of this experiment is to study Amplitude Modulation and Demodulation through fiber optic analog link.

THEORY:

Amplitude Modulation

In Amplitude Modulation the amplitude of high frequency sine wave (carrier) is varied in accordance with the instantaneous value of the modulating signal. Refer Fig.9.1. Consider a sine signal $v_m(t)$ with frequency f (Fig.9.2).

$$V_m(t) = B \cdot \sin(2\pi f \cdot t)$$

And another sine signal $v_c(t)$ is called modulating signal, the signal $v_c(t)$ is called carrier signal.

$$V_c(t) = A \cdot \sin(2\pi F \cdot t)$$

The signal $v_m(t)$ is called modulating signal; the signal $v_c(t)$ is called carrier signal.

Vary the amplitude of the carrier $v_c(t)$ adding the modulating signal $v_m(t)$ to A . You obtain a signal $v_M(t)$ amplitude modulated, which can be expressed by:

$$v_M(t) = [A + k \cdot B \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t) = A \cdot [1 + m \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t)$$

With k = constant of proportionality.

Percentage modulation signal is defined as the value:

$$m = \frac{K \cdot B}{A} \cdot 100$$

with reference to FIG.9.2, the modulation index m can be calculated in this way

$$m = \frac{H-h}{H+h} \cdot 100\%$$

Envelope Detector

The extraction of the modulating signal from an AM signal can be carried out using an envelope detector. Consider the AM signal shown in Fig. 9.2, and note that the modulating signal constitutes the envelope of the waveform reported.

The most common envelope detector consists of a diode followed by a RC filter. Its operation is analogous to the half-wave rectifier, as the output voltage follows the maximum values of the carrier. As the amplitude of the carrier is variable, by properly choosing R and C , the output of the detector can faithfully reproduce faithfully these variations.

Distortions Of The Detected Signal

The demodulated signal can have two kinds of distortions:

If the time constant $R \cdot C$ is too small with respect to the period of the carrier, the envelope is approximated to a waveform affected by ripple, the more evident the lower is the value of $R \cdot C$

If the time constant $R \cdot C$ is too high with respect to the period of the modulating signal, the detected signal always follows the behavior of the envelope, but sometimes it takes an exponential decreasing law (distortion by diagonal cutting).

The distortion presented by the detected signal is known as Distortion by diagonal cutting.

The maximum value of $R \cdot C$ is calculated supposing that the capacitor C discharges on the resistor R to a higher or equal rhythm to the one in which the envelope of the modulated signal drops. The detected condition is as follows:

$$R \cdot C \leq \frac{1}{f_{\max}} \cdot \frac{\sqrt{1-m \cdot 2}}{m}$$

with $m =$ modulation index
 $f_{\max} =$ maximum frequency of the modulating signal.

Note that for $m=1$ the condition cannot be satisfied and so the detected signal will be certainly distorted.

Detection Efficiency

- The efficiency η of the diode envelope detector is defined as the ratio between the amplitude of the output signal of the detector and the amplitude of the envelope of the input AM signal. Supposing that at the carrier frequency, the reactance of the capacitor $[1/(2\pi \cdot F \cdot C)]$ is much smaller than the resistance R (hypothesis checked if $RC \gg 1/F$ to reduce the ripple), the efficiency η practically depends on the relation between R and the differential resistance r_d of the diode.

•

Last Considerations On The Choice Of $R \cdot C$ In The Envelope Detector

- To obtain high detection efficiency the resistance R must be much higher than the differential resistance r_d of the diode ($R \gg r_d$).
- To minimize the ripple in the detected signal the product $R \cdot C$ must be much higher than the period T of the carrier ($R \cdot C \gg T$).
- To avoid distortion by diagonal cutting the product $R \cdot C$ must be smaller than or equal to a value, which depends on the modulating frequency and on the modulation index. In case of the modulating signal with variable amplitude and frequency, the maximum values of the frequency as well as of the modulation index must be considered.

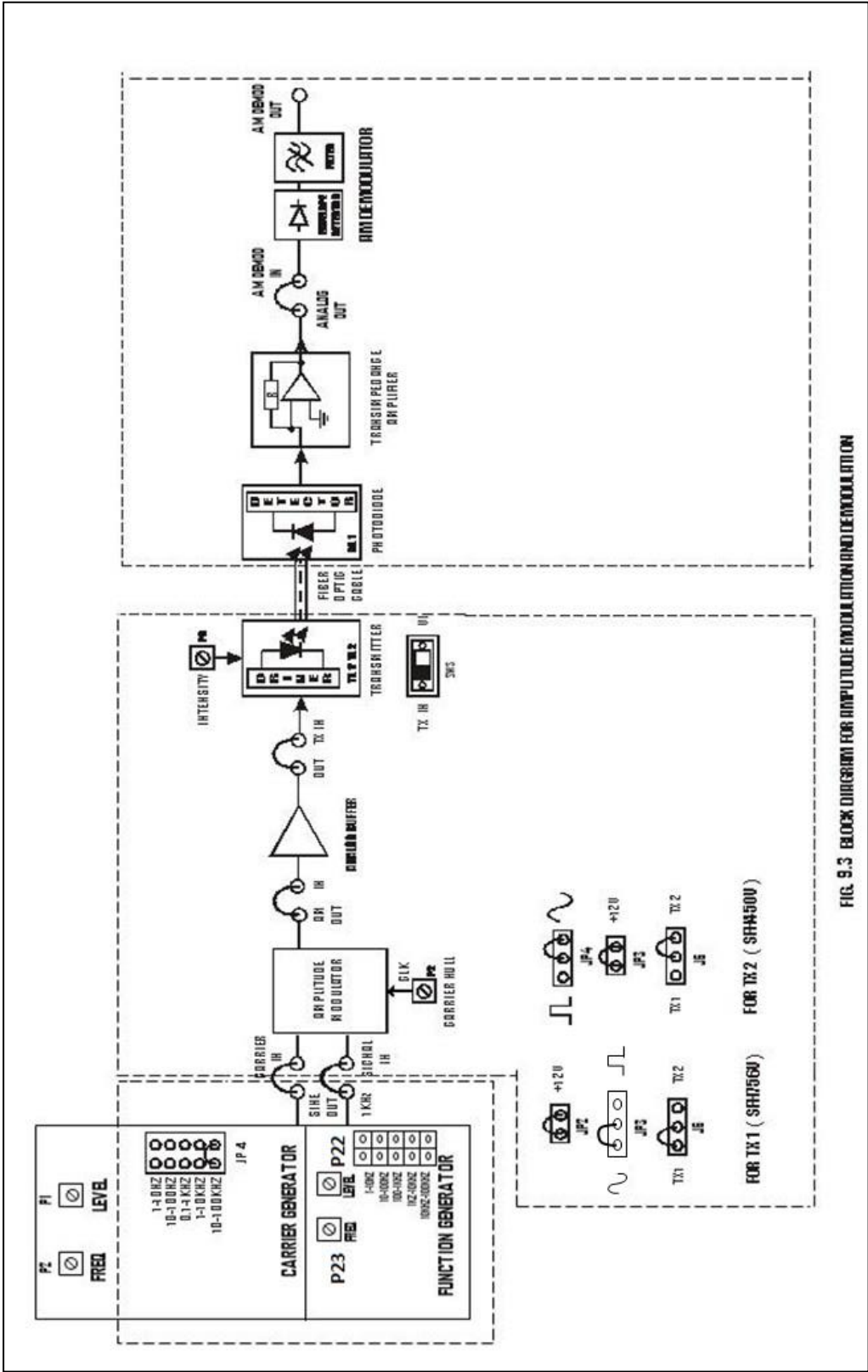
EQUIPMENTS:

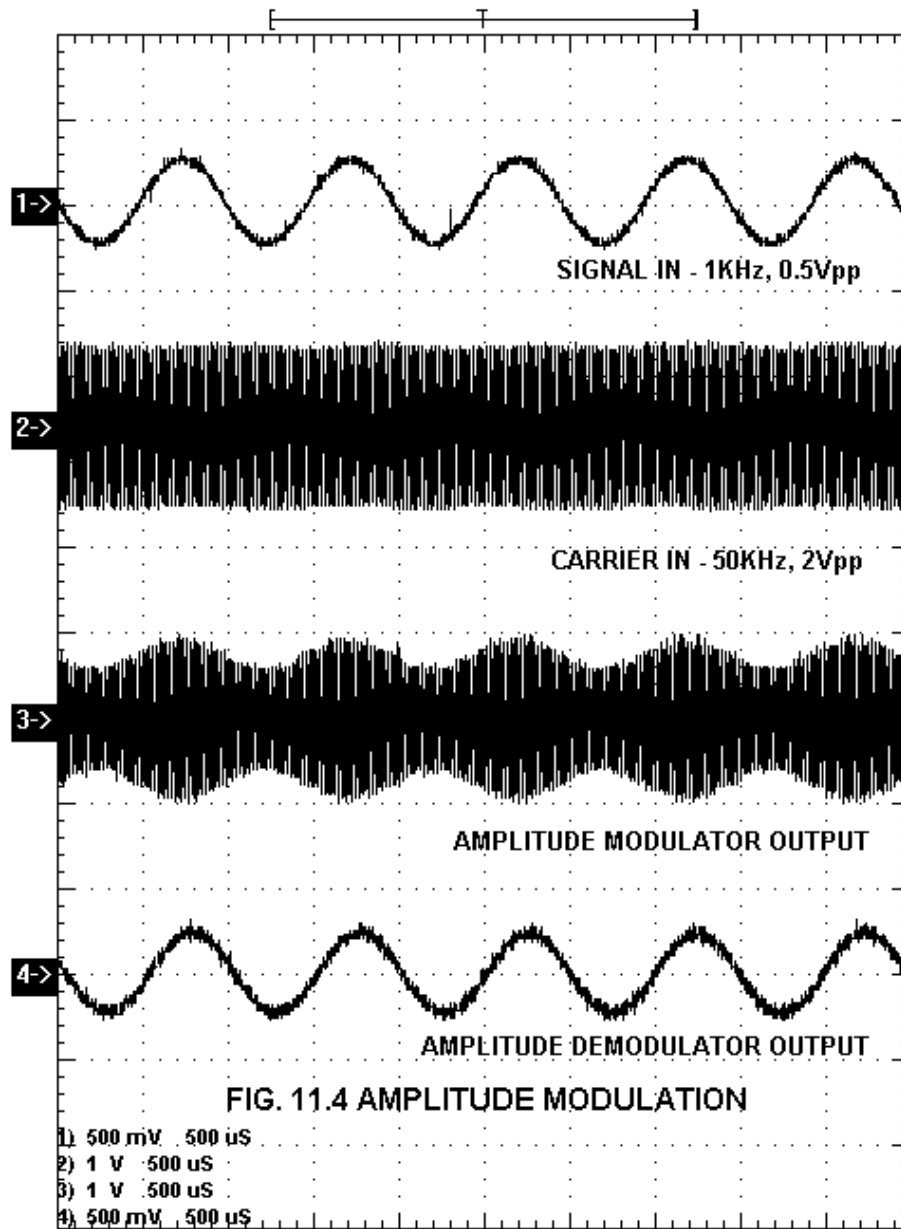
- FOL-B-P
- MHz Dual Channel Oscilloscope
- 1 meter Fiber cable
- Patch chords
- Power supply

PROCEDURE:

Refer to **FIG 9.3** and make the following connections.

- Connect the power supply with proper polarity FOL-B-P. While connecting this, ensure that the power supply is OFF.
- Keep the jumpers **JP2** on **+12V** & **JP3** on **1-2**.
- Keep switch SW3 in TX IN position.
- Switch on the power supply.
- Connect the **1KHz, 0.5Vpp** Signal from Function Generator as a constant signal to the **SIGNAL IN** post of Amplitude Modulator section.
- Keep **CARRIER** signal of about **50KHz** with amplitude of **2Vpp**.
- Connect the post **CARRIER** of Carrier Generator to post **CARRIER IN** of Amplitude Modulator section.
- Connect the output of Amplitude Modulator section post **AM OUT** to post **IN** of Analog Buffer . Keep pot **CARRIER NULL** fully clockwise position.
- Connect the output of Analog Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1-meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect the **ANALOG OUT** post from TRANSIMPEDENCE AMPLIFIER section to **AM DEMOD IN** post from AM DEMODULATION section.
- Observe AM Signal and Amplitude Demodulated signals at the output of AM Modulator & Demodulator Section respectively.





**EXPERIMENT
NO. 10**

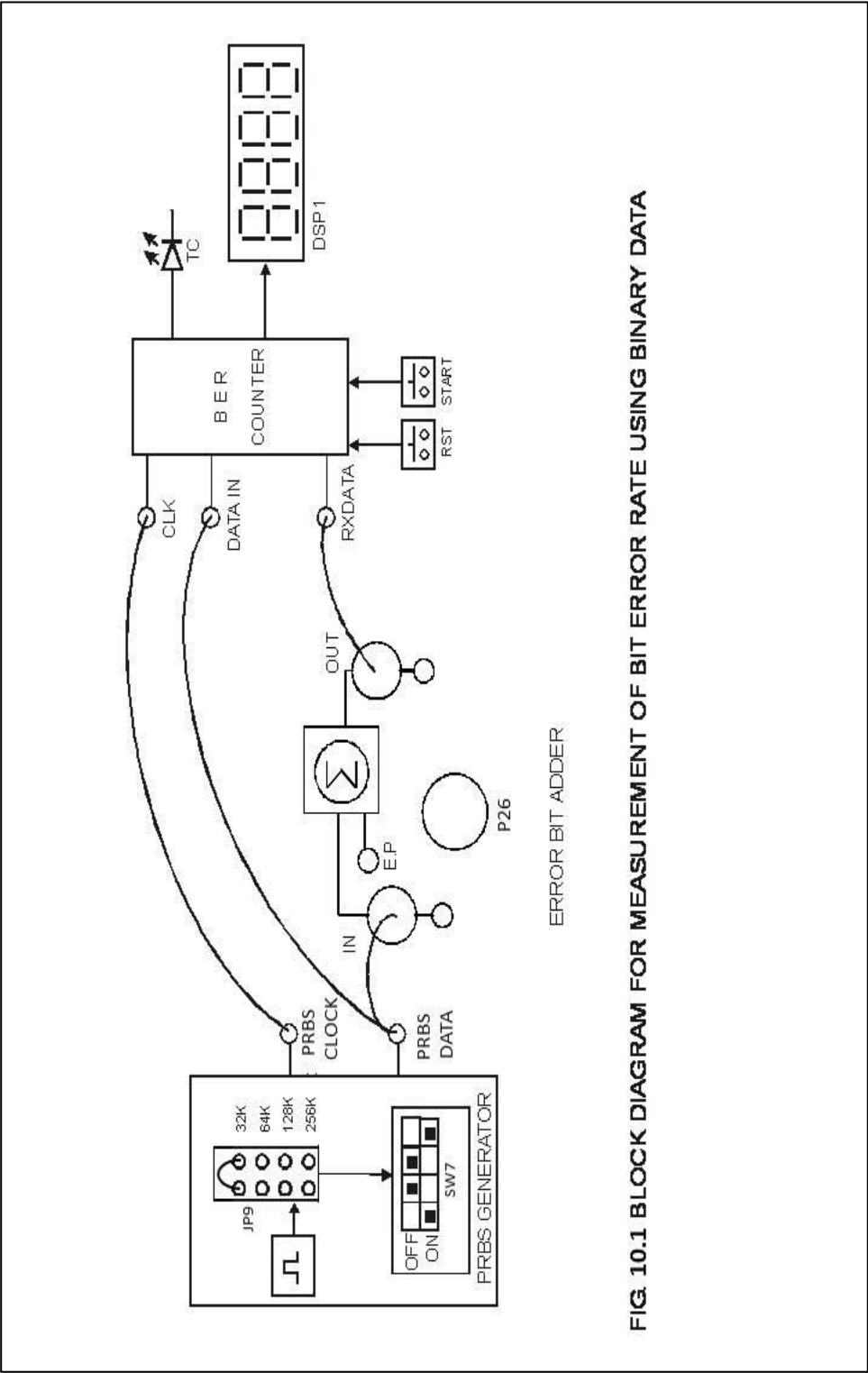


FIG 10.1 BLOCK DIAGRAM FOR MEASUREMENT OF BIT ERROR RATE USING BINARY DATA

EXPERIMENT NO: 10

NAME:

Measurement of Bit Error Rate Using Binary Data

OBJECTIVE:

To measure bit error rate.

THEORY:

In telecommunication transmission, the bit error rate (BER) is a ratio of bits that have errors relative to the total number of bits that have errors relative to the total number of bits received in a transmission. The BER is an indication of how often a packet or other data unit has to be retransmitted because of an error. Too high a BER may indicate that a slower data rate would actually improve overall transmission time for a given amount of transmitted data since the BER might be reduced, lowering the number of packets that had to be resent.

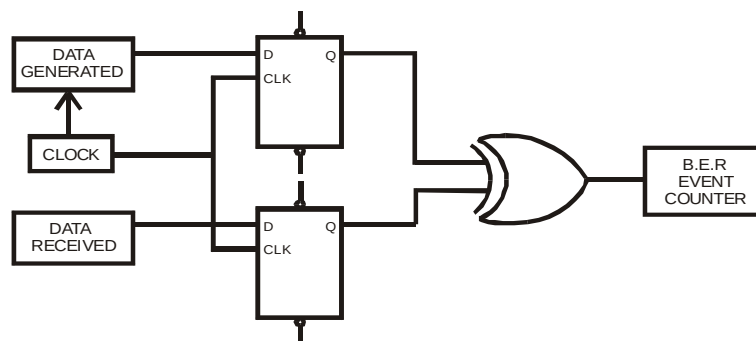
Measuring Bit Error Rate

A BERT (Bit Error Rate Tester) is a procedure or device that measures the BER for a given transmission. The BER, or quality of the digital link, is calculated from the number of bits received in error divided by the number of bits transmitted.

$$\text{BER} = (\text{Bits in error}) / (\text{Total bits transmitted})$$

Using a bench test setup, this is easily measured by means of a comparator in which the transmitted bits are matched in an XOR gate with the received bits.

Fig shows the schematic of the device used for the following measurements.



BIT ERROR RATE TESTER (B.E.R.T.)

If the bits are alike at the XOR gate inputs when clocked in from the D flip-flop, the output is low. If they are different, the XOR output goes high, causing an event count.

A random character generator and white noise source should be used for these measurements.

The number of bit errors is dependent upon the amount of noise entering the system.

- The white noise, or background noise, has an average or RMS value that is exceeded periodically by peaks that rise many times that level. This peak exists only for a short

period of time. When the peak equals or exceeds the signal level, that is noise energy = bit energy, there is a 50/50 chance of error. The peak time periods can be calculated statistically from the error function.

EQUIPMENTS:

- FOL-B-P kit
- Connecting Chords
- Power supply
- 20MHz Dual Trace Oscilloscope

PROCEDURE:

- Refer to the block diagram (Fig.10.1) and carry out the following connections and switch settings.
- Connect power supply in proper polarity to the **FOL-B-P-II** Kit and switch it on.
- Keep the position of the switch **SW7** of the **PRBS GENERATOR** as shown in the Fig.5.1 for the generation of the PRBS pattern.
- Select the **PRBS** clock frequency to **32 KHz** using jumper **JP9**.
- Connect the **PRBS DATA** post to the **IN** post of **ERROR BIT ADDER**.
- Observe the noise pulses at **OUT** post of **ERROR BIT ADDER** by varying the pot **P26**.
- Connect the **OUT** post of the Adder to the **RX DATA** post of the BER meter.
- Keep the pot **P26** of the **ERROR BIT ADDER** at minimum position.
- Connect the **PRBSCLK** post to the **CLK** post of the BER meter.
- Connect the **DATA OUT** post of **PRBS GENERATOR** to **DATA IN** post of the **BER METER**.
- Press the **RST** switch to reset the counter to zero.
- Press the **START** switch to start the counting.
- Observe the error count on the display after 10 seconds upon pressing **START** switch.
- Observe that when the pot is kept at minimum position the bit error count is very low. The error count will increase linearly as we increase the pot to the maximum (clockwise) position.

BER MEASUREMENT

The noisy data from the OUT post of the Adder is connected to the RX DATA IN of the BER meter. As BER is the ratio of error bits (E_b) to total bits transmitted (T_b) in a period time t sec i.e.

$$\text{BER} = E_b/T_b$$

E.g. If PRBS data is transmitted at 32Kbits per sec for a period of 10 sec.

The total bits transmitted in 10 sec (T_b) = 320 bits.

The TTL out data with noise is fed to BER counter, which compares the two data input at each clock input.

The counter gives the 10 bit binary from error count (E_b) which is converted into decimal form and displayed.

E. g.

(0000001010 = 10 (decimal form))

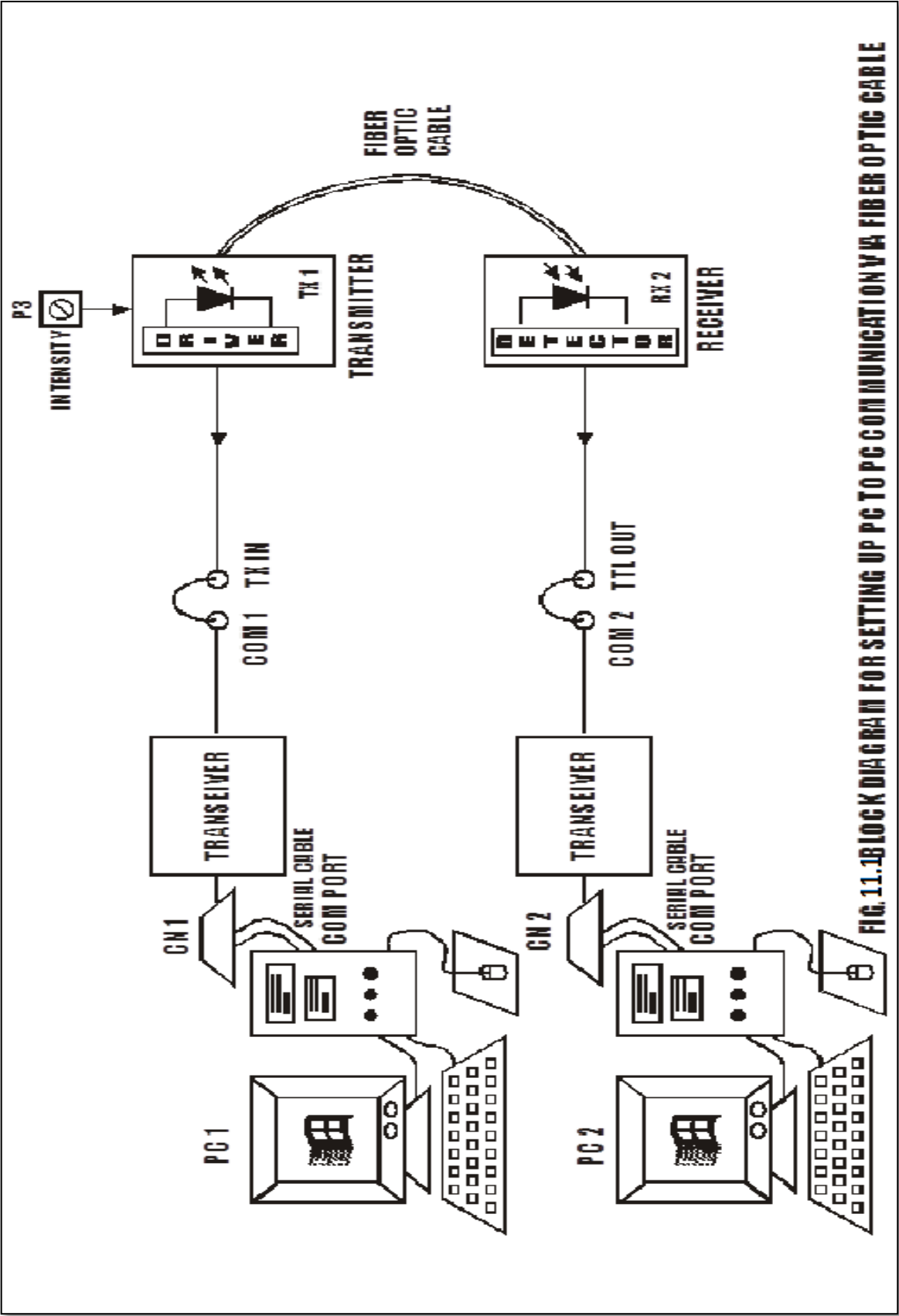
BER RATIO then becomes:

$$\text{BER} = 10/320 \times 10^3$$

$$= 0.00003125$$

i.e. the channel Bit Error Rate ratio is 3.1×10^{-5} (3/100000) or in other words we can say that out of 100000 bits transmitted through the channel, the channel gives 3 bits in error.

**EXPERIMENT
NO. 11**



EXPERIMENT NO.11

NAME:

Study of RS-232 Serial Communication between Two Computers Using Fiber Optic Digital Link

OBJECTIVE:

The objective of this experiment is to connect the RS-232 ports of two computers using Optical Fiber Digital Link, transmit data from one computer over this link and receive the same data on the other computer.

THEORY:

Microprocessor is a parallel device. It transfers the 8,16 or 32 bit of data simultaneously over the data lines. The number of data lines depends upon the type of microprocessor used in the system. This is parallel I/O mode of the data transfer.

However in many situations the parallel data transfer is either impractical or impossible. This is very expensive and noisy especially when the distances are large. Also some devices such as CRT or CTD are not designed for parallel I/O. Moreover in many scientific and industrial process control applications, the devices under control are at the site or plant, which may be long enough from control room. In these situations, the serial I/O mode is used where only one bit at a time is transferred over a single cable. This cable may be a normal cable or an optical fiber.

Very important advantage of serial mode of data transfer is that it is inexpensive. Also the data is accurately transferred and received in the link. It is daily practice to put checks for the data and framing it. Uninterrupted data transfer is greatest advantage of serial mode and exactly this is the reason behind the fact that serial mode is preferred in many applications. This plays vital role in many applications like PC-to-PC data communication, Industrial Process controls, Robotics, CNC and DNC (Distributed Numerical Control) and many more.

So it is necessary to have some system, which will perform serial I/O operation between PC and outside device using optical fiber link. And exactly Fiber FOL-B-P fulfills this need. It provides the simplest and powerful way for serial communication through optical fiber. It is very easy to install and use. Also one can enhance its flexibility through software.

Function Of MAX 232 (RS-232C Transceiver)

The computer communicates from serial COM port, which is at RS232C levels i.e. at 12V. Transceiver MAX 232 performs the function of converting RS232C signals to TTL levels & vice versa.

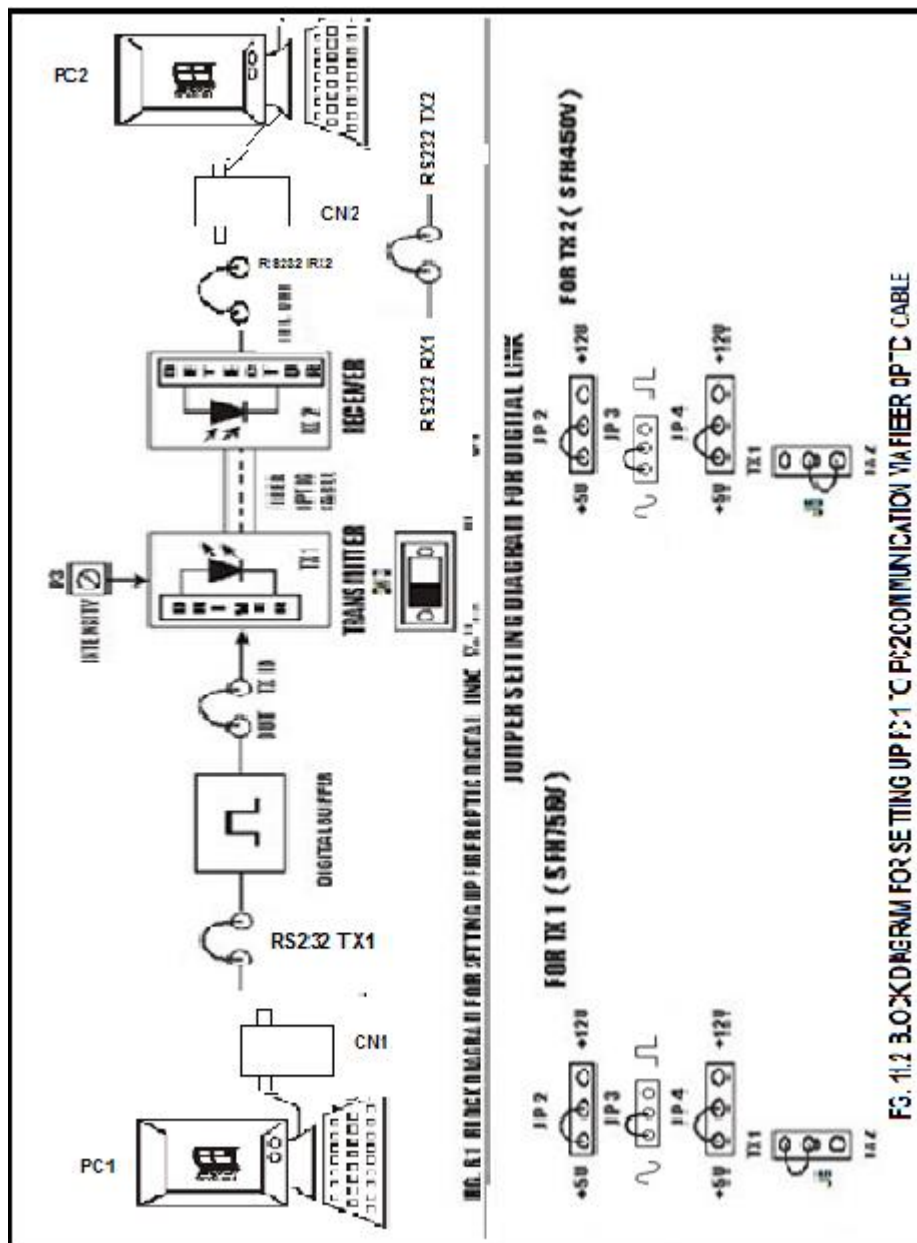


FIG. 4.1.2 BLOCK DIAGRAM FOR SETTING UP F3:1 C PC2 COMMUNICATION VIA FIBER OPTIC CABLE

Hardware Settings

Switch off the power supply of PC.

To perform this expt. the COM ports of PC are used. On board 9-pin D-type (female) connectors are provided for interfacing with the PC. Two 10-pin FRC cables with one end clamped with 10-pin FRC connector and the other end connected to 9-pin D-type male connectors are provided with this kit. Connect FRC connector end of one cable to one of the COM ports of PC and the 9-pin D-type connector end to CN1 (9-pin D-type connector). Similarly connect other FRC cable to other port & CN2.

EQUIPMENTS:

- FOL-B-P KIT
- 1-Meter Fiber Cable
- USB Type A TO B Cables – 2 No's
- Computers – PC, PC/XT, 386 or 486 –two No's (Minimum Configuration)
- Power Supply

PROCEDURE:

1. Slightly unscrew the cap of LED SFH756V (660 nm) on kit. Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
2. Make the jumper settings and connection diagrams as shown in **FIG. 11.1** & **FIG. 11.2**.
3. Refer to **FIG. 11.2** of this experiment and make the necessary connections or connect one end of the 9 to 9 pin cable to PC **COM1** port and other end to **CN1** connector on kit then connect second 9 to 9 pin cable one end to second PC **COM1** port and other end to **CN2** connector on kit. Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF.
4. Connect **CN 1** post on the KIT (RS-232 section) to **IN** post of Digital Buffer Section.
5. Connect **CN 2** post on the KIT (RS-232 section) to **TTL OUT** post of Receiver Section.
6. Switch on the PCs and the power supply of the Kit.
7. After putting ON one of the PC, go to START MENU, PROGRAMS, ACCESSORIES, COMMUNICATION and then Click on HYPER TERMINAL.
8. A new Window will open, where in you Double Click on HYPERTRM, Two Windows will open, one at the background and another (small window) with title Connection Description which will be Active.
9. Enter the name in the box by which you would like to store your connection, for e.g. (PC2PC), and Click OK. Also you could select the Icon provided below. The background window title will change to the name provided by you.

Connect To ? X

 PC2PC

Enter details for the phone number that you want to dial:

Country code: India (91)

Area code:

Phone number:

Connect using: Direct to Com1

OK Cancel

COM1 Properties ? X

Port Settings

Bits per second: 115200

Data bits: 8

Parity: None

Stop bits: 1

Flow control: Xon / Xoff

Advanced... Restore Defaults

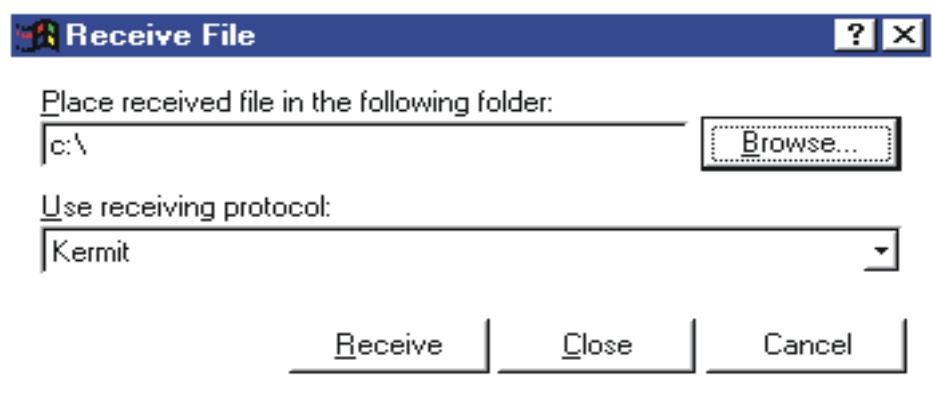
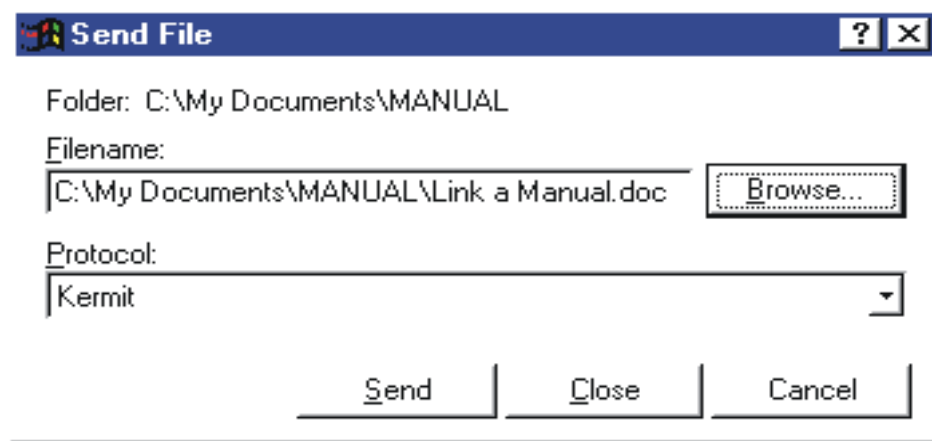
OK Cancel Apply

10. Then specify connect using: by selecting **Direct to COM1** or port where your cable is connected and then click on **OK**. See **FIG. 11.3**.
(NOTE: Please check the Port you have selected and the Ports you are connecting).
11. Now Window with Title **COM 1 Properties** will appear, where Port Setting should be done as shown below and click on **OK**. See **FIG. 11.4**.

BITS PER SECOND	UP TO 115200
Data Bits	8
Parity	None
Stop Bits	1
Flow Control	Xon / Xoff

(NOTE: For Bits persecond setting you could select them for different speeds. Do not exceed it above 115200 bps).

12. After the above settings you click **OK**. The Background window will become Active.
13. Click on **File, Save As**, and save it in the Directory, which you want.
Perform the same procedure (from 8. to 14.) on the computer to with whom you want to communicate.
14. To start communicating between the two PCs Click on the **TRANSFER** Menu and again click on **Send File**. A window will be prompted having title Send File with **File Name** and **Protocol**. See **FIG. 11.5**.
15. Select **Browse** for the file, which you would like to send to the PC connected, select the file and Click on **Open**, the file name and address will be displayed in the small window. Then select the **Kermit** Protocol, (optional use protocols are X modem, Y modem and 1K Xmodem.)
16. To receive the file on the PC Click on the **TRANSFER** Menu and again click on **Receive File**. A window will be prompted having title Receive File with Location at which you want to store the Received file and Receiving Protocol. See **FIG. 11.6**.
Select **Browse** for the location where you would like to store the received file, select the folder and Click **OK**, the folder name and address will be displayed in the small window. Protocol to be selected should be **Kermit** and same as file transmitting PC.
17. On the PC from which the selected file to be transmitted, Click on **SEND** .A window will open showing file transfer status. Immediately at the Receiving PC Click **Receive** (otherwise Time Out Error will be displayed and communication will fail) .You will see a window showing file is begin received in the form of packets. See **FIG. 11.7**.
18. After file is transferred both the windows in the (transmitting & receiving PCs) will close. Check for the received file in the folder where the file is stored.
You can do this procedure vice-versa to transfer the file.



NOTE

To change the bits per second rate (baud Rate) goes to **properties** in the Hyper Terminal window, Click on **File Menu, Properties and configure**, in the bits per second select other rates. From 2400 to 115200 bps. You will observe that the rate at which the file is transferred will vary with the selected baud rate.

Also you can observe and check that by removing fiber from detector packet increment and file transfer stops.

Also ensure that at the left corner of the hyper terminal window connected time must be ON, if disconnected label is there go to call option and press call.

NOTE

If HyperTerminal is not installed in your computer then, on PC Go to My Computer , Control Panel , Add Remove Programs, Windows Setup, Communication, Check Hyper Terminal and Click OK

Zmodem with Crash Recovery file send for pc2pc			
Sending: D:\datasheet\FIBER DATASHEET\detector\detector.pdf			
Last event:	Sending	Files:	1 of 1
Status:	Sending	Retries:	0
File:		90k of 284K	
Elapsed:	00:00:08	Remaining:	00:00:17
		Throughput:	115200 bps
		Cancel	cps/bps

Zmodem with Crash Recovery file receive for pc2pc			
Receiving: AD673.PDF			
Storing as:	C:\Program Files\Accessories\HyperTermi	Files:	1 of 1
Last event:	Receiving	Retries:	
Status:	Receiving		
File:		113k of 279K	
Elapsed:	00:00:10	Remaining:	00:00:15
		Throughput:	112340 bps
		Cancel	Skip file
		cps/bps	

EXPERIMENT NO. 12

EXPERIMENT NO.12

NAME:

Study of Digital Panel Meter using Fiber Optic Link

OBJECTIVE:

The objective of this experiment is to study Digital Panel Meter using Fiber optic link, in which it measure current flowing through Fiber optic Transmitter (SFH756_TX1) as pot P3 gets varied.

THEORY:

Function of ICL7107S

ICL7107S is high performer, low power, 3 ½ digit AD converters. Including seven segment decoders, display driver, reference & clock. This IC content active circuitry for Digital Panel Meter on a single chip. To build a high performance panel meter with auto zero & auto polarity feature.

EQUIPMENTS:

- FOL-B-P KIT
- 1-Meter Fiber Cable
- Connecting links
- Power Supply

PROCEDURE:

- 1 Slightly unscrew the cap of LED SFH756V (660 nm) on kit. Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
- 2 Make sure INTENSITY pot should be on minimum position
- 3 Slightly unscrew the cap of SFH250V on kit. Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
- 4 Now connect **ANALOG OUT** post to **+ve** terminal of Digital Panel Meter (DPM) & **-ve** terminal to **ground**.
- 5 Keep jumper **JP2** on **+12V**, **JP3** on **SINE** position & **J6** on **TX1**.
- 6 Also keep switch SW3 on VI.
- 7 Once all connections are done DPM will display **.000** reading
- 8 As pot P3 gets varied from minimum to maximum slowly reading will display on DPM
- 9 DPM will display current flowing through SFH756V